Virtuoso® Layout Pro:T4 Advanced Commands

Course Version IC 25.1

Lecture Manual **Revision 1.0**

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Virtuoso Layout Pro: T4 Advanced Commands

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Version IC 25.1

Revision **1.0**

Estimated time: **1 Day**

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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have already completed the following courses:

- Virtuoso Layout Pro: T1 Environment and Basic Commands
- Virtuoso Layout Pro: T2 Create and Edit Commands
- Virtuoso Layout Pro: T3 Basic Commands

Or you need to have

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux

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Course Objectives

In this course, you

- Generate clones as Free Objects, Grouped Objects, and Synchronized family
- Generate synchronous copy clones of wires
- Examine synchronous behavior in the synchronous copy clones of wires created by the copy command
- Generate synchronous copy clones of modgen
- Examine synchronous behavior in the synchronous copy clones of modgen created by the copy command
- Generate mutant clones with exact connectivity turned ON/OFF with relax match options
- Create a repetitive layout cell using the clone and the copy commands
- Implement an Engineering Change Order (ECO)
- Update the net and the pin names

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Course Agenda

- About This Course
- Generating Clones
  - Exploring the Types of Clones
  - Generating Mutant Clones
  - Cloning and Copying in VLS-EXL
- Updating Connectivity and Nets
  - Implementing an Engineering Change Order (ECO)
  - Updating the Net and the Pin Names
- Course Conclusions
- Next Steps

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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/85090.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.

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  - Expand career opportunities
- Professional credibility
  - Stand apart from your peers
- For more information, go to www.cadence.com/training or email es_digitalbadge@cadence.com.

How do I register to take the exam?

- Log in to our Learning Management System, click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- Credly validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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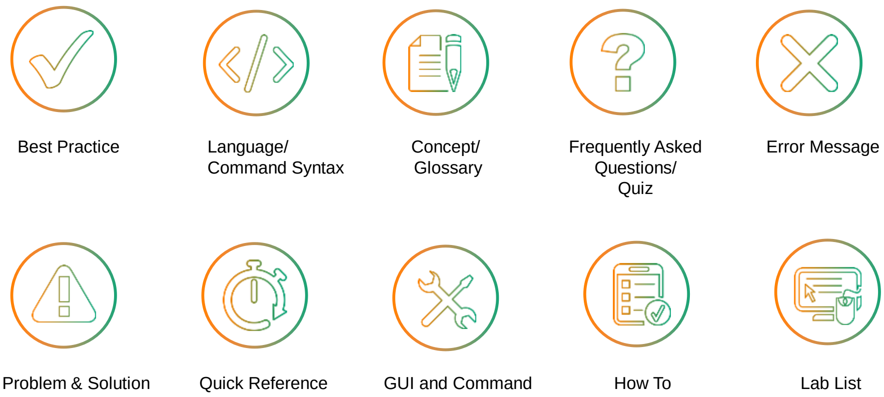
Icons Used in This Class



Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.

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Module 2

Generating Clones

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Module Objectives

In this module, you

- Generate clones as Free Objects, Grouped Objects, and Synchronized family
- Generate synchronous copy clones of wires and examine its synchronous behavior
- Generate synchronous copy clones of modgen and examine its synchronous behavior
- Generate mutant clones with exact connectivity turned on / off with relax match options
- Analyze how to effectively reuse existing structures in your layout using the clone and the copy commands in VLS-EXL

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Some Useful Acronyms and Terms Used
Virtuoso Layout Pro: T4 Advanced Commands Course

Edit In Place (EIP)	You can edit the master cell of an instance while viewing the instance in the current cellview. This is called edit in place (EIP).
Generate Selected From Source (GSFS)	Populates components yet unplaced from the schematic to the layout automatically.
Define Device Correspondence (DDC)	Use the Define Device Correspondence form to view and change the device correspondence for the instances and terminals in your design.
Check Against Source (CAS)	Use the Connectivity – Check – Against Source (CAS) command to check the differences between the schematic and the layout.
Update Components and Nets (UCN)	Use the Update Components and Nets form to automatically update your layout to take account of the instances, pins, and connectivity you have changed in the schematic.

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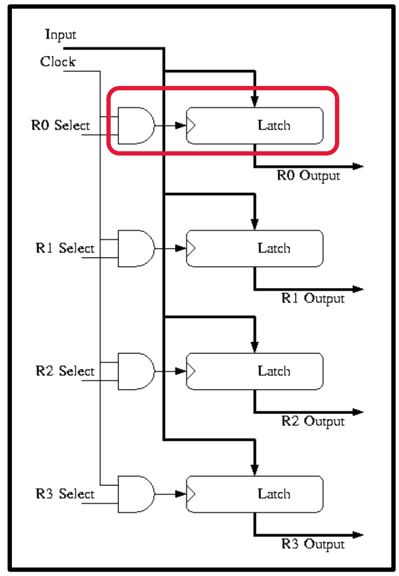
What Is Cloning?

Cloning is the ability to replicate a section of the layout, associated with a section of the schematic.

While cloning, the new piece of layout material can be placed at more than one location, with each part preserving the hierarchical structure of the design.



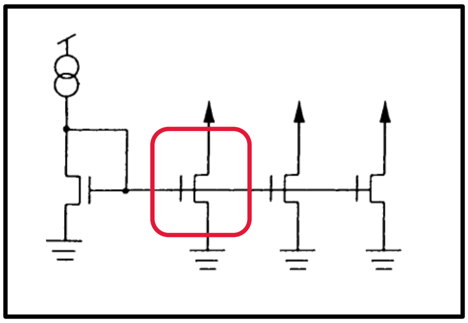
What Is Cloning?

Cloning is the ability to replicate a section of the layout, associated with a section of the schematic.

While cloning, the new piece of layout material can be placed at more than one location, with each part preserving the hierarchical structure of the design.

In the design canvas, choose the Connectivity, Generate, Clones menu command, or click the Generate Clones icon in the Layout EXL toolbar, to start the Clone command.

When to Clone

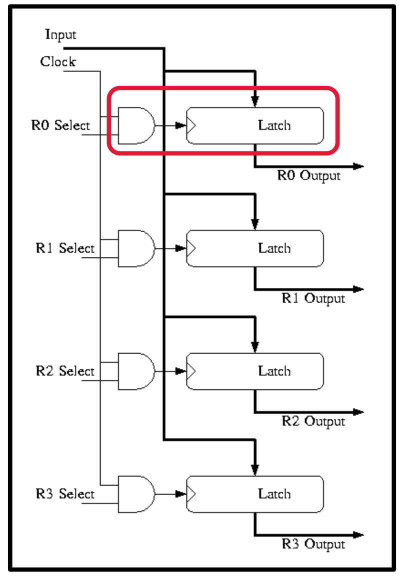


If a transistor is represented in the schematic with a multiplicity factor larger than 1, it is always possible to build it using cloning.

Repetitive layout structures are ideal cloning candidates. They have the same features:

- Connectivity
- Device parameters

The branches of a current mirror can be easily cloned. They have the same connectivity and the same finger dimension.



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Let us see when to clone.

If a transistor is represented in the schematic, with a multiplicity factor larger than 1, it is always possible to build it using cloning.

Decide how the number of fingers have to be grouped.

Layout the “building block” and delete the rest of the fingers.

Use cloning to get the basic building block “copied”.

The branches of a current mirror can be easily cloned. They have the same connectivity and the same finger dimension.

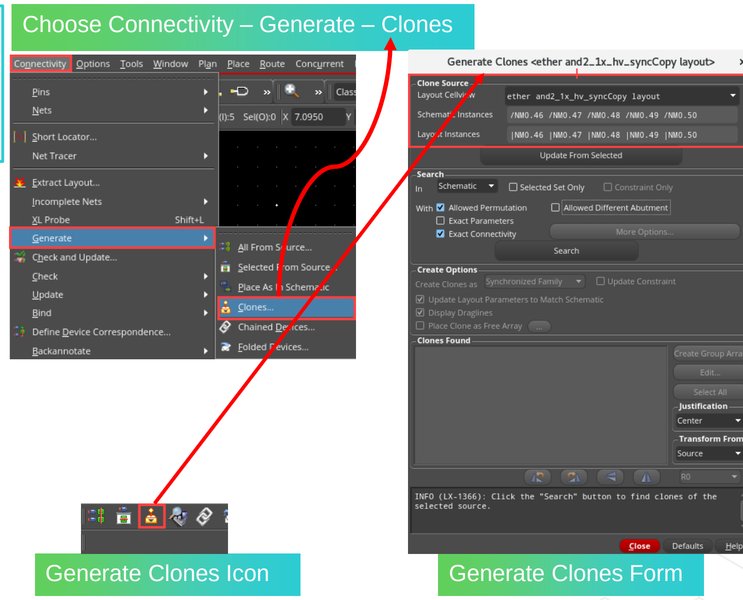
Repetitive layout structures are ideal cloning candidates. They have the same features of connectivity and device parameters.

Invoking the Generate Clones Form

In the layout canvas, choose the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command.

Generate Clones:

You can use the **Generate Clones form** to replicate the section of the layout that is associated with a section of the schematic in such a way that the new piece of layout material can be placed at more than one location, with each part preserving the hierarchical structure of the design.



Let us see how to invoke the Generate Clones Form.

In the design canvas, choose the **Connectivity – Generate – Clones** menu command, or click the **Generate Clones** icon in the Layout EXL toolbar, to open the Generate Clones form.

You can use the **Generate Clones** form, to replicate the section of the layout, that is associated with a section of the schematic, in such a way that the new piece of layout material can be placed at more than one location, with each part preserving the hierarchical structure of the design.

To select the clone source, in the design canvas, you can select the individual component, or components within an area, that need to be cloned, and invoke the **Generate Clones** command.

Alternatively, in the schematic cellview, you can select the individual component, or components within an area, that need to be cloned. Then, in the layout cellview, in which the clone needs to be generated, invoke the **Generate Clones** command.

Invoking the Generate Clones Form (continued)

To select the clone source:

- In the layout canvas, select the individual component or components within an area that need to be cloned, and invoke the **Generate Clones** command.
- Alternatively, in the schematic cellview, select the individual component or components within an area that need to be cloned. Then, in the layout cellview in which the clone needs to be generated, invoke the **Generate Clones** command.

After selecting the required instances that you want to clone, you can use either the menu command or the icon to invoke the Generate Clones form.

After selecting the required instances which you want to clone, you can use either the menu command or the icon to invoke the **Generate Clones** form.

Using the Generate Clones Form

You can use the *Clone Source*, *Search In*, *Search With*, *Create Options*, and *Clones Found* sections in the **Generate Clones** form to specify the required inputs for generating the clones.

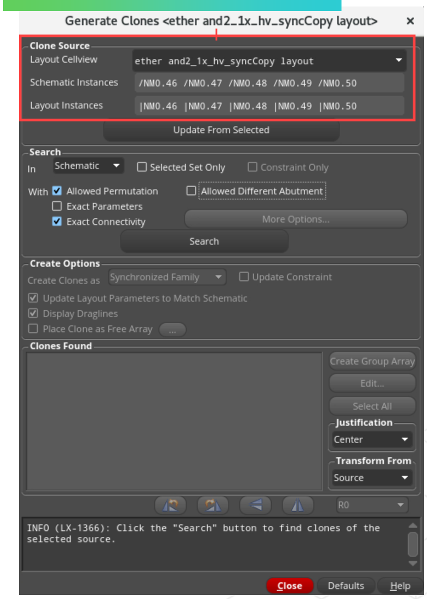
When using the **Generate Clones** form, you can use the *Clone Source*, *Search In*, *Search With*, *Create Options*, and *Clones Found* sections in the **Generate Clones** form to specify the required inputs for generating the clones.

Generate Clones Form: Clone Source Section

Clone Source shows the schematic and layout devices to be used as the clone source. You must specify a clone source in order to enable the other sections of the form.

Layout Cellview	Lists the layout cellviews opened in Layout EXL that have a connectivity reference available, including those opened in the read-only mode.
Schematic Instances	Shows the names of the schematic clone source devices that are bound to the instances in the layout clone source.
Layout Instances	Shows the names of the layout clone source instances to be cloned.

Generate Clones Form



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Let us see the **Clone Source** Section in the **Generate Clones** Form.

Clone Source shows the schematic and layout devices, to be used as the clone source. You must specify a clone source in order to enable the other sections of the form.

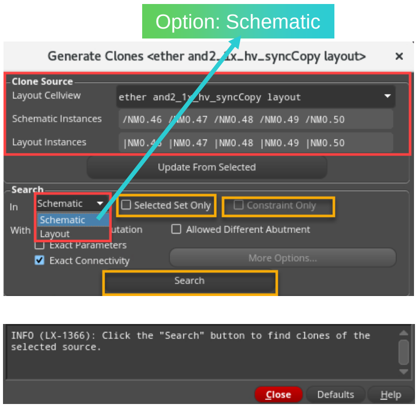
Layout Cellview lists the layout cellviews opened in Layout XL, that have a connectivity reference available, including those opened in the read only mode.

Schematic Instances shows the names of the schematic clone source devices, that are bound to the instances, in the layout clone source.

Layout Instances shows the names of the layout clone source instances to be cloned.

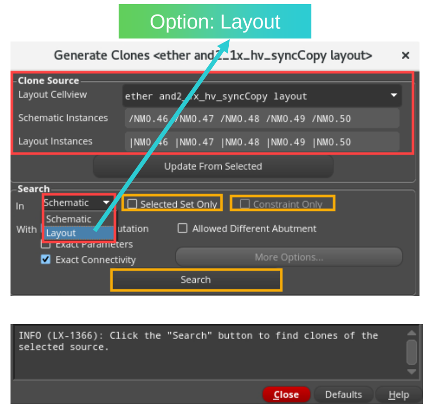
In the design canvas, choose the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout XL toolbar to open the Generate Clones form.

Generate Clones Form: Search In Section: *Schematic/Layout*



You can search the entire schematic to find the matching target structures.

You can search the layout cellview for matching target structures that exist outside the PR Boundary.



Let us see the **Search In** section in the **Generate Clones** form.

You can use the **Search In** section to control the scope and criteria used when searching for the target structures that match the clone source structure. Set the options you want and click the Search button to find the potential clones.

You can search the entire schematic to find the matching target structures.

You can search the layout cellview for matching target structures that exist outside the PR Boundary.

If **Selected Set Only** option is enabled, the cloning engine searches for clone targets only within the selected set in the schematic or layout.

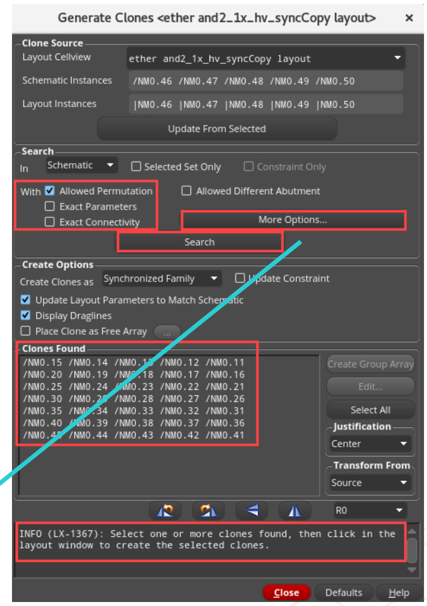
Constraint Only option searches for matching target structures only within the other Clone Defs that belong to the same Clone Family constraint as the source.

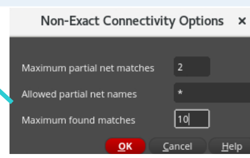
Generate Clones Form: Search With Section

Clone Source → Search In → Search With → Create Options → Clones Found

In the **Generate Clones** form, in the **Search With** section, you can define how precisely the clones have to match the reference layout.

Allowed Permutation	Considers pin permutability when searching for matching target structures.
Exact Parameters	Requires that the parameter values on the components in a target structure match exactly the parameter values on the components in the clone source.
Exact Connectivity (Exact Match)	Requires that the connectivity of the components in a target structure matches exactly the connectivity of the components in the clone source.
Non-Exact Connectivity (Non-Exact Match)	When **Exact Connectivity** option is switched OFF, the **More Options** button becomes active, and Layout XL/EXL reports target structures where the set of instances are the same but are connected differently.





Let us see the **Search With** section in the **Generate Clones** form.

In the **Generate Clones** form in the **Search With** section, you can define how precisely the clones have to match the reference layout.

The options available in the **Search With** section are: **Allowed Permutation**, **Exact Parameters**, and **Exact Connectivity**.

Allowed Permutation option: considers pin permutability when searching for matching target structures.

Exact Parameters option: requires that the parameter values on the components in a target structure match exactly the parameter values on the components in the clone source.

Exact Connectivity option: requires that the connectivity of the components in a target structure matches exactly the connectivity of the components in the clone source.

Non Exact Connectivity is that: when **Exact Connectivity** option is switched OFF, the **More Options** button becomes active and Layout XL/GXL reports target structures where the set of instances are the same but are connected differently.

Generate Clones Form: Create Options Section

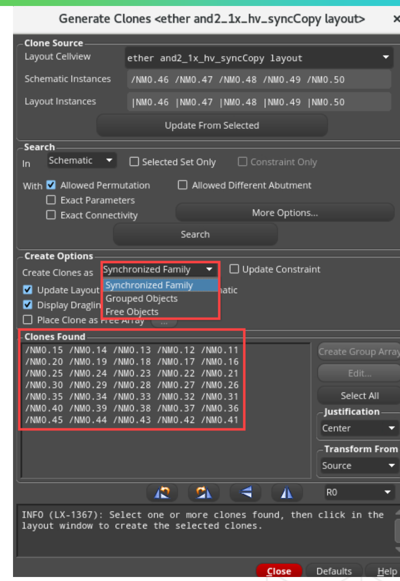
Clone Source → Search In → Search With → Create Options → Clones Found

Create Options section specifies the types of clones that are generated from the target structures found. This section is enabled only if there are target structures reported in the *Clones Found* pane.

You have the capability to **create clones** as:

Synchronized Family	• If a clone is synchronous, then edits to any member of the family are automatically applied to the other members of the family group. • When editing synchronous clones and groups, you are required to use the **Edit – Hierarchy – Edit In Place** command.
Grouped Objects	After placement, they form a figure of groups.
Free Objects	After placement, the objects are free and ungrouped.

When you create clones as free objects, you can edit the instances at the top level.



Let us see the **Create Options** section in the **Generate Clones** form.

Create Options section specify the types of clones that are generated from the target structures found.

This section is enabled only if there are target structures reported in the *Clones Found* pane.

You have the capability to **create clones** as *Synchronized Family*, *Grouped Objects*, and *Free Objects*.

Synchronized Family is that: If a clone is synchronous, then edits to any member of the family are automatically applied to the other members of the family group.

When editing synchronous clones and groups, you are required to use the **Edit – Hierarchy – Edit In Place** command.

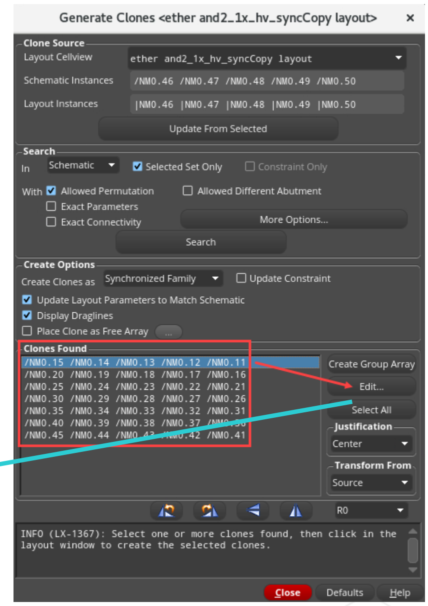
Grouped Objects is that: After placement, they form a figure of groups.

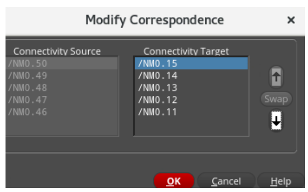
Free Objects is that: After placement, the objects are free and ungrouped.

When you create clones as free objects, you can edit the instances at the top level.

Generate Clones Form: Clones Found Section

- **Clones Found** lists the target structures that can be generated as clones in the layout cellview.
- You can select one of the structures and move your cursor into the layout canvas to generate a clone for that structure.
- You can use the **Edit** button to open the **Modify Correspondence** form, where you can change the correspondence between the instances in source and target structures before you generate a clone.





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Let us see the **Clones Found** section in the **Generate Clones** form.

Clones Found section lists the target structures that can be generated as clones in the layout cellview.

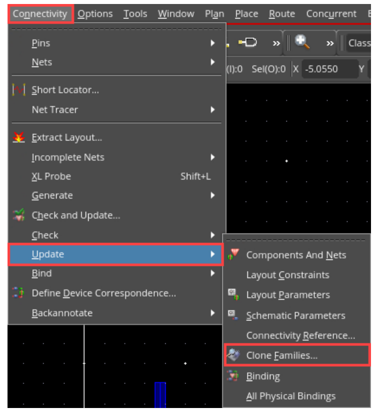
You can select one of the structures and move your cursor into the layout canvas to generate a clone for that structure.

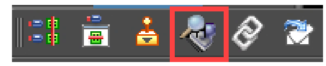
You can use the **Edit** button to open the **Modify Correspondence** form where you can change the correspondence between the instances in source and target structures before you generate a clone.

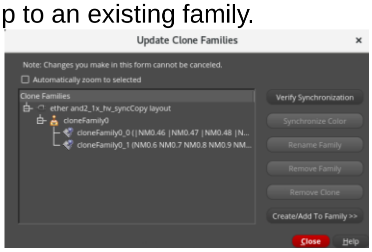
Updating the Clone Families

In the layout canvas, choose the **Connectivity – Update – Clone Families** menu command or click the **Update Clone Families** icon in the Layout EXL toolbar to invoke the Update Clone Families form. 

- You can use the **Update Clone Families** form to manage the families of synchronized clones in your design.
- You can update the status of a clone family, rename a clone family, remove a clone from a family, create and remove clone families, and add a group to an existing family.







Caution: Changes that you make in this form take effect immediately and cannot be cancelled.

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Let us see how to update the Clone Families.

In the design canvas choose the **Connectivity** – **Update** – **Clone Families** menu command or click the **Update Clone Families** icon in the Layout XL toolbar to invoke the Update Clone Families form.

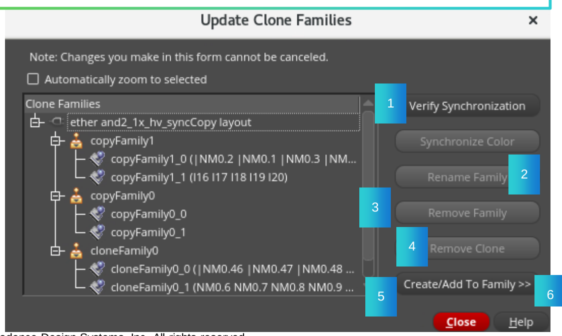
You can use the **Update Clone Families** form to manage the families of synchronized clones in your design.

You can update the status of a clone family, rename a clone family, remove a clone from a family, create and remove clone families, and add a group to an existing family.

Note that, the changes that you make in this form take effect immediately and cannot be cancelled.

Using the Update Clone Families Form

You can use the Update Clone Families form to manage the families of synchronized clones in your design. You can update the status of a clone family, rename a clone family, remove a clone from a family, create and remove clone families, and add a group to an existing family.



You can perform the following tasks in the **Update Clone Families** form:

1. Verifying a clone family
2. Renaming a clone family
3. Removing a clone from a family
4. Removing a clone
5. Creating a new clone family
6. Adding a group to an existing family

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1. Verifying a Clone Family

If you edit one or more members of a clone family in an application other than Layout EXL, you might need to verify the family to ensure its members are consistent with each other.

You can verify one or more clone families with a single button push.

Layout EXL checks the consistency of the clones in each family and takes the following actions:

- Any clone that is out of sync with the other clones in the same family is removed from the family.
- If two or more clones are out of sync with the family but in sync with each other, they are moved to a new clone family. In this case, both the original family and the new family are renamed to follow the *orginalFamilyName_index* nomenclature.
- If all the clones in the family are different, the family is removed.

Let us see how to verify a clone family.

If you edit one or more members of a clone family in an application other than Layout XL, you might need to verify the family to ensure its members are consistent with each other.

You can verify one or more clone families with a single button push.

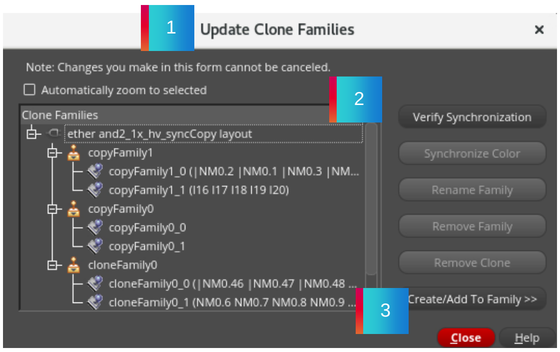
Layout XL checks the consistency of the clones in each family, and takes the following actions:

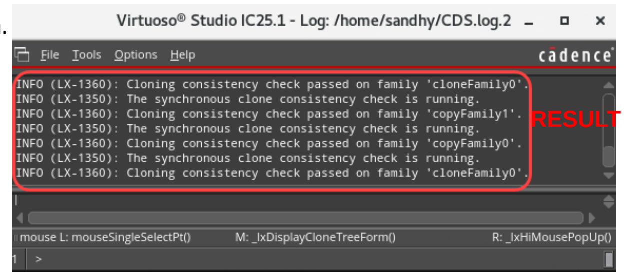
- Any clone that is out of sync with the other clones in the same family is removed from the family.
- If two or more clones are out of sync with the family, but in sync with each other, they are moved to a new clone family.
- In this case, both the original family, and the new family are renamed to follow the *orginalFamilyName_index* nomenclature.
- If all the clones in the family are different, then the family is removed.

1. Verifying a Clone Family (continued)

To verify that a clone family is synchronized:

1. Invoke the **Update Clone Families** form.
2. Select the clone family you want to verify and click **Verify Synchronization**. Or to verify that all the clone families are synchronized, click **Verify Synchronization** with none of the families selected.
   Result: Layout EXL modifies the clone families as outlined here and prints the results to the CIW and CDS.log file.
3. Click **Close** to dismiss the **Update Clone Families** form.





CDS.log

```
INFO (LX-1350): The synchronous clone consistency check is running.
INFO (LX-1360): Cloning consistency check passed on family 'copyFamily1'.
INFO (LX-1350): The synchronous clone consistency check is running.
INFO (LX-1360): Cloning consistency check passed on family 'copyFamily0'.
INFO (LX-1350): The synchronous clone consistency check is running.
INFO (LX-1360): Cloning consistency check passed on family 'cloneFamily0'.
```

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To verify that a clone family is synchronized, you can do the following steps.

Firstly, invoke the *Update Clone Families* form.

Then select the clone family you want to verify and click the *Verify Synchronization* button.

Or to verify that all the clone families are synchronized, click the *Verify Synchronization* button with none of the families selected.

Layout XL modifies the clone families as outlined here and prints the results to the CIW and the CDS.log file.

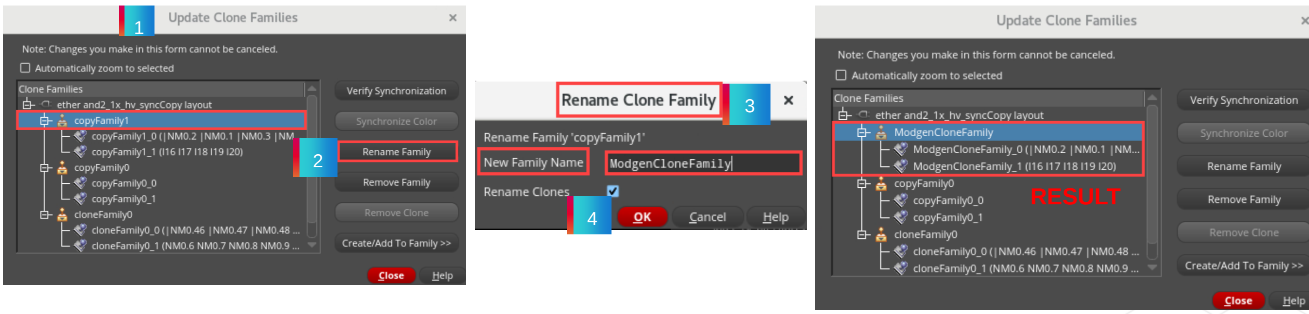
Click the *Close* button to dismiss the *Update Clone Families* form.

2. Renaming a Clone Family

To rename a clone family:

1. Invoke the **Update Clone Families** form.
2. Select the clone family to be renamed and click **Rename Family**.
3. In the **New Family Name** field, type a new name for the clone family.
4. Select the **Rename Clones** checkbox if you also want to rename all the clone members using the new family name. Alternatively, set the environment variable, *renameClonesWhenFamilyIsRenamed*, to **t** and Click **OK** to rename the clone family.

Result: The selected clone family and the clone members (if set for renaming), display the new clone family base name.



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Let us see how to rename a clone family.

To rename a clone family, you can do the following steps.

Firstly, invoke the *Update Clone Families* form.

Then select the clone family to be renamed and click the *Rename Family* button.

In the *New Family Name* field, type a new name for the clone family.

Select the *Rename Clones* checkbox if you also want to rename all the clone members using the new family name.

Alternatively, you can set the environment variable to *t* and Click *OK* to rename the clone family.

You can observe that the selected clone family and the clone members (if set for renaming), display the new clone family base name.

3. Removing a Clone Family

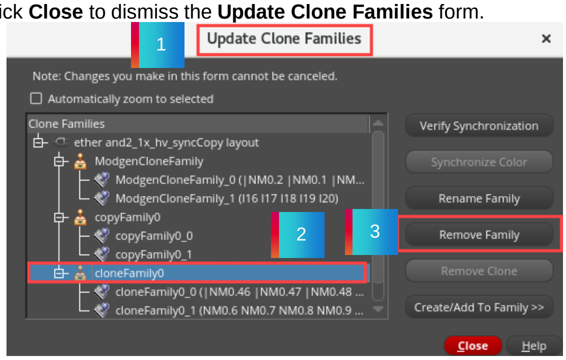
To remove a clone family from your design:

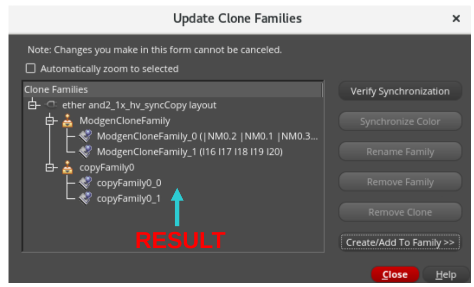
1. Invoke the **Update Clone Families** form.
2. Select the family you want to remove
3. Click **Remove Family**.

Result: The clone family is removed. The clones it contained are still present in the layout but they are no longer synchronized.

Note: You cannot remove a family if one of its members is currently being edited in place. You need to exit the **Edit In Place** command first.

Note: Click **Close** to dismiss the **Update Clone Families** form.





Let us see how to remove a clone family.

To remove a clone family from your design, you can do the following steps.

Firstly, invoke the *Update Clone Families* form.

Then select the family you want to remove and click the *Remove Family* button.

You can see that the clone family is removed.

The clones it contained are still present in the layout but they are no longer synchronized.

Note that, you cannot remove a family if one of its members is currently being edited in place.

You need to exit the *Edit In Place* command first.

Click the *Close* button to dismiss the *Update Clone Families* form.

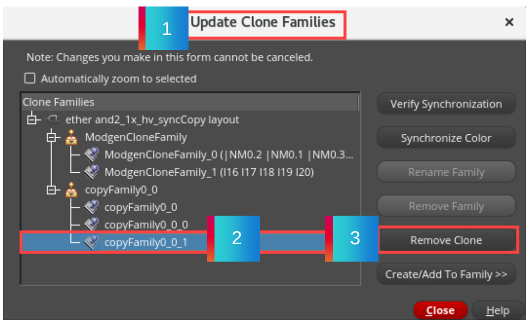
4. Removing a Clone from a Family

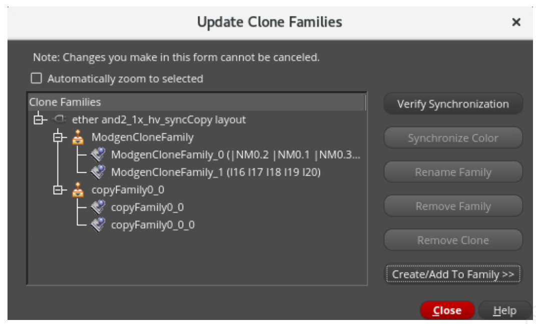
To remove a clone from a family:

1. Invoke the **Update Clone Families** form.
2. Select the clone to be removed
3. click **Remove Clone**.

Result: The selected clone is removed from the family. It is still present in the layout but is no longer synchronized with the other members of the family. If you remove all except a single clone from a family, the family is automatically removed as well.

- **Note:** You cannot remove a clone if a member of its clone family is currently being edited in place. You need to exit the **Edit In Place** command first. Click **Close** to dismiss the **Update Clone Families** form.





Let us see how to remove a clone from a family.

To remove a clone from a family, you can do the following steps.

Firstly, invoke the *Update Clone Families* form.

Then select the clone to be removed and click the *Remove Clone* button.

You can observe that the selected clone is removed from the family.

It is still present in the layout but is no longer synchronized with the other members of the family.

If you remove all except a single clone from a family, the family is automatically removed as well.

Note that, you cannot remove a clone if a member of its clone family is currently being edited in place.

You need to exit the *Edit In Place* command first.

Click the *Close* button to dismiss the *Update Clone Families* form.

5. Creating a New Clone Family

To create a new synchronized clone family containing two or more of the layout groups currently in your design:

1. Invoke the **Update Clone Families** form.
2. Click **Create/Add** To Family.

   The form is expanded to include the options.
3. Ensure there is no existing family selected in the **Clone Families** pane, and then select at least two groups from the list of **Layout groups**.

   The **Synchronization** reference drop-down and the **Create Family** button are enabled.
4. From the **Synchronization** reference drop-down, choose the reference group with which the others are to be synchronized.
5. Click **Create Family**.

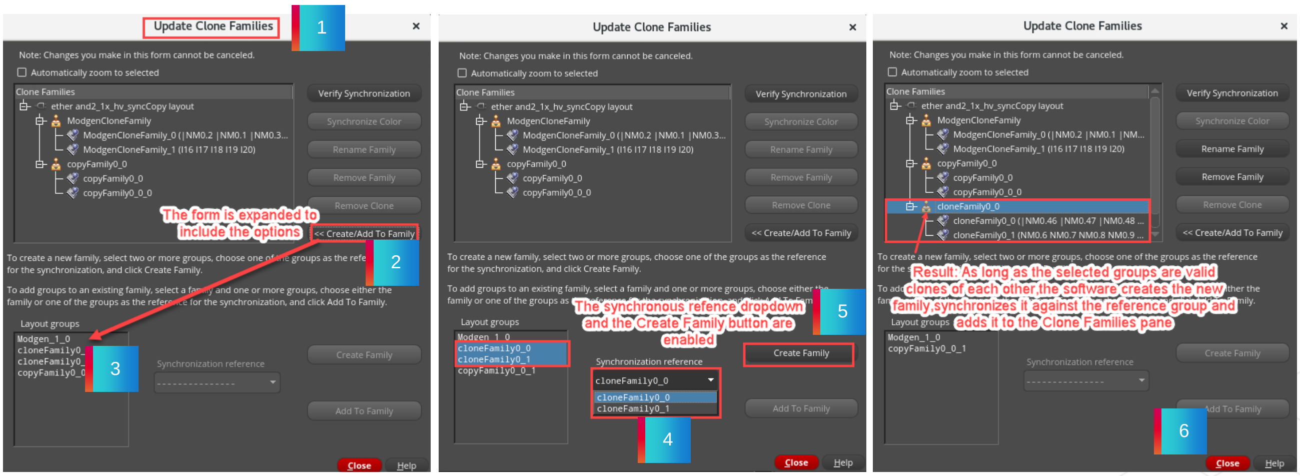
   Result: As long as the selected groups are valid clones of each other, the software creates the new family, synchronizes it against the reference group, and adds it to the **Clone Families** pane.
6. Click **Close** to dismiss the **Update Clone Families** form.

Let us see how to create a new clone family.

To create a new synchronized clone family containing two or more of the layout groups currently in your design, you can do the following steps.

5. Creating a New Clone Family (continued)

Result: As long as the selected groups are valid clones of each other, the software creates the new family, synchronizes it against the reference group, and adds it to the **Clone Families** pane.



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To create a new synchronized clone family, firstly, invoke the *Update Clone Families* form.

Then click the *Create/Add To Family* button.

The form is expanded to include the options.

Ensure that there is no existing family selected in the *Clone Families* pane, and then select at least two groups from the list of Layout groups.

The *Synchronization reference* drop-down and the *Create Family* button are enabled.

From the *Synchronization reference* drop-down, choose the reference group with which the others are to be synchronized.

Then click the *Create Family* button.

You can see that as long as the selected groups are valid clones of each other, the software creates the new family, synchronizes it against the reference group, and adds it to the *Clone Families* pane.

Click the *Close* button to dismiss the *Update Clone Families* form.

6. Adding a Group to an Existing Family

To add a layout group to an existing family:

1. Invoke the **Update Clone Families** form.
2. Click **Create/Add** To Family.

The form is expanded to include the options.

3. In the **Clone Families** pane, select the family you want to update.
4. In the **Layout groups** pane, select the group to be added to the family.

The **Synchronization** reference drop-down and the **Add To Family** button are both enabled.

5. From the **Synchronization** reference drop-down, choose one of the following:
   - The reference group with which all other members of the family are to be synchronized.
   - The existing family, if the new group is to be changed to match existing family members.
6. Click **Add To Family**.

Result: As long as the selected group can be considered a clone of the other members in the family you selected, it is added to the family.

7. Click **Close** to dismiss the **Update Clone Families** form.

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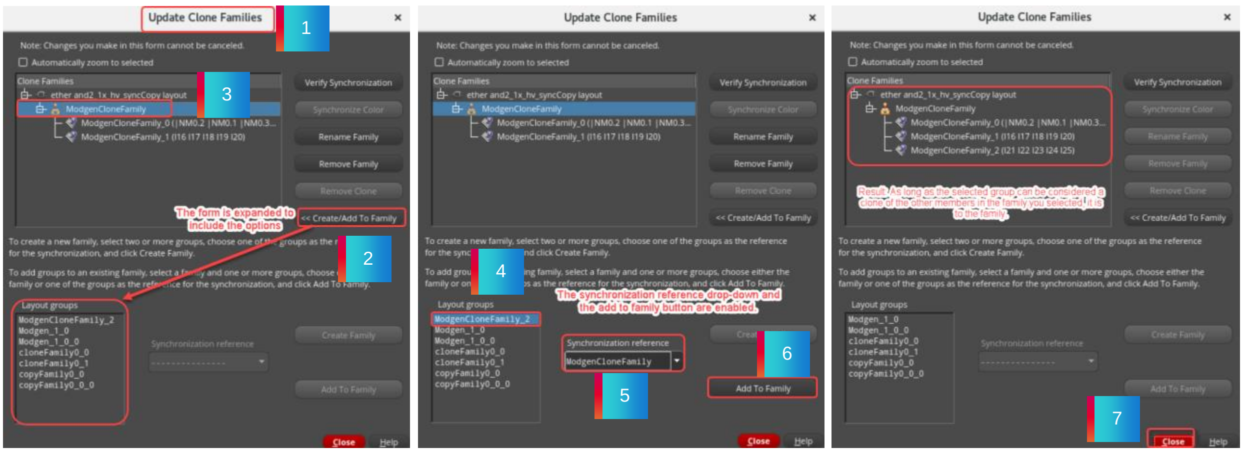
Let us see how to Add a Group to an Existing Family.

To add a layout group to an existing family, you can do the following steps.

6. Adding a Group to an Existing Family (continued)

Result: As long as the selected group can be considered a clone of the other members in the family you selected, it is added to the family.



Firstly, invoke the *Update Clone Families* form.

Then click the *Create/Add To Family* button.

The form is expanded to include the options.

In the *Clone Families* pane, select the family you want to update.

From the *Layout groups* pane, select the group to be added to the family.

The *Synchronization reference* drop-down and the *Add To Family* button are both enabled.

From the *Synchronization reference* drop-down, you can choose one of the following options.

The reference group with which all other members of the family are to be synchronized.

Or the existing family, if the new group is to be changed to match existing family members.

Then click the *Add To Family* button.

You can see that as long as the selected group can be considered a clone of the other members in the family you selected, it is added to the family.

Click the *Close* button to dismiss the *Update Clone Families* form.

How to Use the Generate Clones Form

You can use the **Generate Clones** form to replicate the section of the layout that is associated with a section of the schematic in such a way that the new piece of layout material can be placed at more than one location with each part preserving the hierarchical structure of the design.

1. In the layout canvas, in the Navigator assistant, click-drag to select the required instances which you want to clone, say for example, **NM0.46** through **NM0.50**.
2. In the layout canvas, choose the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command.
3. In the Generate Clones form, click the **Search** button to find the matches in the schematic.

Result: The matches are returned in the *Clones Found* pane from where they can be selected and placed in the layout canvas.

Let us see how to use the **Generate Clones** form.

You can use the **Generate Clones** form to replicate the section of the layout that is associated with a section of the schematic in such a way that the new piece of layout material can be placed at more than one location with each part preserving the hierarchical structure of the design.

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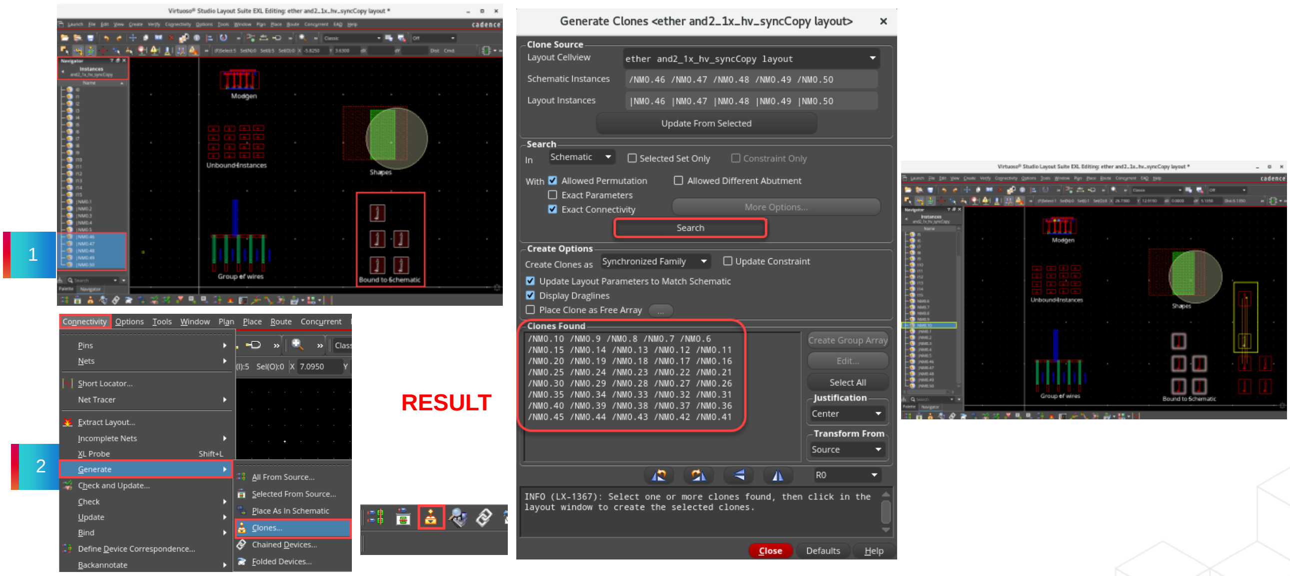
Then in the design canvas, choose the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command.

In the Generate Clones form, click the **Search** button to find the matches in the schematic.

You can observe that the matches are returned in the *Clones Found* pane from where they can be selected and placed in the layout canvas.

How to Use the Generate Clones Form (continued)

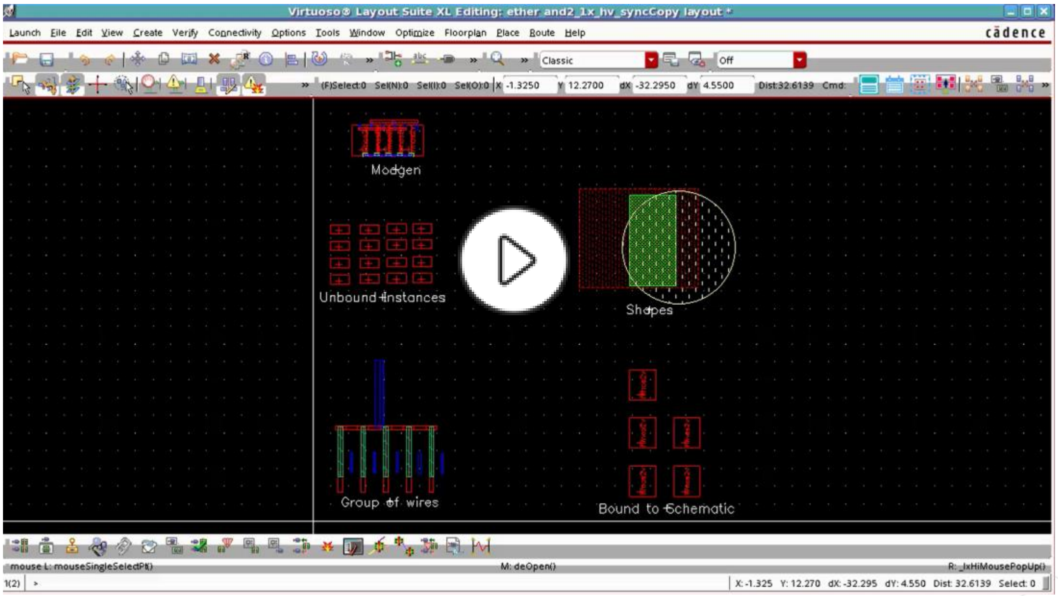
Result: The matches are returned in the *Clones Found* pane from where they can be selected and placed in the canvas.



After selecting the required instances which you want to clone, open the **Generate Clones** form, and click the **Search** button to find the matches in the schematic.

You can observe that the matches are returned in the *Clones Found* pane, from where they can be selected and placed in the layout canvas.

Demo: How to Generate Clones as Free Objects



Video Play Time: 2.54 minutes

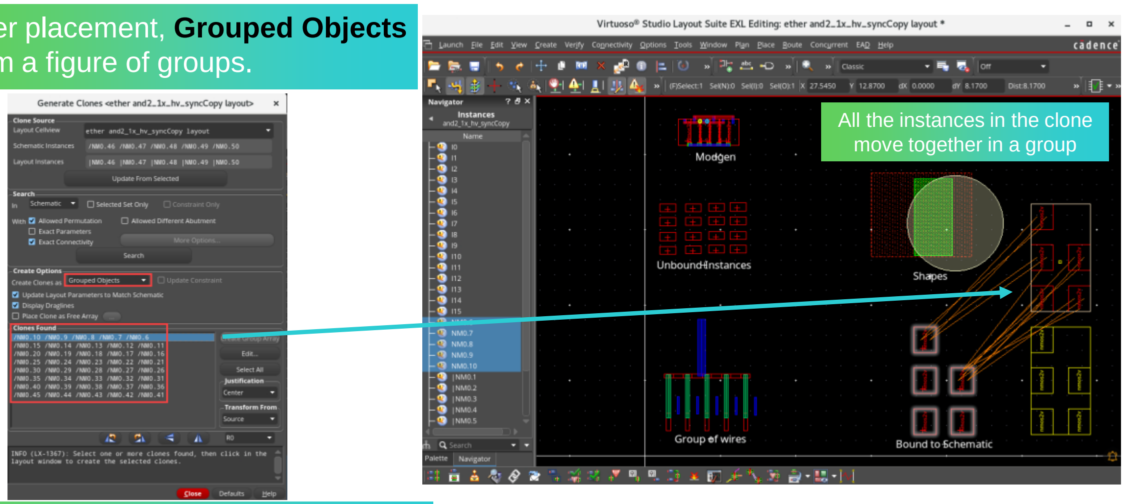
Click the Play button to start the video.

Click the link How to Generate Clones as Free Objects to play the video on support.cadence.com.

https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgvFEAQ&pageName=ArticleContent

How to Generate Clones as Grouped Objects

After placement, **Grouped Objects** form a figure of groups.

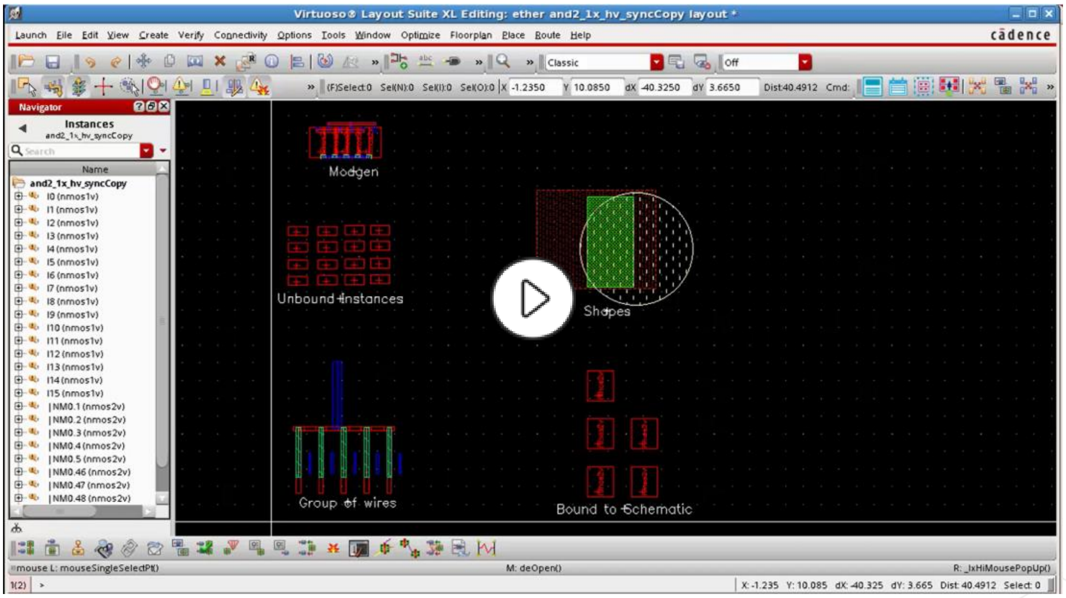


Select **Grouped Objects** in the **Create Clones** as group box in the **Create Options** section.

Select **Grouped Objects** in the **Create Clones as** group box in the **Create Options** section.

After placement, **Grouped Objects** form a figure of groups.

Demo: How to Generate Clones as Grouped Objects



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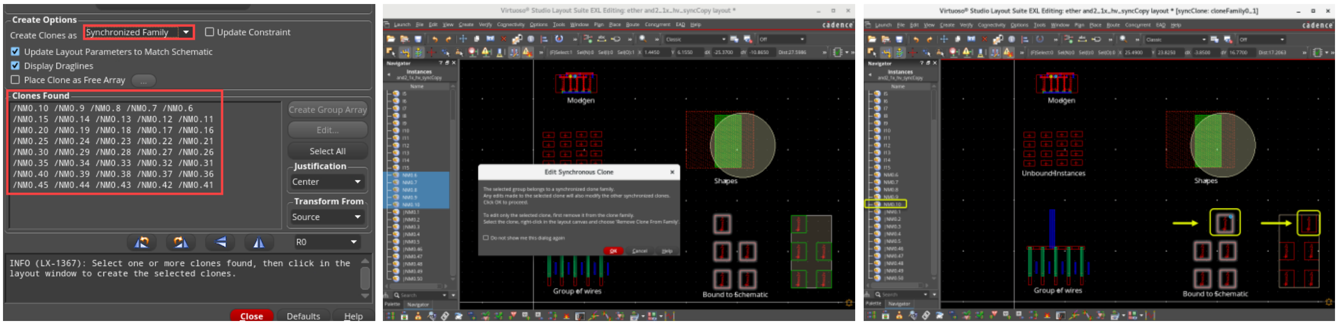
Video Play Time: 2.37 minutes

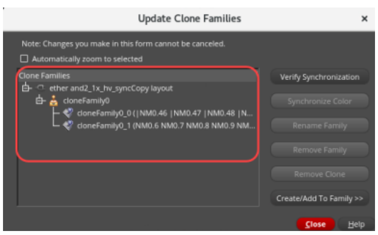
Click the Play button to start the video.

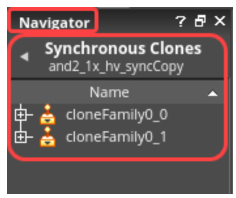
Click the link [How to Generate Clones as Grouped Objects]to play the video on support.cadence.com.

https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgvKEAQ&pageName=ArticleContent

How to Generate Clones as Synchronized Family







If a clone is synchronous, then edits to any member of the family are automatically applied to the other members of the family group.

Select **Synchronized Family** in the **Create Clones** as group box in the **Create Options** section.

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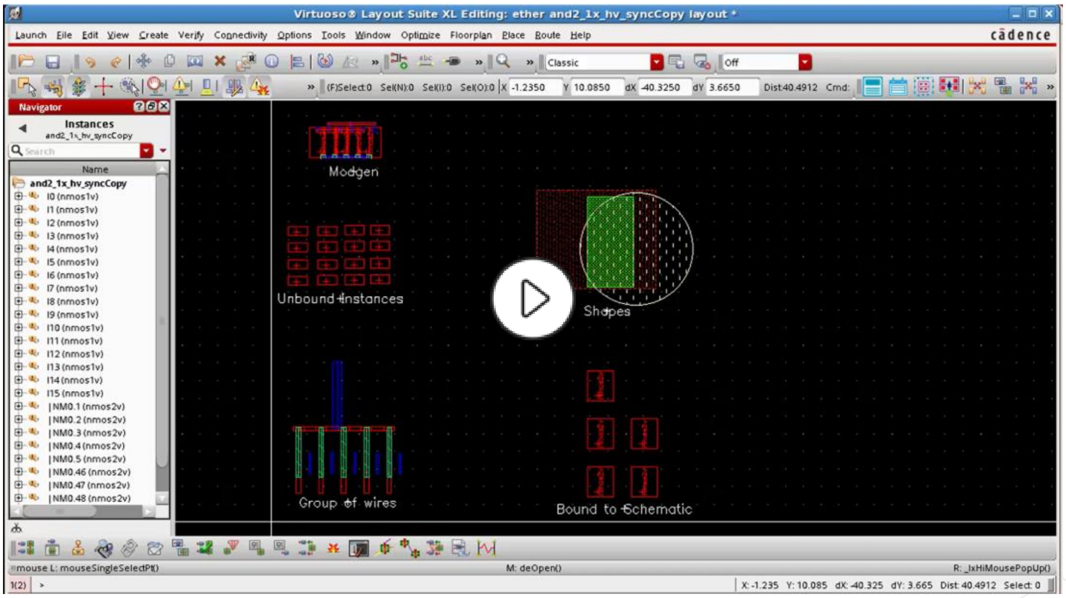
Select **Synchronized Family** in the **Create Clones as** group box in the **Create Options** section.

If a clone is synchronous, then edits to any member of the family are automatically applied to the other members of the family group.

Demo: How to Generate Clones as Synchronized Family



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Video Play Time: 3.21 minutes

Click the Play button to start the video.

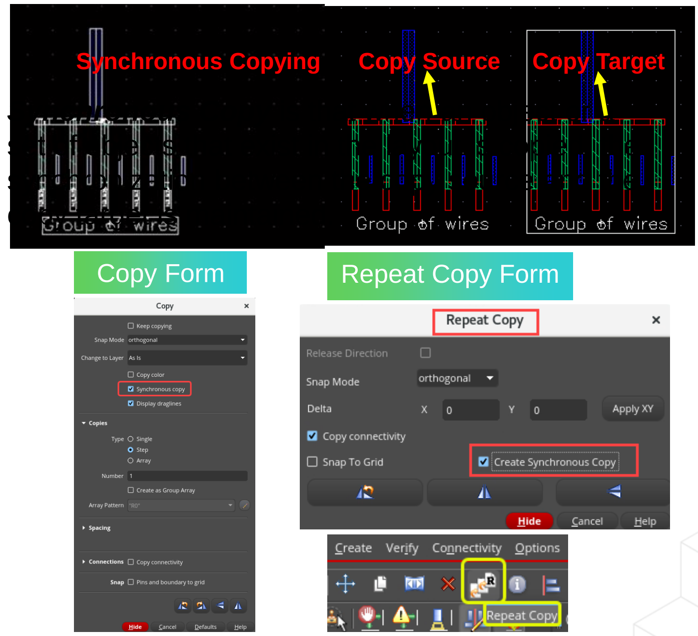
Click the link How to Generate Clones as Synchronized Family to play the video on support.cadence.com.

https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgvUEAQ&pageName=ArticleContent

What Is Synchronous Copy?

Synchronous Copy is that when an object or a set of objects that exist in the layout is copied, the synchronization between the copy source and the copy target is maintained.

- The copy source and the copy target are generated as a group in the layout view.
- Each group is called a **synchronous copy** and a set of such groups is called a **synchronized family.**
- Any changes made to the copy source are automatically reflected in the copy target and vice versa.
  - For example, if you edit the routing in the copy source or the copy target, the change is replicated in the other group.



Use the "**Synchronous copy**"/"**Create Synchronous Copy**" checkboxes available in the **Copy / Repeat Copy** forms to create synchronous copies.

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What Is Synchronous Copying?

Synchronous Copy is that when an object or a set of objects that exist in the layout is copied, the synchronization between the copy source and the copy target is maintained.

The copy source and the copy target are generated as a group in the layout view.

Each group is called a synchronous copy and a set of such groups is called a synchronized family.

Any changes made to the copy source are automatically reflected in the copy target and vice versa.

For example, if you edit the routing in the copy source or the copy target, the change is replicated in the other group.

Use the *Synchronous copy* and *Create Synchronous Copy* checkboxes available in the *Copy* and *Repeat Copy* forms to create synchronous copies.

Cloning differs from synchronous copying in a way that the cloned structure maintains its schematic correspondence and inherits the connectivity information whereas a synchronous copy does not maintain any schematic correspondence.

What Is Synchronous Copy (continued)

- Cloning (Synchronous mode) has been available in the Virtuoso® Layout Suite EXL for a number of releases and is widely used.
- However, Virtuoso Layout Suite users find its interface unfamiliar, and there are some use model restrictions. Before IC6.1.5, it was only possible to clone objects that were in the schematic.
- The new *Synchronous copy* / *Create Synchronous Copy* functionalities remove this restriction and let you create synchronous groups of objects whether they exist in the schematic or not.
- This new functionality is presented as options in the *Copy* and *Repeat Copy* commands. It extends the available functionality, so when you are making the transition to E*XL*, you will not have to learn a new interface.

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Background of *Synchronous copy* and *Create Synchronous Copy* features.

Synchronous Cloning has been available in Virtuoso Layout Suite EXL for a number of releases and is widely used.

However, Virtuoso Layout Suite users find its interface unfamiliar and there are some use model restrictions.

Before IC6.1.5, it was only possible to clone objects that were in the schematic.

The new *Synchronous copy* and *Create Synchronous Copy* functionalities remove this restriction and lets you create synchronous groups of objects whether they exist in the schematic or not.

This new functionality is presented as options in the *Copy* and *Repeat Copy* commands.

It extends the available functionality so when you are making the transition to E*XL*, you will not have to learn a new interface.

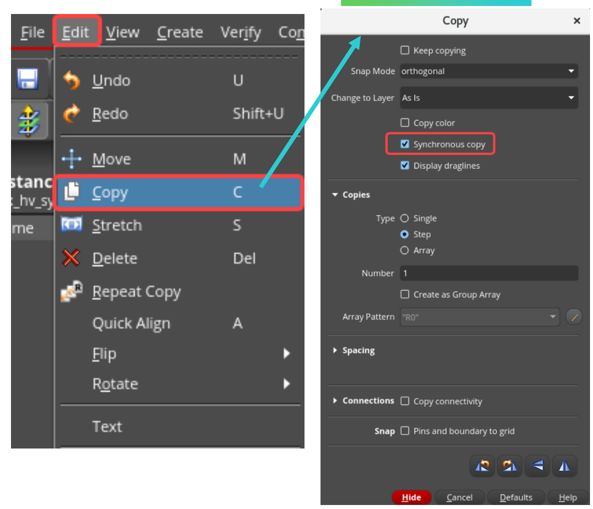
Synchronous Copy Feature in Copy Form

To invoke the Copy form in the layout canvas:

1. Choose the **Edit – Copy** menu command.
2. Or click the **Copy** icon in the Edit toolbar.
3. Or press the bindkey **C**.

Use the **Synchronous copy** checkbox available in the **Copy** form to create synchronous copies.

Copy form



Synchronous copy feature is available in the **Copy** form in the Layout XL/EXL tiers.

Let us see the *Synchronous copy* feature in the *Copy* form.

To invoke the Copy form, in the layout canvas, choose the *Edit – Copy* menu command or click the *Copy* icon in the Edit toolbar or press the bindkey *C*.

You can use the *Synchronous copy* checkbox available in the *Copy* form to create synchronous copies.

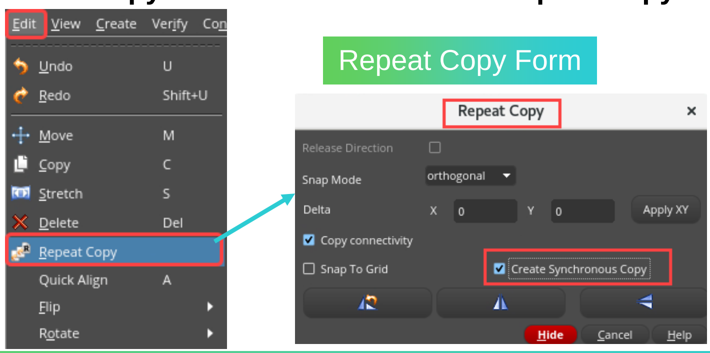
Note that *Synchronous copy* feature is available in the *Copy* form in Layout XL and EXL tiers.

Create Synchronous Copy Feature in Repeat Copy Form

To invoke the Repeat Copy form in the layout canvas:

1. Choose the **Edit – Repeat Copy** menu command.
2. Or click the **Repeat Copy** icon in the Edit toolbar.

Use the **Create Synchronous Copy** checkbox available in the **Repeat Copy** form to create synchronous copies.



Create Synchronous Copy feature is available in the **Repeat Copy** form in Layout XL/EXL tiers.

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Let us see the *Create Synchronous Copy* feature in the *Repeat Copy* form.

To invoke the Repeat Copy form, in the layout canvas, choose the *Edit – Repeat Copy* menu command or click the *Repeat Copy* icon in the Edit toolbar.

You can use the *Create Synchronous Copy* checkbox available in the *Repeat Copy* form to create synchronous copies.

Note that *Create Synchronous Copy* feature is available in the *Repeat Copy* form in Layout XL and EXL tiers.

Creating a Synchronous Copy

New functionality in the Virtuoso Layout Suite EXL.

- Does **NOT** replace cloning. It is a complementary functionality.
  - Used model is easier to understand for users migrating from the L tier.
  - Removes the requirement for searching the schematic for matches.
  - Does not bind copied layout instances to schematic instances.
- Allows copying a group of physical objects that exist in the layout maintaining the synchronization between the copied groups.
- It is available as options in the *Copy* and *Repeat Copy* forms.
- Creates synchronous groups that are managed exactly the same way as Synchronous Clones via *Update Clone Families*.

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Synchronous Copy is a new functionality in Virtuoso Layout Suite XL and EXL.

It does not replace cloning. It is a complementary functionality.

This use model is easier to understand for users who are migrating from the XL tier.

It removes the requirement for searching the schematic for matches.

It does not bind copied layout instances to schematic instances.

It allows copying a group of physical objects that exist in the layout maintaining the synchronization between the copied groups.

It is available as options in the *Copy* and *Repeat Copy* forms.

It creates synchronous groups that are managed exactly the same way as Synchronous Clones via *Update Clone Families*.

After the clones are created, the Synchronous Clone synchronization mechanism is reused to keep the physical clones synchronized.

The synchronous copy generated by using the *Copy* or *Repeat Copy* forms cannot be differentiated from the synchronous clones generated using the *Generate Clones* form. These clones differ only in the way they are generated.

Generating a Synchronous Copy

Designers want to synchronously copy some routing groups. However, this is not possible with cloning because these are *physical* objects that exist only in the layout.

Copy and **Repeat Copy** commands with **Synchronous Copy** and **Create Synchronous Copy** options can be used.

1. In the layout canvas or in the Navigator assistant, select the copy source for which you want to create a synchronous copy.
2. In the design canvas, invoke the **Copy** form and select the **Synchronous Copy** checkbox to create a copy that is synchronized with the source.
3. Select the area in the layout canvas and place the selected copy source.

Result: The copy source and the copy target are added to a synchronized family which can be accessed by using the Navigator assistant.

Let us see how to generate synchronous copies.

Designers want to synchronously copy some routing groups.

However, this is not possible with cloning, because these are physical objects that exist only in the layout.

Copy and *Repeat Copy* commands with *Synchronous Copy* and *Create Synchronous Copy* options can be used.

To generate synchronous copies, in the layout canvas or in the Navigator assistant, select the copy source for which you want to create a synchronous copy.

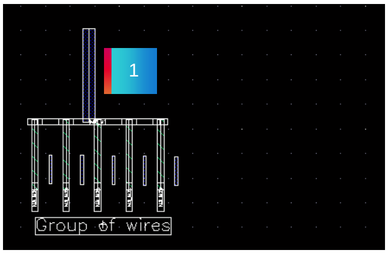
In the design canvas, invoke the *Copy* form and select the *Synchronous Copy* checkbox to create a copy that is synchronized with the source.

Select the area in the layout canvas and place the selected copy source.

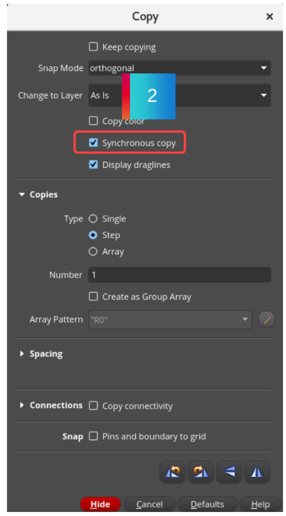
You can observe that the copy source and the copy target are added to a synchronized family which can be accessed by using the Navigator assistant.

Generating a Synchronous Copy (continued)

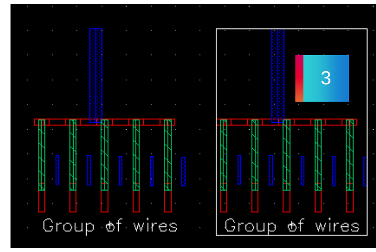
Select the **Group of wires** in the layout canvas.

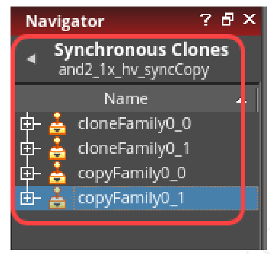


Invoke the **Copy** command and select **Synchronous copy** option.



Both the source and target of the copy are created as synchronous clones.





RESULT

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Generating Synchronous Copy of Wires

The cloning functionality requires that at least one instance in the source be in the schematic. Users have requested the capability to create synchronous groups that **ONLY** contain physical objects, meaning objects that do **NOT** exist in the schematic.

Starting with the release of IC 6.1.5, this has been possible by using the *Synchronous Copy* / *Create Synchronous Copy* commands, which are available in the Virtuoso software.

Let us see how to generate synchronous copy clones of wires and examine its synchronous behavior.

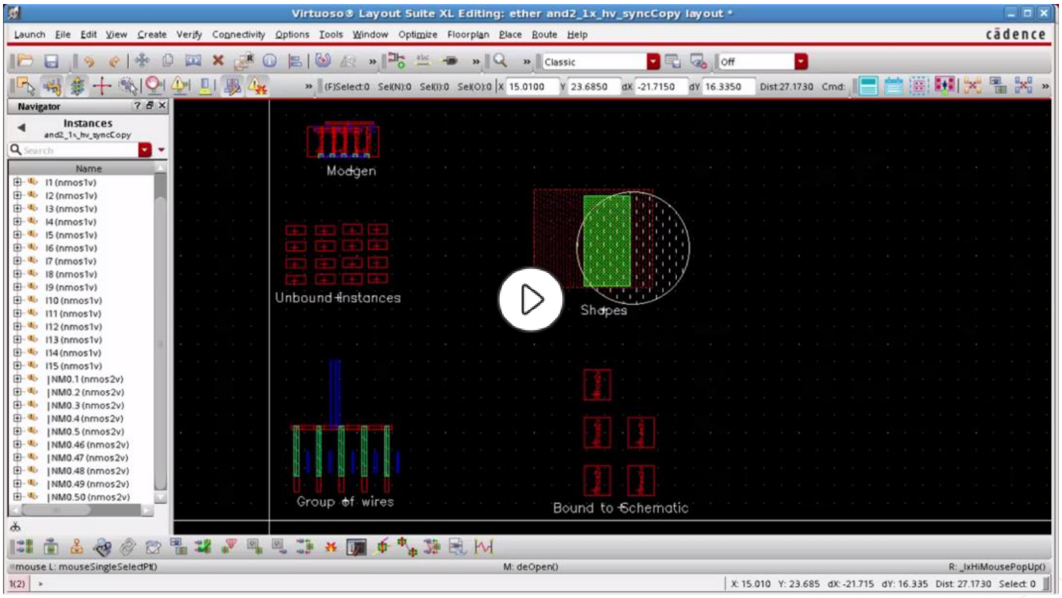
Select the "Group of wires" in the layout canvas.

Invoke the *Copy* command and select *Synchronous copy* option.

After placing the copy target both the source and the target of the copy are created as Synchronous Clones.

The copy source and the copy target are added to a synchronized family which can be accessed by using the Navigator assistant.

Demo: How to Generate Synchronous Copy Clones of Wires and Examine Its Synchronous Behavior



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Video Play Time: 3.37 minutes

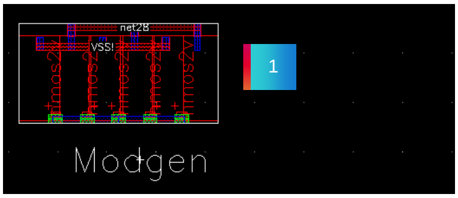
Click the Play button to start the video.

Click the link How to Generate Synchronous Copy Clones of Wires and Examine its Synchronous Behavior to play the video on support.cadence.com.

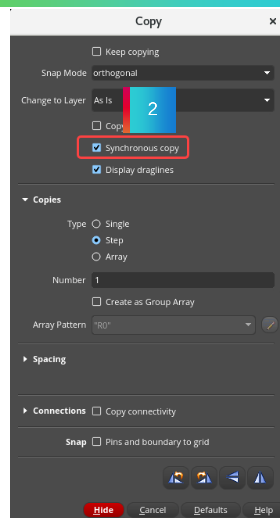
https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgvZEAQ&pageName=ArticleContent

How to Generate Synchronous Copy Clones of Modgen and Examine Its Synchronous Behavior

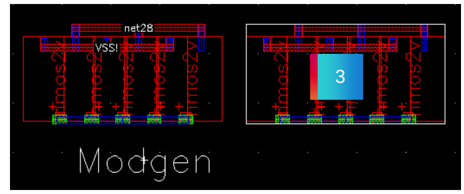
Select the **modgen** in the layout canvas.

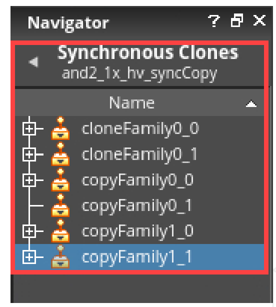


Invoke the **Copy** command and select **Synchronous copy** option.



Both the source and target of the copy are created as Synchronous Clones.





RESULT

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Let us see how to generate synchronous copy clones of modgen and examine its synchronous behavior.

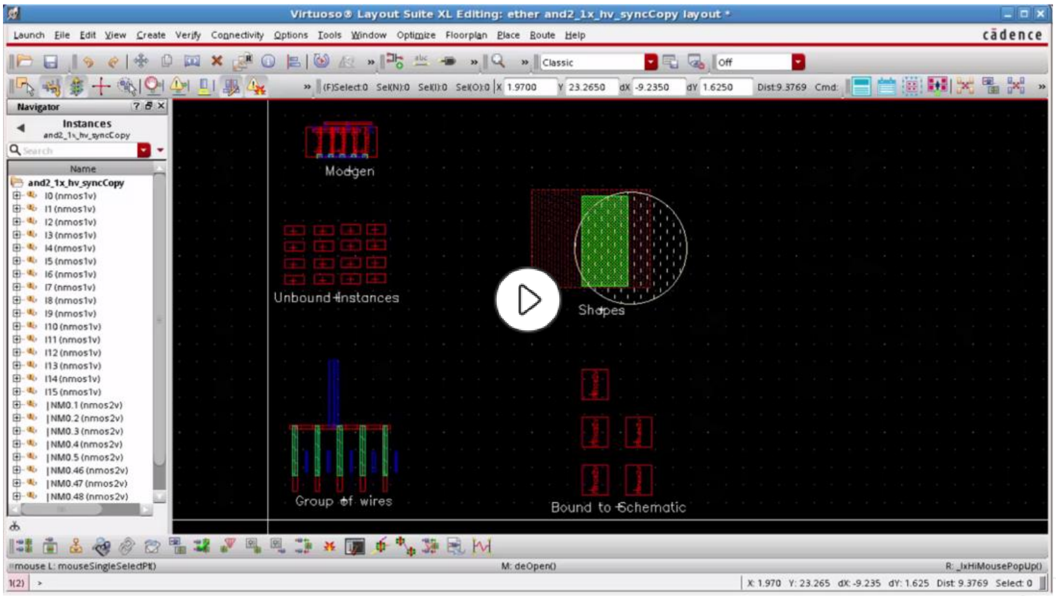
Select the modgen in the layout canvas.

Invoke the *Copy* command and select *Synchronous copy* option.

After placing the copy target both the source and the target of the copy are created as Synchronous Clones.

The copy source and the copy target are added to a synchronized family which can be accessed by using the Navigator assistant.

Demo: How to Generate Synchronous Copy Clones of Modgen and Examine Its Synchronous Behavior



Video Play Time: 2.28 minutes

Click the Play button to start the video.

Click the link How to Generate Synchronous Copy Clones of Modgen and Examine its Synchronous Behavior to play the video on support.cadence.com.

https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgveEAA&pageName=ArticleContent

Generating Mutant Clones with Exact Connectivity Turned ON/OFF with Relax Match Options

A target that also matches the source in terms of connectivity is considered an exact match. A target that tolerates slight differences in connectivity is a non-exact match, also known as a mutant clone.

All three groups of gates share the same structure even if there are some differences in the connectivity.

Group 1

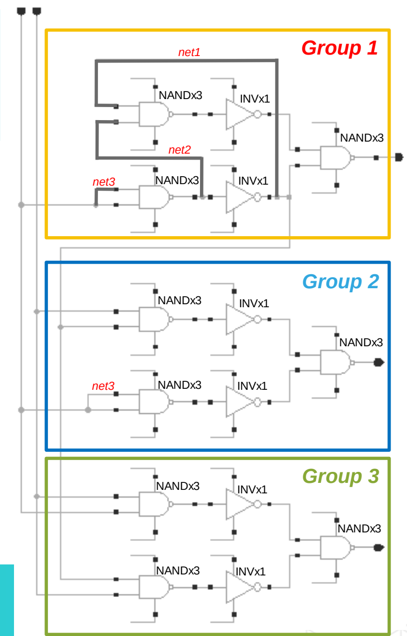
- This will be used as the “reference” object.
- The layout for these 5 standard cells has already been created.

Group 2

- net1 and net2 are not present (2 connectivity differences).

Group 3

- net1, net2, and net3 are not present (3 connectivity differences).



There is no grouping of the gates in the schematic; all standard cells are instantiated “flat” at the same schematic level.

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Mutant Clones

Mutants are clones that may not exactly match the source in connectivity. So, when searching for clone targets on the *Generate Clones* form with the *Exact Connectivity* option set to false, you might find mutant targets in your search result. These mutants are displayed as mutant (number) *instanceName1*, *instanceName2*, and so on.

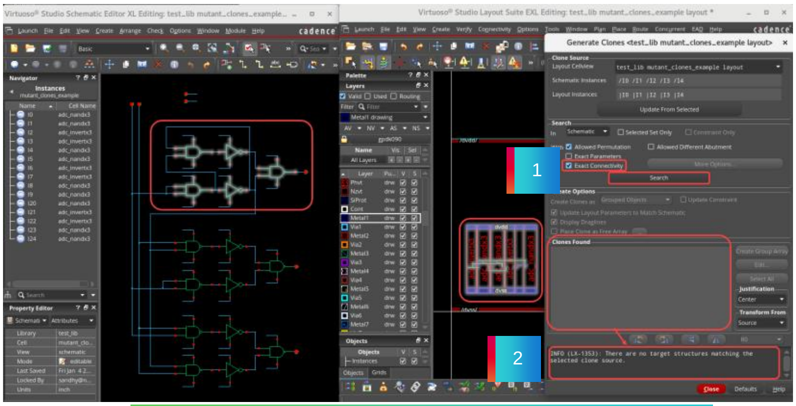
To be considered a valid target, the instance master referenced by the components in the target must exactly match the instance master referenced by the components in the source. A target that also matches the source in terms of connectivity is considered an *exact match*. A target that tolerates slight differences in connectivity is a *non-exact match*, also known as a *mutant clone*.

When cloning structures with connectivity differences, always try to keep the tolerance as small as possible. In this example, allowing *3 differences* in the beginning will lead to the identification of a false positive where a mix of instances from *group2* and *group3* are identified.

Generating Mutant Clones with Exact Connectivity Turned ON/OFF with Relax Match Options (continued)

Generating Mutant Clones with *Exact Connectivity* turned ON

1. When generating the mutant clones with the *Exact Connectivity* option turned **ON** in the Generate Clones form,
2. **No matching target is found** in the *Clones Found* pane.



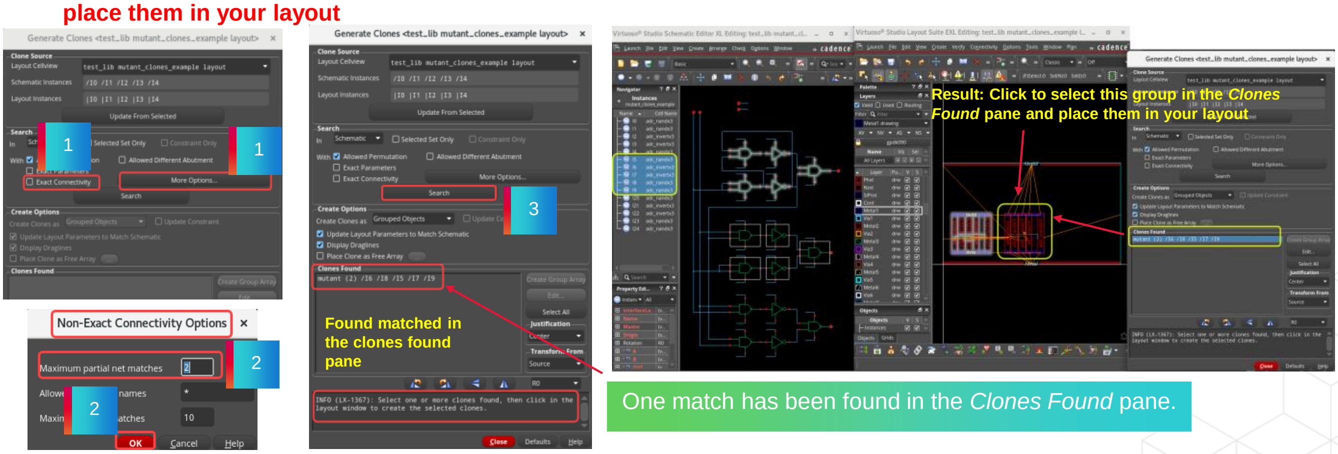
No matching target is found in the *Clones Found* pane.

When generating the mutant clones with the *Exact Connectivity* option turned **ON** in the *Generate Clones* form, **no** matching target is found in the *Clones Found* pane.

Generating Mutant Clones with Exact Connectivity Turned ON/OFF with Relax Match Options (continued)

Generating Mutant Clones with *Exact Connectivity* turned OFF with Relax Match as 2

1. Turn the *Exact Connectivity* option **OFF**. Select **More options** .
2. In the **Non Exact Connectivity Option** select **2** as a **Maximum partial net matches** and click **Ok**.
3. click the **Search** button again to find the matches. **Result: Click to select this group in the *Clones Found* pane and place them in your layout**



It is possible to “relax” the matching condition by turning the *Exact Connectivity* option **OFF**.

In this case, it is also important to specify how many “differences” you allow.

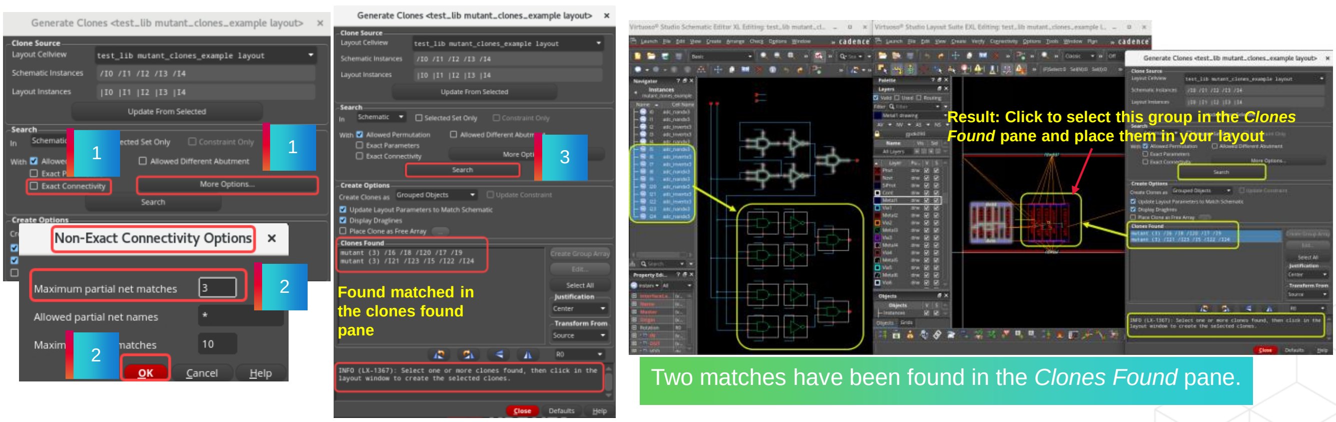
If there are too many differences, then the clones probably do not correspond.

When generating the mutant clones with the *Exact Connectivity* option turned **OFF** in the Generate Clones form and specifying *2* as *Maximum partial net matches* in the *Non-Exact Connectivity Options* form, you can see that **one** match has been found in the *Clones Found* pane.

Generating Mutant Clones with Exact Connectivity Turned ON/OFF with Relax Match Options (continued)

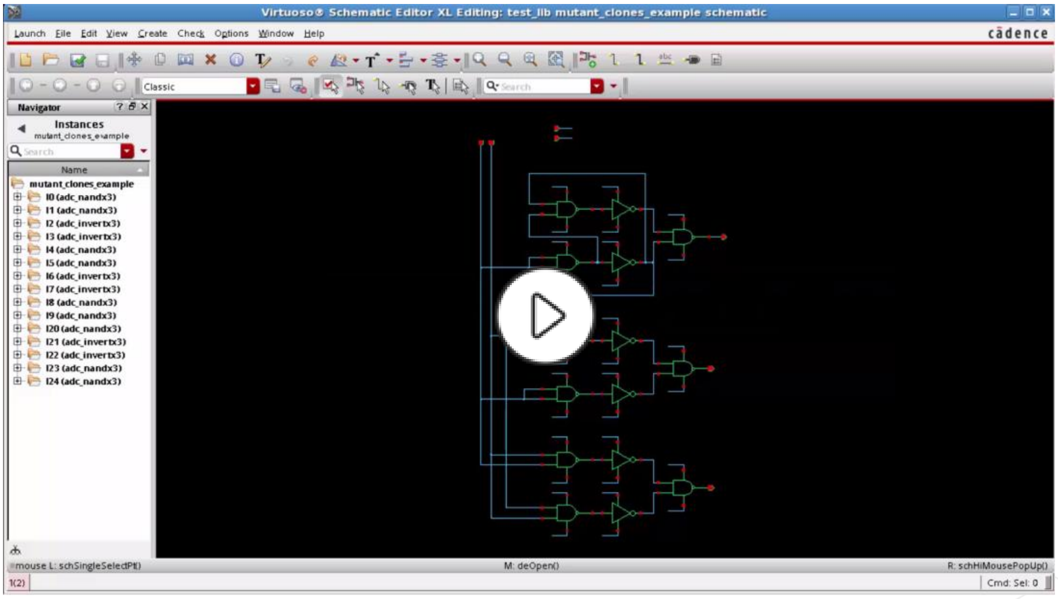
Generating Mutant Clones with *Exact Connectivity* turned OFF with Relax Match as 3

1. Turn the *Exact Connectivity* option **OFF**. Select **More options** .
2. In the **Non Exact Connectivity Option** select **3** as a **Maximum partial net matches** and click **Ok**.
3. click the **Search** button to find the matches. **Result: Click to select this group in the *Clones Found* pane and place them in your layout**



When generating the mutant clones with the *Exact Connectivity* option turned **OFF** in the Generate Clones form and specifying *3* as *Maximum partial net matches* in the *Non-Exact Connectivity Options* form, you can see that **two** matches have been found in the *Clones Found* pane.

Demo: How to Generate Mutant Clones with Exact Connectivity Turned ON/OFF with Relax Match Options



Video Play Time: 3.07 minutes

Click the Play button to start the video.

Click the link How to Generate Mutant Clones with Exact Connectivity Turned ON/OFF with Relax Match Options to play the video on support.cadence.com.

https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgvjEAA&pageName=ArticleContent

Clone Non-Checked Objects

Labels and Text Displays	Are copied but not synchronized. Not all editing made in one clone are synchronized in the other clones. These objects are not synchronized and checked by the synchronous clone consistency check. They are called *Clone Non-Checked Objects*. *Copy*/*Repeat Copy* will copy them but won't be synchronized.
Steiners and Guides	Are *Clone Non-Checked Objects*. These objects are not supported by the *Copy*/*Repeat Copy* commands. If the copy source contains a steiner or a guide, the copy operation fails and an appropriate warning message is generated, indicating that the selected source type is unsupported.

Clone Non-Checked Objects:

Clone non-checked objects are non-synchronized objects such as labels and text displays. When only clone non-checked objects are selected as the copy source, synchronous copies are not created. However, these objects, except some, such as *prBoundary*, are copied by the *Copy*/*Repeat Copy* commands.

For example, if you select an instance and a label as the copy source, both the objects are synchronously copied when you run a *Copy* or *Repeat Copy* command with the *Synchronous Copy*/*Create Synchronous Copy* checkbox selected. However, if you edit, move, or delete the label in one copy, the changes are not reflected in the corresponding label in the other copy because the labels in the two copies are not synchronized.

Let us see about the Clone Non Checked Objects.

Not all editing made in one clone are synchronized in the other clones.

These objects are not synchronized and checked by the Synchronous Clone Consistency Check.

They are called *Clone Non-Checked Objects*.

Copy and *Repeat Copy* commands will copy them but won't be synchronized.

Examples are *labels* and *text displays*. They are copied but not synchronized.

Steiners and *Guides* are *Clone Non-Checked Objects*.

These objects are not supported by the *Copy* and *Repeat Copy* commands.

If the copy source contains a steiner or a guide, the copy operation fails and an appropriate warning message is generated, indicating that the selected source type is unsupported.

Copied Objects

Bound Instances	The copied objects will not be bound by the physical clone engine. However, they might be bound in some cases by the incremental connectivity aware binder engine.
P&R Objects	Some of these objects cannot be copied (such as the P&R Boundary and the Snap Boundary). If *P&R Boundary*/*Snap Boundary* is selected among the copy source, the *Copy* command issues a warning message and the *P&R Boundary*/*Snap Boundary* is not copied. If a figure is not copied for any reason, this figure will not be included inside the source synchronous clone.
Non-Basic Groups	NOT supported. Non-basic groups include groups such as custom row and placement area which are not supported.

Let us see about the copied objects.

For bound instances, the copied objects will not be bound by the physical clone engine.

However they might be bound in some cases by the incremental connectivity aware binder engine.

For P&R Objects, some of these objects cannot be copied such as the P&R Boundary and the Snap Boundary.

If P&R Boundary or Snap Boundary is selected among the copy source, the Copy command issues a warning message and the P&R Boundary or Snap Boundary is not copied.

If a figure is not copied for any reason, this figure will not be included inside the source synchronous clone.

Non-Basic Groups are not supported.

Non-basic groups include groups such as custom row and placement area which are not supported.

More About Copied Objects

Hierarchical Synchronous Clones	NOT supported. If more than one synchronous clone is selected as the copy source, you will get this message: Caution (LX-2379): Hierarchical synchronous clones are not supported.
ROD Objects	Supported. ***Synchronization might be slow*** because when a rod object is edited, the ROD engine deletes the object and recreates a new modified object.
Mosaics	Are copied by the *Copy*/*Repeat Copy* commands. The source mosaic will be added inside the clone source figGroup and the target mosaic will be added inside the clone target figGroup. However, since mosaics are only partially supported by cloning, there may be some issues during synchronization.
PinFig Objects	Pins are copied by the *Copy*/*Repeat Copy* commands. When a pin figure (*pinFig*) is copied, a new pin is created for this *pinFig* and the pin is connected to the same terminal as the source pin.

Let us see some more about copied objects.

Hierarchical synchronous clones are not supported.

If more than one synchronous clone is selected as the copy source, you will get this message: Caution: Hierarchical synchronous clones are not supported.

Rod Objects are supported. Synchronization might be slow because when a rod object is edited, the ROD engine deletes the object and recreates a new modified object.

Mosaics are copied by the *Copy* and *Repeat Copy* commands.

The source mosaic will be added inside the clone source figGroup and the target mosaic will be added inside the clone target figGroup.

However, since mosaics are only partially supported by cloning, there may be some issues during synchronization.

PinFig Objects. Pins are copied by the *Copy* and *Repeat Copy* commands.

When a pin figure is copied, a new pin is created for this pinFig and the pin is connected to the same terminal as the source pin.

Generating Selected from Source(GSFS)

GSFS command to incrementally generate schematic instances and pins in the layout view. You can select the components to be generated either before or after you launch the command. You can select the components to be generated either from the schematic canvas or from the *Ungenerated instances* list in the Navigator assistant.

Use *GSFS* command to populate the design elements from the schematic to an empty or partially populated layout canvas.

1. Select the components to be generated, say, for example, **NM0** in the schematic window or in the Navigator assistant.
2. Generate selected from the source by choosing either the **Connectivity − Generate − Selected From Source** menu command or by clicking the **Generate Selected From Source** (**GSFS**) icon in the Layout EXL toolbar to open the Generate Selected Components form.
3. Keep the default values in the Generate Selected Components form and place the components inside the layout canvas.

Result: In this example, the 16 transistors are placed inside the *prBoundary* in the layout canvas.

Let us see how to use the *Generate Selected From Source* command.

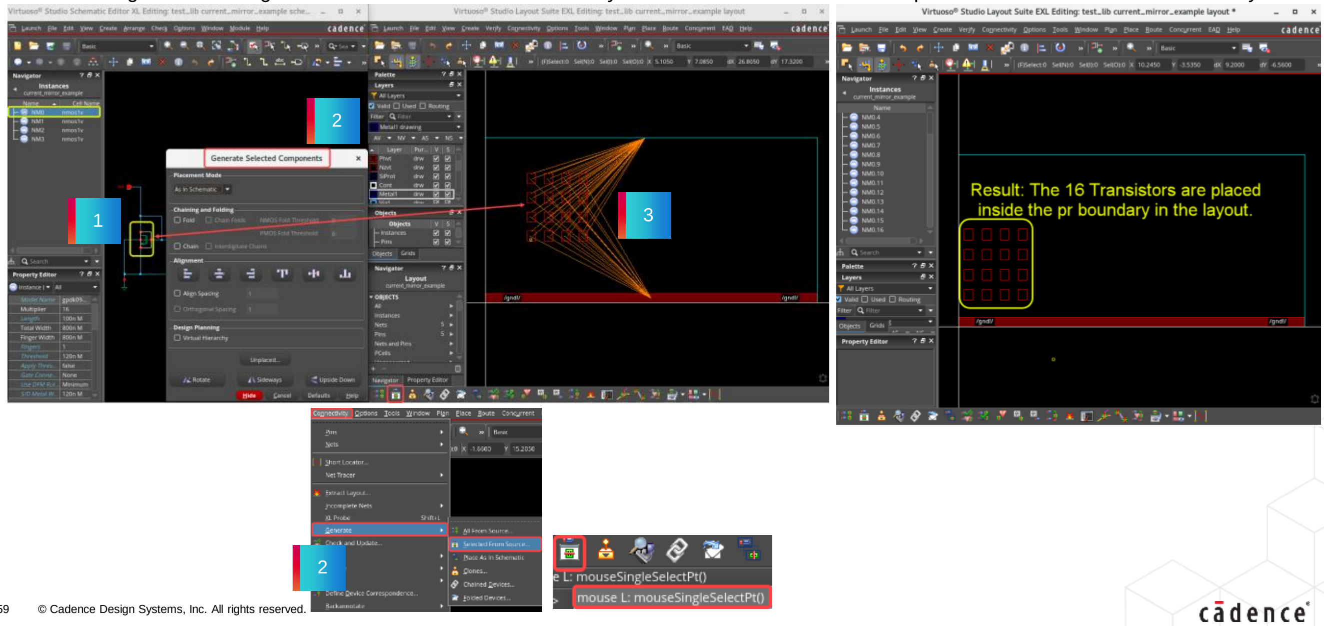
You can use the Generate Selected From Source command to incrementally generate schematic instances and pins in the layout view.

You can select the components to be generated either before or after you launch the command.

You can select the components to be generated either from the schematic canvas or from the *Ungenerated instances* list in the Navigator assistant.

Generating Selected from Source (continued)

The instances generated using the *GSFS* command are automatically bound to their schematic counterparts and have the correct connectivity.



You can use the *GSFS* command to populate the design elements from the schematic to an empty or partially populated layout canvas.

Select the components to be generated, say, for example, *NM0* in the schematic window or the Navigator assistant.

Generate selected from the source by choosing either the *Connectivity – Generate – Selected From Source* menu command or by clicking the *Generate Selected From Source* icon in the Layout XL toolbar to open the Generate Selected Components form.

Keep the default values in the Generate Selected Components form and place the components inside the layout canvas.

You can see that in this example, the 16 transistors are placed inside the *prBoundary* in the layout canvas.

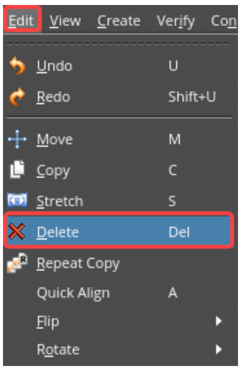
The instances generated using the *GSFS* command are automatically bound to their schematic counterparts and have the correct connectivity.

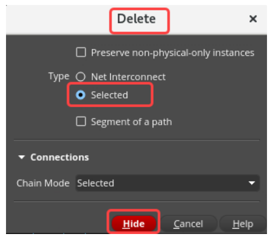
Deleting the Instances

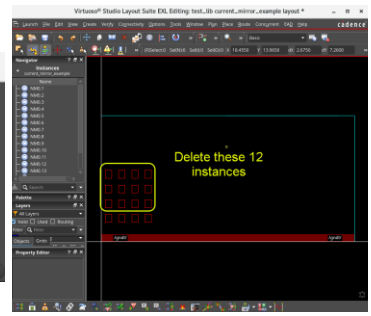
Use the **Delete** form to delete net interconnect, path segments, or device chains.

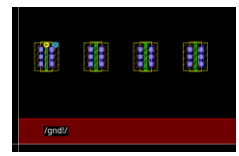
To invoke the Delete form in the layout canvas:

1. Choose the **Edit – Delete** menu command.
2. Or click the **Delete** icon in the Edit toolbar.
3. Or press the bindkey **Del**.









Because you intend to use a block of "4-fingers" as your building tile, you can delete the remaining 12 instances and only keep the 4 you need.

Let us see how to delete the instances.

In the design canvas, choose the *Edit – Delete* menu command or click the *Delete* icon in the Edit toolbar or press the bindkey *Delete* to invoke the Delete form.

You can use the *Delete* form to delete the net interconnect, the path segments, or the device chains.

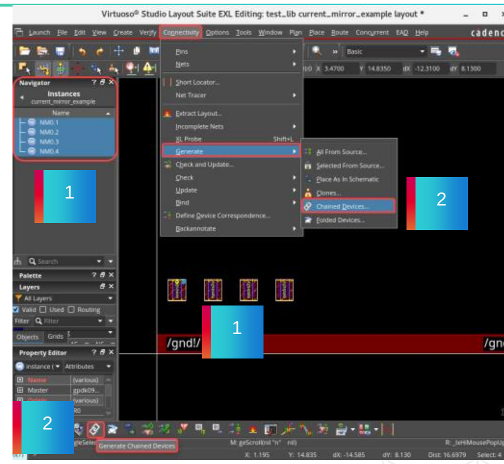
In this example, because you intend to use a block of "*4-fingers*" as your building tile, you can delete the remaining *12* instances and only keep the *4* you need.

Abutting and Chaining the Devices

You can use the **Generate Chained Devices** form to select and chain specific devices in your design.

Select the devices you want to chain directly in the layout canvas or in the Navigator assistant and click **Apply** in the form to chain them.

1. Select the devices to be chained, say for example, select all 4 transistors in the layout canvas or in the Navigator assistant.
2. To chain the devices, in the design canvas, choose the **Connectivity – Generate – Chained Devices** menu command or click the **Generate Chained Devices** icon in the Layout EXL toolbar.



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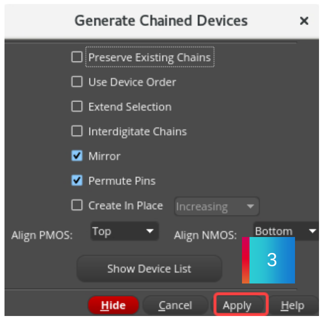
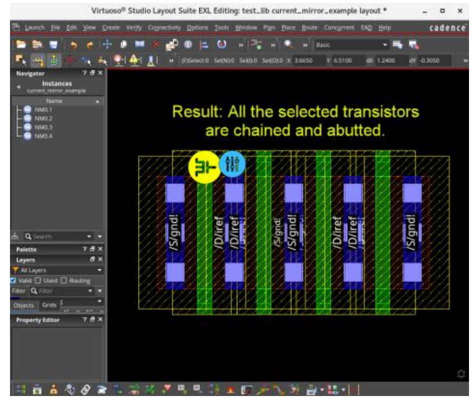
You can use the *Generate Chained Devices* form to select and chain specific devices in your design.

Select the devices you want to chain directly in the layout canvas or in the Navigator assistant and click the *Apply* button in the form to chain them.

Abutting and Chaining the Devices (continued)

3. In the Generate Chained Devices form, click the **Apply** button and place the devices to chain them.
   - If the majority of the selected devices are oriented vertically, the generated chain is also oriented vertically.
   - The devices to be chained need not be bound to a schematic instance.





Result: All the selected 4 transistors are chained and abutted.

Select the devices to be chained, say, for example, select all the 4 transistors in the layout canvas, or in the Navigator assistant.

To chain the devices, in the design canvas, choose the *Connectivity – Generate – Chained Devices* menu command or click the *Generate Chained Devices* icon in the Layout XL toolbar.

In the Generate Chained Devices form, click the *Apply* button and place the devices to chain them.

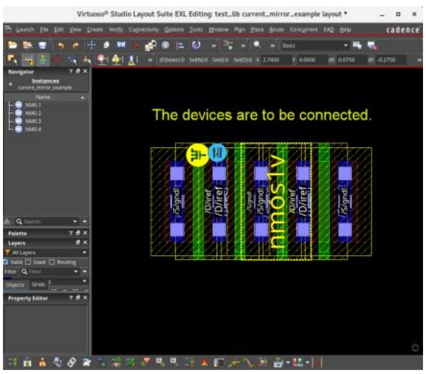
You can observe that in this example, all the selected transistors are chained and abutted.

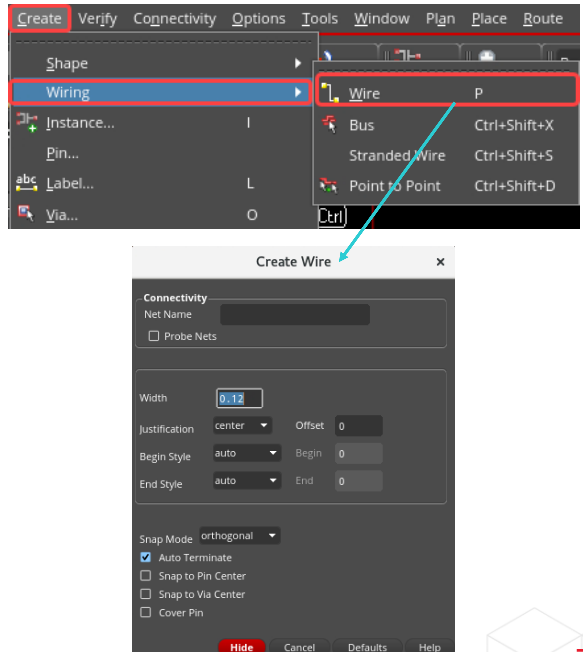
If the majority of the selected devices are oriented vertically, the generated chain is also oriented vertically.

The devices to be chained need not be bound to a schematic instance.

Wiring the Devices

To create wiring for the devices in the layout canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.





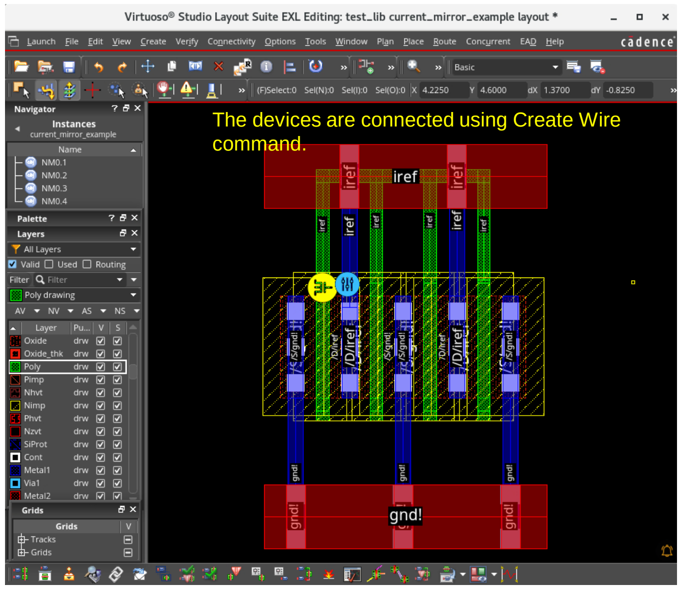
To create wiring for the devices, in the design canvas choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire form.

Using the *Create Wire* form, you can create pathSegs and vias in the routes.

In this example, the devices are connected using the *Create Wire* command.

Wiring the Devices (continued)

- Using the *Create Wire* form, you can create pathSegs and vias in routes. Use Bindkey **P** to use the wire command frequently.



To create wiring for the devices, in the design canvas choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire form.

In this example, the devices are connected using the *Create Wire* command.

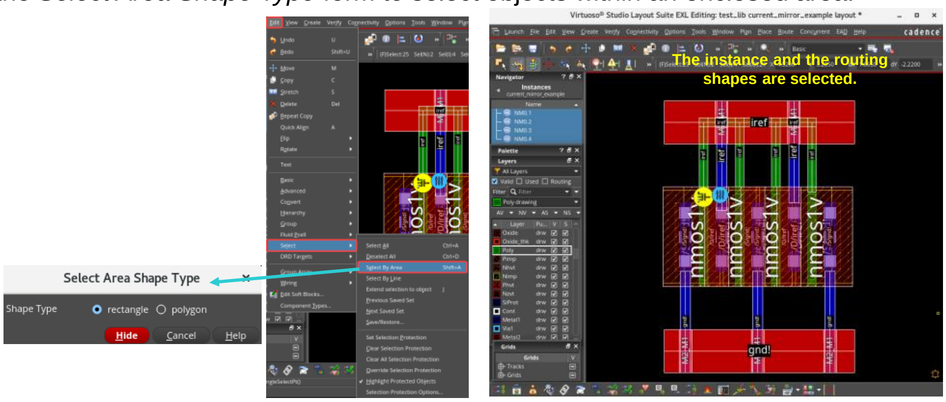
Using the *Create Wire* form, you can create pathSegs and vias in the routes.

Use Bind-key "P" to use the wire command frequently.

Selecting Objects by Defining an Area

In the layout canvas, choose the **Edit – Select – Select By Area** menu command or press the bindkey **Shift+A** to invoke the Select Area Shape Type form.

Use the *Select Area Shape Type* form to select objects within an enclosed area.



Use this command to select all the 4 instances and the routing shapes in the design.

Let us see how to select the objects by defining an area.

In the design canvas, choose the *Edit – Select – Select By Area* menu command or press the bindkey *Shift+A* to invoke the Select Area Shape Type form.

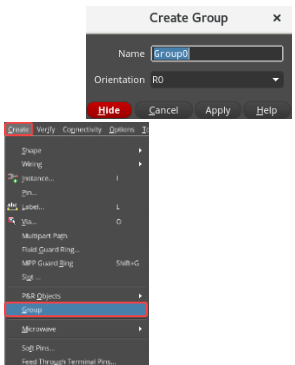
You can use this form to select the objects within an enclosed area selecting the shape type as either rectangle or polygon.

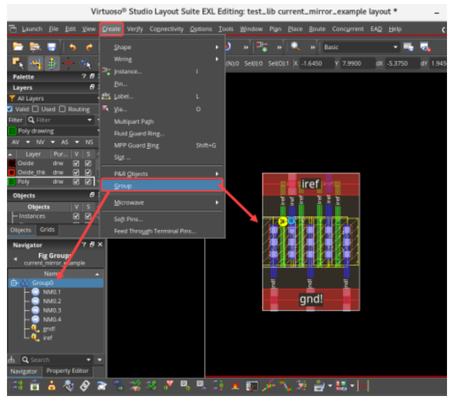
Use this command to select all the *4* instances and the routing shapes in the design.

Creating the Groups

In the layout canvas, choose the **Create – Group** menu command to open the **Create Group** form.

Use the *Create Group* form to create the groups.





The new group "*Group0"* is displayed in the Navigator assistant.

This group of *4-fingers* will be the building block you use to create all the transistors in your layout cell.

Let us see how to create the Groups.

In the design canvas, choose the *Create – Group* menu command to open the Create Group form.

You can use this form to create the groups.

In this example, after selecting the *4* instances and the routing shapes in the design, when you apply the Create Group command they are formed as "*Group0*".

The new group "*Group0*" is displayed in the Navigator assistant.

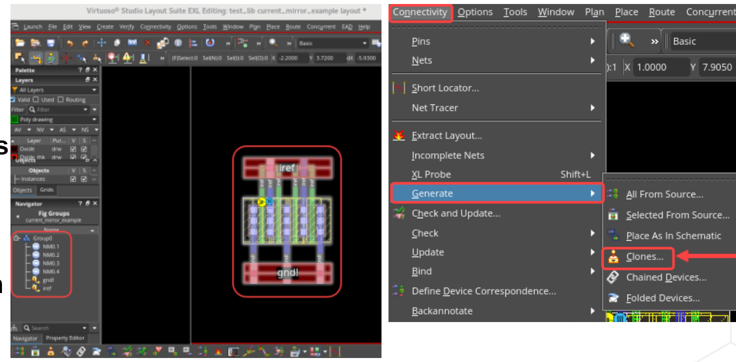
This group of *4-fingers* will be the building block you use to create all the transistors in your layout cell.

Generating the Clones for the Group

Generate Clones form to replicate the section of the layout that is associated with a section of the schematic in such a way that the new piece of layout material can be placed at more than one location with each part preserving the hierarchical structure of the design.

You can use the Generate Clones form to generate the clones for the group.

1. In the layout canvas, select the **Group0** and choose the **Connectivity – Generate – Clones** menu command or choose the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command.
2. Leave the default settings and click the **Search** button in the Generate Clones form.



Result: This action populates the Clones Found list with 12 instances. These are the same 12 instances you deleted previously.

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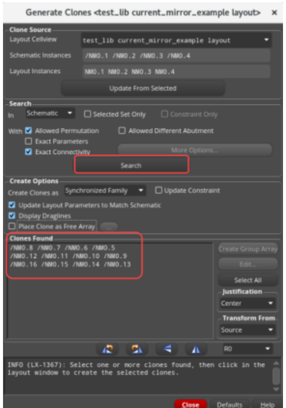
You can use the *Generate Clones* form to replicate the section of the layout that is associated with a section of the schematic in such a way that the new piece of layout material can be placed at more than one location with each part preserving the hierarchical structure of the design.

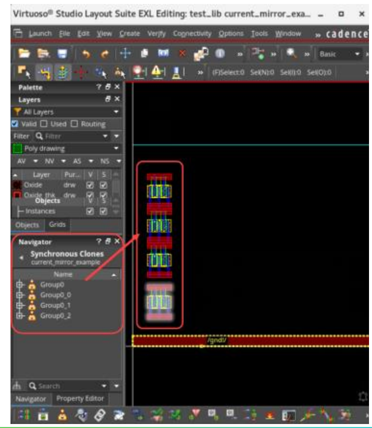
You can use this form to generate the clones for the group.

Generating the Clones for the Group (continued)



This action populates the *Clones Found* list with **12** instances. These are the same **12** instances you deleted previously.

Keep these settings in the Generate Clones form.



Keep selecting all these matches found, place these three groups of four transistors one on top of the other which forms a column automatically.

Let us see how to Generate the Clones.

In the design canvas, after selecting the newly created group, *Group0*, start the Clone command.

Leave the default settings and click the Search button in the Generate Clones form.

This action populates the *Clones Found* list with *12* instances.

These are the same *12* instances you deleted previously.

Now you will place these matches found in the layout canvas.

Keep these settings in the Generate Clones form.

Keep selecting all these matches found, place these three groups of four transistors, one on top of the other which forms a column automatically.

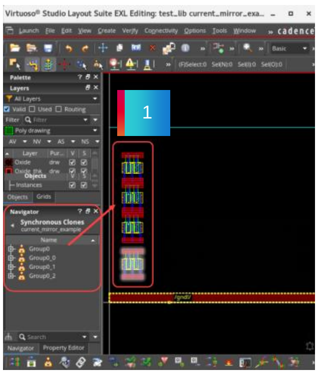
The four clone groups *Group0*, *Group0_0*, *Group0_1*, and *Group0_2*, are now "roughly" aligned in a column, but their placement is not ideal.

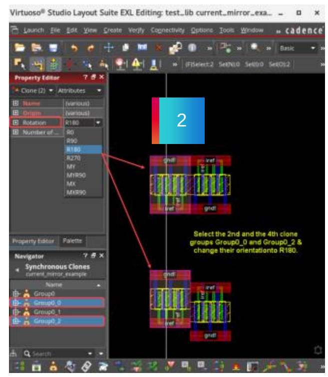
Mirroring the Devices

The four clone groups *Group0*, *Group0_0*, *Group0_1*, and *Group0_2* are now “roughly” aligned in a column, but their placement is not ideal.

You can “mirror” some of the transistors to have a better connection between the **VSS** rails.

1. Select the **2nd** and the **4th** clone groups *Group0_0* and *Group0_2* in the layout window or in the Navigator assistant.
2. Change their orientation to **R180**.





Result: The **2nd** and the **4th** transistors are mirrored now.

Let us see how to mirror the devices.

You can “mirror” some of the transistors to have a better connection between the **VSS** rails.

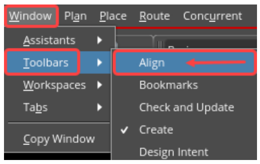
Select the **2nd** and the **4th** clone groups *Group0_0* and *Group0_2* and change their orientation to **R180**.

You can observe that the **2nd** and the **4th** transistors are mirrored now.

Aligning Objects by Using the Align Toolbar

For performing quick alignment operations, you can use the **Align** command.

In the layout window, choose the **Window – Toolbars – Align** menu command to display it in case it is hidden.



Align toolbar

In the layout window, set these options in the *Alignment* toolbar.

1. Set the *Align Spacing* to **0**.
2. Select *Group0* & *Group0_2*.
3. Click the **Align Right** button.

Result: *Group0* & *Group0_2* are aligned to the right now.

1. Set the *Align Spacing* to **0**.
2. Select *Group0_0* & *Group0_1*.
3. Click the **Align Left** button.

Result: *Group0_0* & *Group0_1* are aligned to the left now.

Use the **Align** toolbar to align the components.

Let us see how to align the objects by using the Align toolbar.

You can use the *Align* command for performing quick alignment operations.

In the design window, choose the *Window – Toolbars – Align* menu command to display it in case it is hidden.

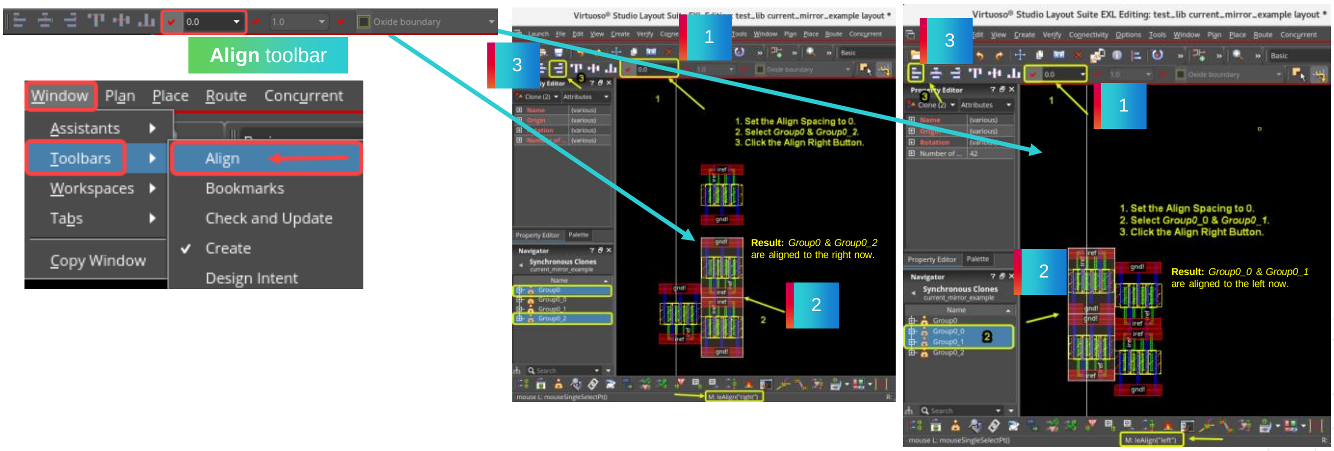
You can use the *Align* toolbar to align the components.

In the design window, set the required options in the *Alignment* toolbar for aligning the objects.

Aligning Objects by Using the Align Toolbar (continued)

Using the **Align** toolbar, you can perform left, right, top, and bottom alignment of the selected objects.

You can also align them vertically or horizontally. The toolbar also provides the option of specifying alignment and orthogonal spacing values.



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Using the *Align* toolbar, you can perform left, right, top, and bottom alignment of the selected objects.

You can also align them vertically or horizontally.

The toolbar also provides the option of specifying alignment and orthogonal spacing values.

Firstly, set the *Align Spacing* to *0* in the Align toolbar.

Then select *Group0* and *Group0_2*.

And click the *Align Right* button.

You can see that *Group0* and *Group0_2* are aligned to the right now.

Next, set the *Align Spacing* to *0*.

Then select *Group0_0* and *Group0_1*.

And click the *Align Left* button.

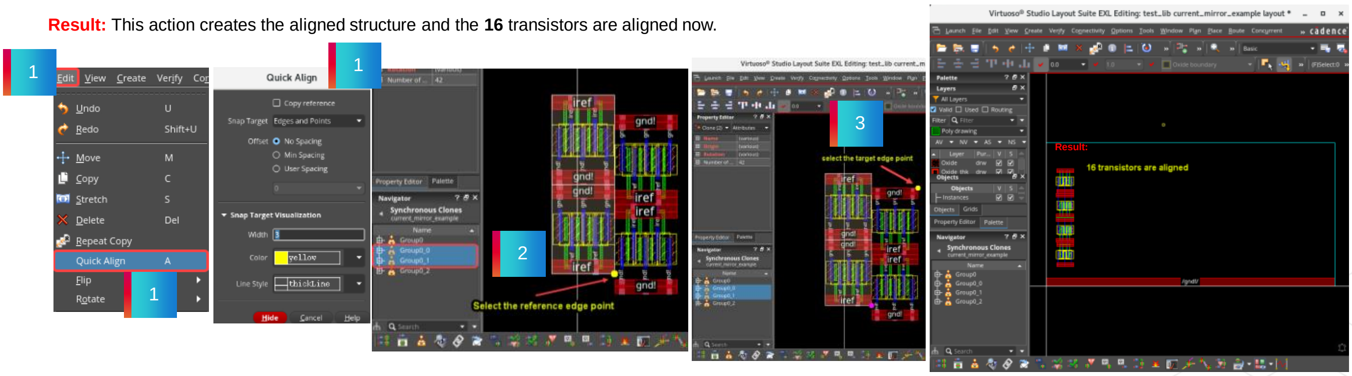
You can observe that *Group0_0* and *Group0_1* are aligned to the left now.

Aligning Objects by Using the Quick Align Command

The **Quick Align** command enables you to use objects, instances, and groups as the reference set and also use them as the target to which the reference set is to be aligned.

1. With the **2 upper clone families** selected, choose the **Edit – Quick Align** menu command or press the bindkey **A** to invoke the Quick Align form.
2. Select the reference edge point WITHIN selection set.
3. Select the target edge point.

Result: This action creates the aligned structure and the **16** transistors are aligned now.



Let us see how to align the objects by using the Quick Align command.

The *Quick Align* command enables you to use objects, instances, and groups as the reference set and also use them as the target to which the reference set is to be aligned.

With the *2 upper clone families* selected choose the *Edit – Quick Align* menu command or press the bindkey *A* to invoke the Quick Align form.

In the Quick Align form, select the reference edge point WITHIN selection set.

Then select the target edge point.

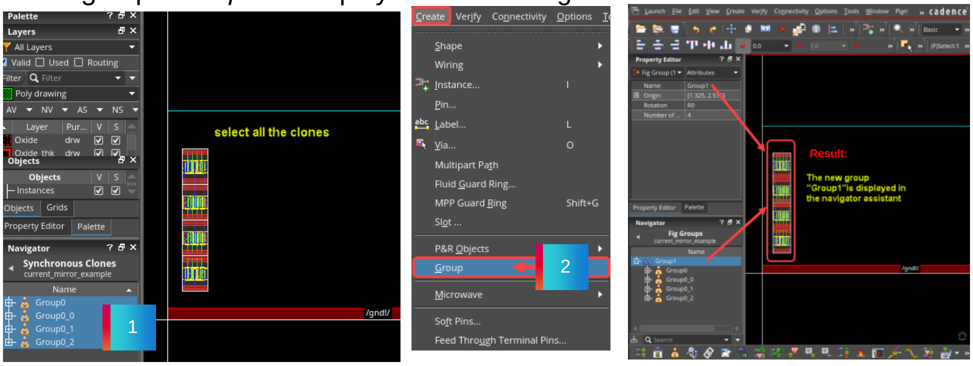
You can observe that this action creates the aligned structure and the *16* transistors are aligned now.

Grouping the Transistors

The **Group** command allows you to group shapes and/or instances together in order for you to manipulate the objects together, rather than individually.

1. Select all the clones in the layout canvas.
2. Choose the **Create – Group** menu command in the layout canvas.

Result: The new group *"Group1"* is displayed in the Navigator assistant.



You can now group together the *16 transistors* you just aligned in the layout canvas.

The *Group* command allows you to group shapes and or instances together in order for you to manipulate the objects together, rather than individually.

Select all the clones in the layout canvas.

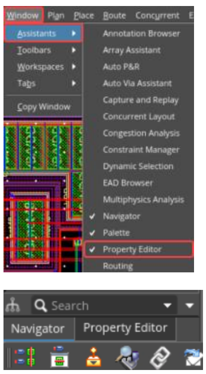
Then choose the *Create – Group* menu command.

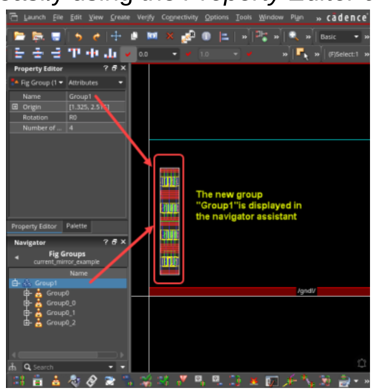
You can observe that the new group "*Group1*" is displayed in the Navigator assistant.

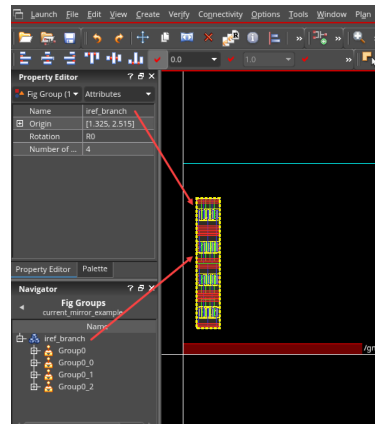
Renaming the Group Using the Property Editor Assistant

In the design canvas, choose the **Window – Assistants** menu command if **Property Editor** assistants is not displaying.

You can rename the groups easily using the *Property Editor* assistant.







The group **Group1** is renamed as **iref_branch** using the **Property Editor** assistant.

In the design canvas, choose the *Window – Assistants* menu command and select Navigator and Property Editor assistants.

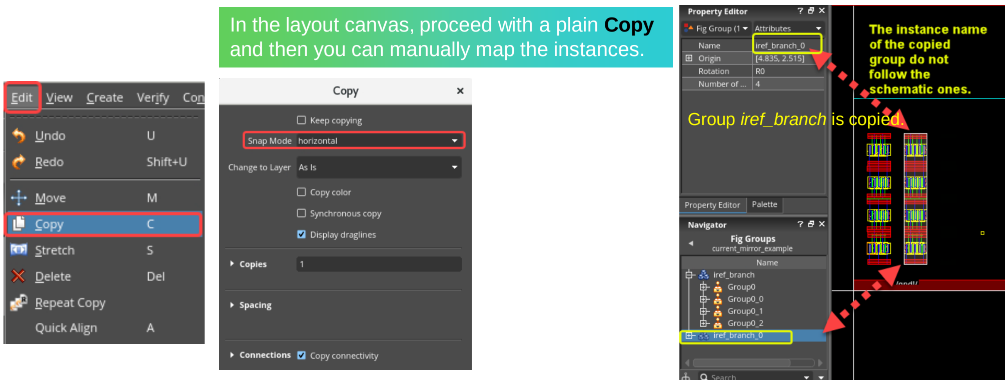
You can rename the groups easily using the Property Editor assistant.

Here the group *Group1* is renamed as *iref_branch* using the Property Editor assistant.

Copying the Group of Devices

In the design canvas, select the object to be copied and choose the **Edit – Copy** menu command or click the **Copy** icon in the Edit toolbar or press the bindkey **C** to invoke the Copy form.

The *Copy* command enables you to place a copy of an object in a cellview. You can copy an object to the same cellview or another cellview.



The resulting copy is not “understood” by Layout-EXL.
You need to manually map the instances.

You need to create the second transistor.

In this case, you cannot clone it because the connectivity is not identical.

The gate connection is different.

You can proceed with a plain *Copy* and then you can manually map the instances.

In the design canvas, select the object to be copied and choose the *Edit – Copy* menu command or click the *Copy* icon in the Edit toolbar or press the bindkey *C* to invoke the Copy form.

The *Copy* command enables you to place a copy of an object in a cellview.

You can copy an object to the same cellview, or another cellview.

The group *iref_branch* is copied as seen here.

The resulting copy is not “understood” by Layout-XL.

You need to manually map the instances.

Defining the Device Correspondence

You can use the **Define Device Correspondence** (**DDC**) form to view and change the device correspondence for the instances and terminals in your design.

Use *DDC* command to manually map the instances.

1. In the layout canvas, choose the **Connectivity – Define Device Correspondence** menu command or click the **Define Device Correspondence** (**DDC**) icon in the Layout EXL toolbar to open the Define Device Correspondence form.
2. Select **NM1** on the schematic side (this is the transistor you are currently building) and all of the "unknown" instances on the layout side.
3. Then click the **Bind** button.

Result: In the DDC form, you can check that the mapping is done between **NM1** in the schematic window and the newly copied instances in the layout window.

You can use the *Define Device Correspondence* form to view and change the device correspondence for the instances and terminals in your design.

You can use the *DDC* command to manually map the copied instances.

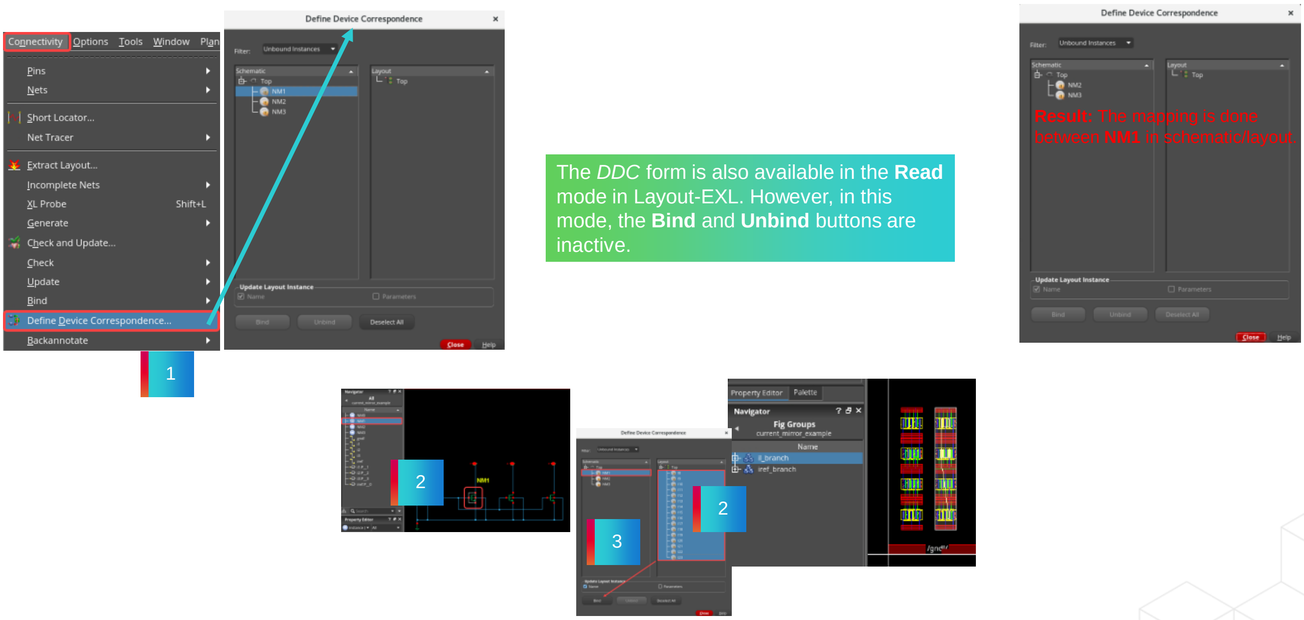
In the design canvas, choose the *Connectivity – Define Device Correspondence* menu command or click the *Define Device Correspondence* icon in the Layout EXL toolbar to open the DDC form.

Select the objects to be mapped in the schematic, in this example, *NM1* on the schematic side which is the transistor you are currently building and all of the "unknown" instances on the layout side.

Then click the *Bind* button.

You can observe that the mapping is done between *NM1* in the schematic window and the newly copied instances in the layout window in the DDC form.

Defining the Device Correspondence (continued)



After opening the DDC form, select the objects to be mapped in the schematic and in the layout and click the Bind button to map the instances.

You can observe that the mapping is done between *NM1* in the schematic window and the newly copied instances in the layout window in the Define Device Correspondence form.

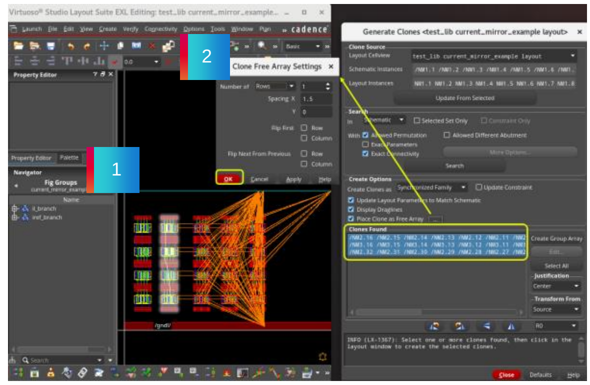
The Define Device Correspondence form is available in the *Read* mode as well in Layout-EXL.

However, in this mode, the *Bind* and *Unbind* buttons are inactive.

Generating the Missing Transistors

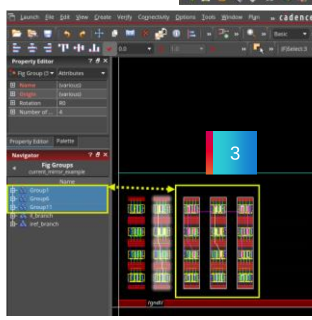
You can clone the copied and mapped structure.

1. Select NM1 and run Search in the Generate Clones form.
2. Set these options in the Generate Clones form.
3. Place the three groups of transistors one close to the other.



Three groups of transistors are found in the *Clones Found* pane

You can observe that 8 instances of *NM3* are missing from the layout



The first 16 instances of *NM2*
The second 16 instances of *NM2* (this is a 32x transistor)
16 of the 24 instances which compose *NM3* (this is a 24x transistor)

You will Generate the Missing Transistors now.

Because you are still missing *8 transistors*, use cloning to generate those devices.

Select the bottom *8 transistors* in the last group.

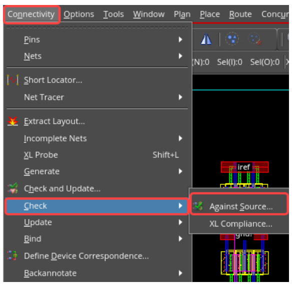
Clone them by invoking the *Generate Clones* form, click the *Search* button and place the matches found in the layout canvas.

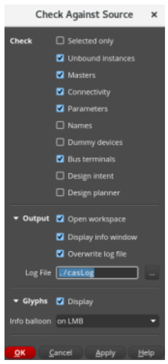
You have the complete layout for your current mirror device.

Checking Against Source

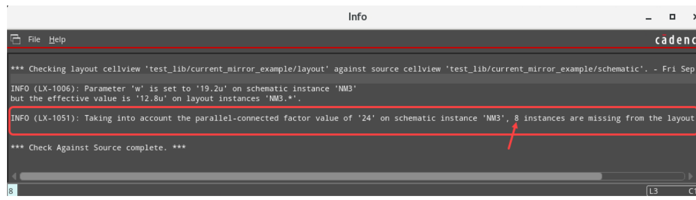
In the design canvas, choose the **Connectivity – Check – Against Source** menu command or click the **Check Against Source (CAS)** icon in the Layout EXL toolbar to open the Check Against Source form.

Use the *CAS* command to check the differences between the schematic and the layout.





Use the *CAS* form to specify the types of layout versus schematic checks you want to run and how the report should be displayed.



The *Info* window shows the differences between the schematic and the layout.
You can check the *casLog* file as well from the current working directory.
You can observe that *8 instances* are missing from the layout.

Let us see how to use the *Check Against Source* command.

In the design canvas, choose the *Connectivity – Check – Against Source* menu command or click the *Check Against Source* icon in the Layout EXL toolbar to open the CAS form.

You can use this command to check the differences between the schematic and the layout.

Use this form to specify the types of layout versus schematic checks you want to run and how the report should be displayed.

When you run this command, the *Info* window shows the differences between the schematic and the layout.

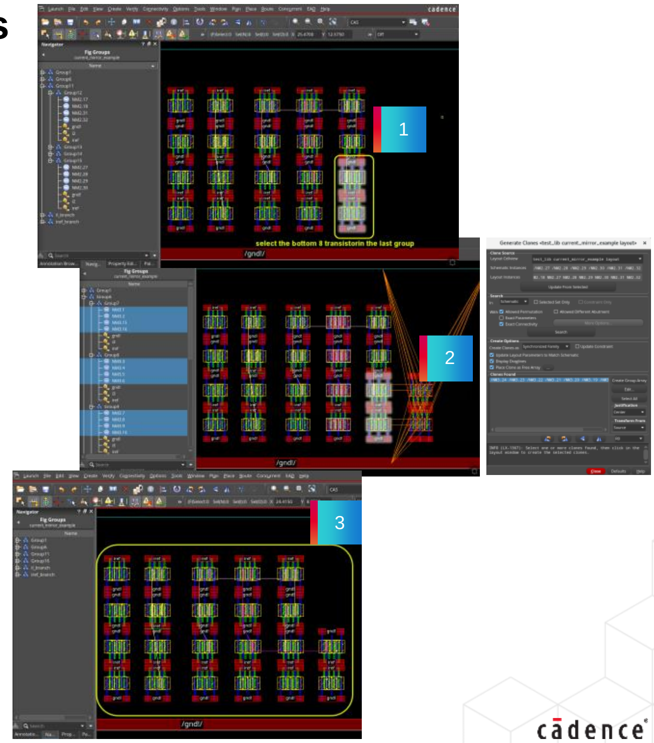
You can check the *casLog* file as well from the current working directory.

You can observe that *8 instances* are missing from the layout.

Generating the Missing Transistors

You are still missing **8 transistors**, use cloning to generate those devices.

1. Select the bottom 8 transistors in the last group.
2. Run Search in the Generate clones form and place the matched found.
3. You have the complete layout for your current mirror device.



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You will Generate the Missing Transistors now.

Because you are still missing *8 transistors*, use cloning to generate those devices.

Select the bottom *8 transistors* in the last group.

Clone them by invoking the *Generate Clones* form, click the *Search* button and place the matches found in the layout canvas.

You have the complete layout for your current mirror device.

Generating the Missing Transistors (continued)

Run **Check Against Source** (**CAS**) again and observe the *Info* window report.

There are no info/warning messages reported now.

Run *Check Against Source* command again and observe the *Info* window report.

You can see that there are no info or warning messages reported now.

Module Summary

In this module, you learned how to

- Generate clones as Free Objects, Grouped Objects, and Synchronized family
- Generate synchronous copy clones of wires and examine its synchronous behavior
- Generate synchronous copy clones of modgen and examine its synchronous behavior
- Generate mutant clones with exact connectivity turned ON/OFF with relax match options
- Analyze how to effectively reuse existing structures in your layout using the clone and the copy commands in VLS-EXL

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This page does not contain notes.

Labs

Lab 2-1 Create different Types of Clones

Lab 2-2 Generating Mutant Clones

Lab 2-3 Cloning and Copying in VLS-EXL

This brings you to lab exercise on Module 2.

In Lab 1, you will explore and create different types of clones.

In Lab 2, you will create clones when the connectivity is not identical, which is called mutant clones.

In Lab 3, you will clone a structure with the “exact” connectivity and copy and map to reuse blocks that do not have matching connectivity.

FAQ: Cloning

What is **Cloning**?

Cloning is the ability to replicate a section of the layout, associated with a section of the schematic.

This page does not contain notes.

FAQ: Generate Clones Form

How do you invoke the **Generate Clones** form?

In the design canvas, choose the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command.

This page does not contain notes.

FAQ: Update Clone Families Form

Name the tasks you can perform using the **Update Clone Families** form.

- You can use the **Update Clone Families** form to manage the families of synchronized clones in your design.
- You can update the status of a clone family, rename a clone family, remove a clone from a family, create and remove clone families, and add a group to an existing family.

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FAQ: Synchronous Copy

How do you generate **Synchronous Copy?**

Synchronous Copy is generated when an object or a set of objects that exist in the layout is copied and the synchronization between the copy source and the copy target is maintained.

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This page does not contain notes.

FAQ: Generating Objects

Which command can be used to incrementally generate schematic instances and pins in the layout view?

You can use the **Generate Selected From Source** (**GSFS**) command to incrementally generate schematic instances and pins in the layout view.

This page does not contain notes.

Quiz: Replicating the Layout Design

Which form can you use to replicate the section of the layout that is associated with a section of the schematic in such a way that the new piece of layout material can be placed at more than one location?

[Select the single correct choice]

A. Mutant Clones

B. Generate Clones

C. Synchronous Copy

D. Update Clone Families

E. None of the above

Answer: B

You can use the **Generate Clones** form.

This page does not contain notes.

Quiz: The Cloning Feature

The cloning feature is available in the following:

[Select all the correct choices]

A. Virtuoso Layout Suite XL

B. Virtuoso Layout Suite EXL

C. Layout Editor

D. All of the above

Answer: A and B

The cloning feature is available in **Virtuoso Layout Suite XL** and **Virtuoso Layout Suite EXL**.

This page does not contain notes.

Quiz: Synchronous Copy Feature

Name the form where "**Synchronous copy**" checkbox is available?

[Select the single correct choice]

A. Move

B. Stretch

C. Group

D. Copy

E. None of the above

Answer: D

"**Synchronous copy**" checkbox is available in the **Copy** form.

This page does not contain notes.

Quiz: Connectivity of the Clones

When the connectivity of the clones are not identical, what are they known as? ------ ------

[Enter the answer in the blank field]

Answer: **Mutant Clones**

When the connectivity of the clones are not identical, they are known as **Mutant Clones**.

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Quiz: Types of Objects

You can edit the instances at the top level, when you create clones as ---- Objects.

[Enter the answer in the blank field]

Answer: **Free**

You can edit the instances at the top level, when you create clones as **Free** Objects.

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Quiz: Analyzing the Differences Between Schematic/Layout

Which command can you use to analyze the differences between the schematic and the layout?

[Select the single correct choice]

A. Connectivity – Update – Components And Nets
B. Connectivity – Update – Connectivity Reference
C. Connectivity – Check – Against Source
D. Connectivity – Define Device Correspondence
E. None of the above

Answer: C

You can use the **Check Against Source** command to analyze the differences between the schematic and the layout.

This page does not contain notes.

Quiz: Update Clone Families Form

Changes made in the **Update Clone Families** form can be cancelled.

[Select the single correct choice from the two choices]

A. True

B. False

Answer: B

Changes that you make in the **Update Clone Families** form take effect immediately and **cannot** be cancelled.

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Quiz: Synchronous Copying

Objects created by “Synchronous Copying” are bound to schematic instances.

[Select the single correct choice from the two choices]

A. True

B. False

Answer: B

Objects created by “Synchronous Copying” will **NOT** be bound to schematic instances.

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Quiz: Create Options Section

You have the capability to **Create Clones as**:

[Select all the correct choices]

A. Non-Grouped Objects
B. Synchronized Family
C. Grouped Objects
D. Free Objects
E. All of the above

Answer: B, C, and D

You have the capability to **Create Clones as** *Synchronized Family*, *Grouped Objects*, and *Free Objects*.

This page does not contain notes.

Quiz: Updating Object Information

Which form can you use to view and change the object information for the instances and terminals in your design?

[Select the single correct choice]

A. Define Device Correspondence (DDC)

B. Check Against Source (CAS)

C. Create Group

D. Update Components and Nets (UCN)

E. None of the above

Answer: A

You can use the **DDC** form to view and change the device correspondence for the instances and terminals in your design.

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Module 3

Updating Connectivity and Nets

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Module Objectives

In this module, you

- Update the Connectivity Reference of the layout using a schematic different from the default schematic
- Examine the differences between the original and the new schematic views
- Verify data between schematic and layout views
- Update components and nets to implement an Engineering Change Order (ECO)
- Update the layout parameters with the new schematic parameters
- Place the new devices within the prBoundary using the Annotation Browser assistant
- Create wiring for the components
- Update the layout to reflect the pin names changes between schematic and layout using the Update Components and Nets (UCN) command

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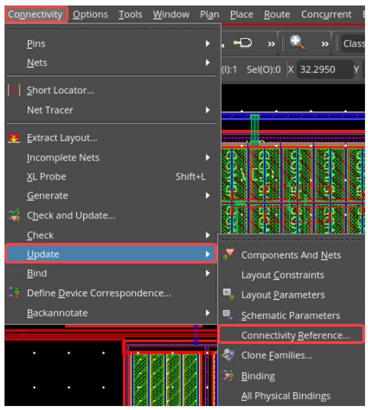
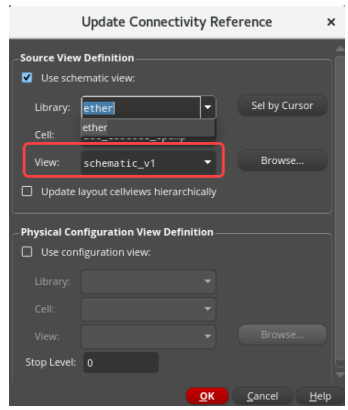
Invoking the Update Connectivity Reference Form

Choose the **Connectivity – Update – Connectivity Reference** menu command to display the Update Connectivity Reference form.

Use the *Update Connectivity Reference* command to update the source schematic for the layout view to change the physical configuration, which specifies how the layout is generated from the schematic.





Let us see how to invoke the *Update Connectivity Reference* form.

To update the connectivity reference, in the layout view, choose the *Connectivity – Update – Connectivity Reference* menu command to display the Update Connectivity Reference form.

You can use the *Update Connectivity Reference* command to update the source schematic for the layout view to change the physical configuration, which specifies how the layout is generated from the schematic.

Using the Update Connectivity Reference Form

Source View Definition lets you specify the view to use as the connectivity source.

Physical Configuration View Definition lets you specify the physical configuration view to use for the schematic-layout pair.

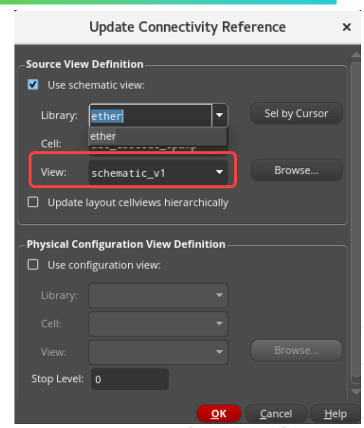
In the *Update Connectivity Reference* form, **Source View Definition** lets you specify the view to use as the connectivity source.

Physical Configuration View Definition lets you specify the physical configuration view to use for the schematic-layout pair.

Update Connectivity Reference Form:
Source View Definition Section

You can use the **Source View Definition** section to specify the view to use as the connectivity source.

Update Connectivity Reference Form



Use schematic view	Tells Layout-EXL to use a schematic view as the connectivity reference. If you uncheck the box, it means there is no connectivity reference.
Update layout cellviews hierarchically	Descends into a hierarchical layout cellview and updates the library setting for any lower-level cellviews in accordance with the library name specified for the top-level cell.

In the *Update Connectivity Reference* form, you can use the *Source View Definition* section to specify the view to use as the connectivity source.

In this section, *Use schematic view* field tells Layout-XL to use a schematic view as the connectivity reference.

If you uncheck the box, it means there is no connectivity reference.

Update layout cellviews hierarchically field descends into a hierarchical layout cellview and updates the library setting for any lower-level cellviews in accordance with the library name specified for the top-level cell.

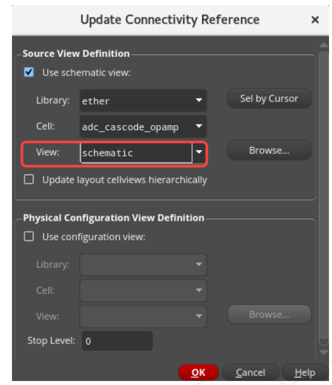
Update Connectivity Reference Form:

Source View Definition Section: Updating the Schematic View

Updating the Schematic View

- In Layout-EXL, the default schematic is always the schematic view with the same cell name as the layout view.
- To update your layout using a different schematic, you need to update the connectivity reference.
- In the *Update Connectivity Reference* form:
  - Check the **Use schematic view** option and select the **Library**, **Cell**, and **View** names of the schematic in the fields provided.
  - Select **Update layout cellviews hierarchically** to update the library setting for any lower-level cellviews in a hierarchical cellview.
  - When clicking **OK**, Layout-EXL sets the connectivity reference to the schematic you selected.

Update Connectivity Reference Form



Let us see how to update the schematic view.

In Layout-EXL, the default schematic is always the schematic view with the same cell name as the layout view.

To update your layout using a different schematic, you need to update the connectivity reference.

In the *Update Connectivity Reference* form:

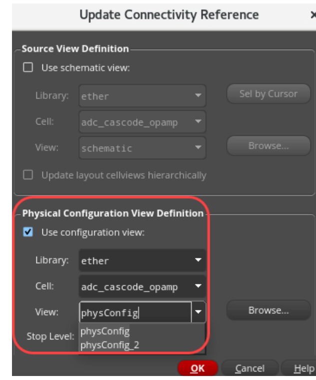
Check the *Use schematic view* option and select the *Library*, *Cell*, and *View* names of the schematic in the fields provided.

Select the *Update layout cellviews hierarchically* option to update the library setting for any lower-level cellviews in a hierarchical cellview.

When clicking the *OK* button, Layout-EXL sets the connectivity reference to the schematic you selected.

Update Connectivity Reference Form:
Physical Configuration View Definition Section

Update Connectivity Reference Form



You can use the **Physical Configuration View Definition** section to specify the physical configuration view to use for the schematic-layout pair.

Use configuration view	Tells Layout-EXL to use a configuration view as the connectivity reference. If you deselect this checkbox, Layout-EXL runs without a physical configuration.

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In the *Update Connectivity Reference* form, you can use the *Physical Configuration View Definition* section to specify the physical configuration view to use for the schematic-layout pair.

In this section, *Use configuration view* field tells Layout-EXL to use a configuration view as the connectivity reference.

If you deselect this checkbox, Layout-EXL runs without a physical configuration.

How to Update the Connectivity Reference

You can use the **Update Connectivity Reference** command to update the source schematic for your layout view and to change the physical configuration which specifies how the layout is generated from the source.

1. In the layout view, choose the **Connectivity – Update – Connectivity Reference** menu command to open the **Update Connectivity Reference** form.
2. In the **Update Connectivity Reference** form, check the **Use schematic view** option and update the ***View*** value from *schematic_v1* to *schematic*.
3. Click **OK** in the form.

Result: The *schematic* view opens which is referenced with *layout_v1* view. The new *schematic* view is now the connectivity reference for *layout_v1*.

Let us see How to Update the Connectivity Reference.

You can use the *Update Connectivity Reference* command to update the source schematic for your layout view and to change the physical configuration which specifies how the layout is generated from the source.

To update the connectivity reference, firstly in the layout view, choose the *Connectivity – Update – Connectivity Reference* menu command to open the Update Connectivity Reference form.

Then in this form, check the *Use schematic view* option and update the *View* value from *schematic_v1* to *schematic*.

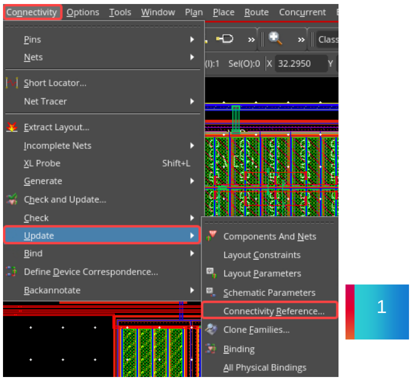
And click the *OK* button in the form.

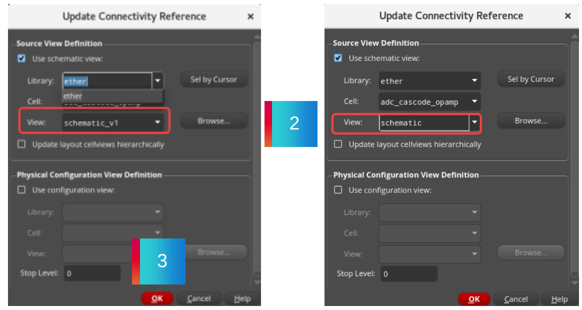
You can observe that the *schematic* view opens which is referenced with *layout_v1* view.

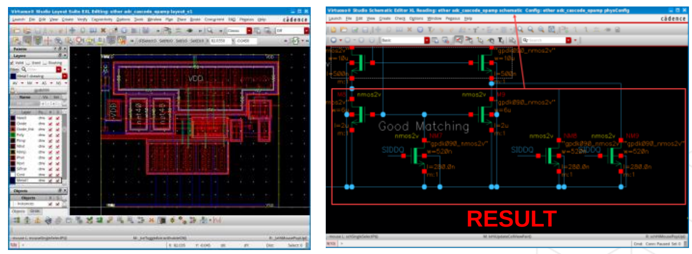
The new *schematic* view is now the connectivity reference for *layout_v1*.

How to Update the Connectivity Reference (continued)

Result: The *schematic* view opens which is referenced with *layout_v1* view. This view is now the connectivity reference for *layout_v1*.







The schematic view is updated from **schematic_v1** to **schematic**.

After invoking the *Update Connectivity Reference* form, with the *Use schematic view* option selected, change the *View* value from *schematic_v1* to *schematic* and click *OK* in the form.

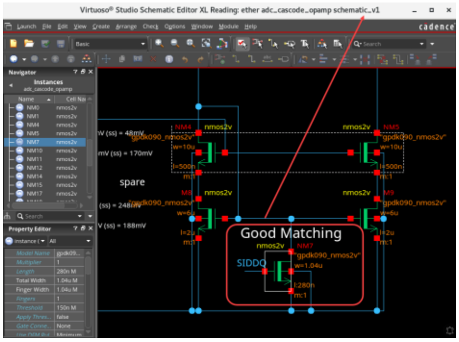
You can observe that the *schematic* view opens which is referenced with *layout_v1* view.

This view is now the connectivity reference for *layout_v1*.

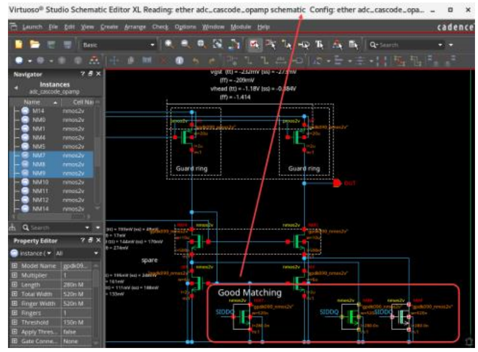
The schematic view is updated from *schematic_v1* to *schematic*.

Examining the Differences Between the Original *schematic_v1* and the New *schematic* Reference

Original *schematic_v1* View



New *schematic* Reference View



In the new *schematic* cellview, there are two additional devices, **NM8** and **NM9**, in the lower right of the schematic. The **NM7** device width size has also changed from **1.04µ** to **0.52µ**.

If you examine the differences between the original *schematic_v1* and the new *schematic* reference, you can observe that the changes are made in the lower right of the new connectivity reference, *schematic* view.

The original *schematic_v1* view remains open.

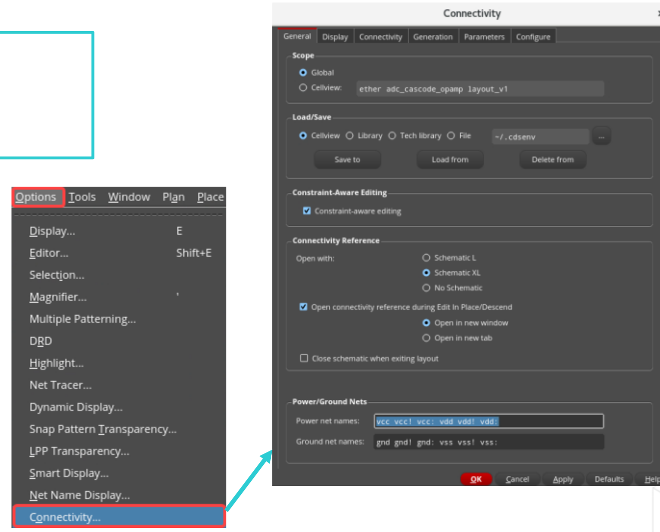
In the new *schematic* cellview, there are two additional devices, *NM8* and *NM9*, in the lower right of the schematic.

The *NM7* device width size has also changed from *1.04µ* to *0.52µ*.

Invoking the Layout XL Options Form

To set Layout XL options in the layout view, choose the **Options – Connectivity** menu command to open the Connectivity form.

Use the **Connectivity** form to set Layout XL options either for the current cellview or for the current Layout XL session.



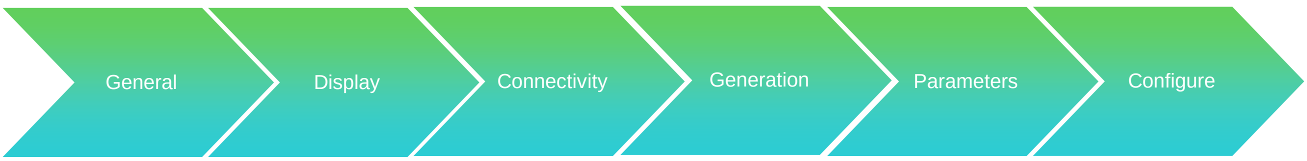
Let us see how to invoke the *Layout XL Options* form.

To set the Layout XL options, in the design view choose the *Options – Connectivity* menu command to open the Layout XL Options form.

You Use the **Connectivity** form to set Layout XL options either for the current cellview or for the current Layout XL session.

Using the Layout XL Options Form

You can use the *General*, *Display*, *Connectivity*, *Generation*, *Parameters*, and Configure sections in the **Layout XL Options** form to specify the required inputs to set Layout XL options.



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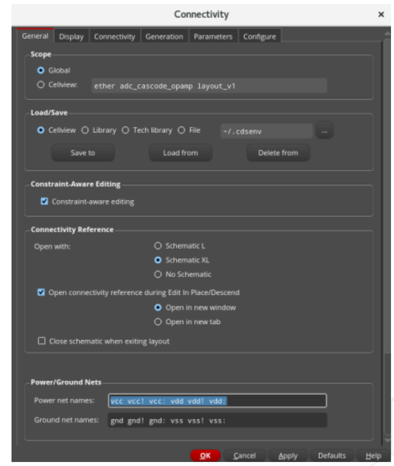
You can use the *General*, *Display*, *Connectivity*, *Generation*, *Parameters*, and *Configure* sections in the *Layout XL Options* form to specify the required inputs to set the Layout XL options.

Layout XL Options Form: General Section

General | Display | Connectivity | Generation | Parameters | Configure

Layout XL Options Form



Use the **General** options to define the scope of the options settings you make in the form and to load and save the settings from and to an options file.

Scope	Specifies whether the option settings apply for the current cellview only or globally for the current cellview and all the cellviews opened subsequently in Layout XL.
Load/Save	Lets you load, save or delete options values to or from a cellview, library, technology library, or options file.
Constraint-Aware Editing	Ensures that the *Copy*, *Move*, *Stretch*, and *Rotate* commands honor the following constraints: *Alignment*, *Fixed*, *Locked*, *Matched Parameters*, *Orientation*, *Matched Orientation*, and *Symmetry*.
Connectivity Reference	**Open with** specifies whether or not the Virtuoso® Schematic Editor is started when you launch Layout XL or GXL. **Open connectivity reference during Edit In Place/Descend** specifies whether or not the source cellview is opened when you use the *Edit In Place* or *Descend* commands.
Power/Ground Nets	**Power/Ground net names** specifies the list of power/ground net names used to achieve optimized chaining results in Layout XL device level schematics.

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Let us see the *General* section in the *Layout XL Options* form.

You can use the *General* options to define the scope of the options settings you make in the form and to load and save the settings from and to an options file.

Scope field specifies whether the option settings apply for the current cellview only or globally for the current cellview and all the cellviews opened subsequently in Layout XL.

Load or Save field lets you load, save, or delete options values to or from a cellview, library, tech library, or options file.

Constraint Aware Editing field ensures that *Copy*, *Move*, *Stretch*, and *Rotate* commands honor the following constraints: namely, *Alignment*, *Fixed*, *Locked*, *Matched Parameters*, *Orientation*, *Matched Orientation*, and *Symmetry*.

In *Connectivity Reference* field, *Open with* section specifies whether or not the Virtuoso Schematic Editor is started when you launch Layout XL or GXL.

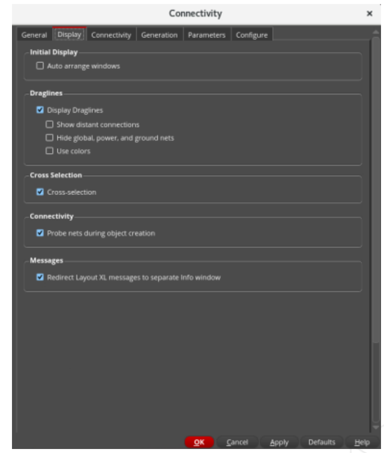
Open connectivity reference during Edit In Place or Descend section specifies whether or not the source cellview is opened when you use the *Edit In Place* or *Descend* commands.

In *Power and Ground Nets* field, *Power and Ground net names* specifies the list of power and ground net names used to achieve optimized chaining results in Layout XL device level schematics.

Layout XL Options Form: Display Section

General → **Display** → Connectivity → Generation → Parameters → Configure

Layout XL Options Form



Use the **Display** options to specify how the Layout XL windows are arranged on your desktop, how draglines are displayed in the layout window, and to enable and disable Cross-selection and net probing.

Initial Display	**Auto arrange windows** controls whether Layout XL automatically rearranges its four windows on your desktop when you launch the application. The default is **ON**.
Draglines	**Display Draglines** toggles the display of draglines when using the *Generate Selected From Source*, *Generate Clones*, *Move*, and *Stretch* commands.
Cross-selection	Turns on Cross-selection between the layout and the schematic. When you select a component in the layout, the corresponding component is selected in the schematic and vice versa, unless the instance is ignored in either view.
Connectivity	**Probe nets during object creation** controls whether nets that are tapped during interactive editing are highlighted in the layout canvas.
Messages	**Redirect Layout XL messages to separate Info window** shows the messages issued by the *Check Against Source*, *Update Layout Parameters*, and *Update Schematic Parameters* commands in a separate info window instead of in the CIW.

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Let us see the *Display* section in the *Layout XL Options* form.

You can use the *Display* options to specify how the Layout XL windows are arranged on your desktop, how draglines are displayed in the layout window, and to enable and disable cross-selection and net probing.

In *Initial Display* field, *Auto arrange windows* controls whether Layout XL automatically rearranges its four windows on your desktop when you launch the application. The default is *ON*.

In *Draglines* field, *Display Draglines* toggles the display of draglines when using the *Generate Selected From Source*, *Generate Clones*, *Move*, and *Stretch* commands.

Cross-selection field turns on cross-selection between layout and schematic.

When you select a component in the layout, the corresponding component is selected in the schematic and vice versa, unless the instance is ignored in either view.

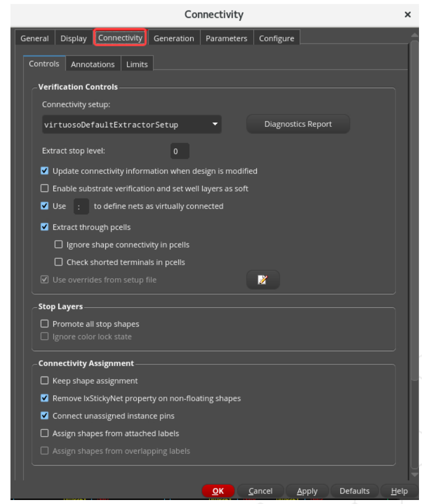
In *Connectivity* field, *Probe nets during object creation* controls whether nets that are tapped during interactive editing are highlighted in the layout canvas.

In *Messages* field, *Redirect Layout XL messages to separate Info window* shows the messages issued by the *Check Against Source*, *Update Layout Parameters*, and *Update Schematic Parameters* commands in a separate info window instead of in the CIW.

Layout XL Options Form: Connectivity Section

General → Display → Connectivity → Generation → Parameters → Configure

Layout XL Options Form



Use the **Connectivity** options to set the Layout XL parameters for the current design. This tab is further split into the following tabs, namely: **Controls**, **Annotations**, and **Limits**.

Controls	**Verification Controls** and **Connectivity Assignment** fields can be used to set the default constraint group from which Layout XL derives the extractable layers in the design and to control the connectivity of the design.
Annotations	Used to report the violations in the design.
Limits	**Maximum number of opens**, **Maximum number of shorts**, and **Maximum number of illegal connections** are specified here.

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Let us see the *Connectivity* section in the *Layout XL Options* form.

You can use the *Connectivity* options to set the Layout XL parameters for the current design.

This tab is further split into the following tabs, namely: *Controls*, *Annotations*, and *Limits*.

Controls tab has *Verification Controls* and *Connectivity Assignment* fields which can be used to set the default constraint group from which Layout XL derives the extractable layers in the design and to control the connectivity of the design.

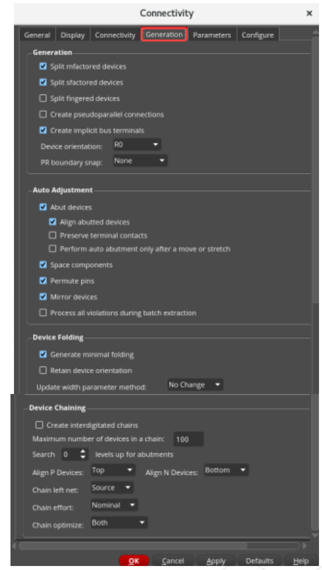
Annotations tab can be used to report the violations in the design.

In *Limits* tab, *Maximum number of opens*, *Maximum number of shorts*, and *Maximum number of illegal connections* can be specified.

Layout XL Options Form: Generation Section

General > Display > Connectivity > **Generation** > Parameters > Configure

Layout XL Options Form



Use the **Generation** options to specify how certain components are handled when they are generated in the layout view.

Generation	Controls how Layout XL places schematic devices.
Auto Adjustment	Controls the abutment of devices, aligning abutted devices, preserving terminal contacts, performing auto abutment only after a move or stretch, spacing components, permuting pins, mirroring devices, and processing all violations during batch extraction.
Device Folding	Controls the number of folded devices, device orientation, and updating the width parameter method.
Device Chaining	Controls the creation of interdigitated chains, Maximum number of devices in a chain, searching levels up for abutments, aligning P/N devices, and chaining the left net.

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Let us see the *Generation* section in the *Layout XL Options* form.

You can use the *Generation* options to specify how certain components are handled when they are generated in the layout view.

Generation field controls how Layout XL places the schematic devices.

Auto Adjustment field controls the abutment of devices, aligning abutted devices, preserving terminal contacts, performing auto abutment only after a move or stretch, spacing components, permuting pins, mirroring devices, and processing all violations during batch extraction.

Device Folding field controls the number of folded devices, device orientation, and updating the width parameter method.

Device Chaining field controls the creation of interdigitated chains, Maximum number of devices in a chain, searching levels up for abutments, aligning P and N devices, and chaining the left net.

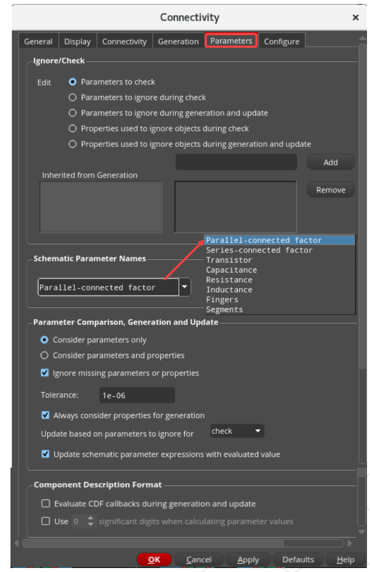
Layout XL Options Form: Parameters Section

General | Display | Connectivity | Generation | **Parameters** | Configure

Use the **Parameters** options to specify which parameters are to be ignored by the generation and check command and how parameters are compared during *Check Against Source*.

Ignore/Check	Lets you specify the names of parameters to be ignored (or checked) during check and update commands and the names of properties that cause objects to be ignored during generation and check.
Schematic Parameter Names	Lets you specify the names of special parameters used in the schematic. Choose the parameter you want to edit from the cyclic field and type the names in the text field.
Parameter Comparison, Generation and Update	Defines how CDF parameters and properties are compared during the *Check Against Source*, *Update Layout Parameters*, and *Update Schematic Parameters*.
Component Description Format	**Evaluate CDF callbacks during generation and update** causes all SKILL callbacks defined on CDF parameters to be evaluated by default when you run the *Generate* and *Update* commands in the Virtuoso Layout Suite XL. **Use significant digits when calculating parameter values** controls the number of significant digits used by Layout XL when calculating the value of layout CDF parameters.

Layout XL Options Form



Let us see the *Parameters* section in the *Layout XL Options* form.

You can use the *Parameters* options to specify which parameters are to be ignored by the generation and check command, and how parameters are compared during *Check Against Source*.

Ignore or Check field lets you specify the names of parameters to be ignored or checked during check and update commands and the names of properties that cause objects to be ignored, during generation and check.

Schematic Parameter Names field lets you specify the names of special parameters used in the schematic.

Choose the parameter you want to edit from the cyclic field and type the names in the text field.

Parameter Comparison, Generation and Update field defines how CDF parameters and properties are compared during the *Check Against Source*, *Update Layout Parameters*, and *Update Schematic Parameters* commands.

In *Component Description Format* field, *Evaluate CDF callbacks during generation and update* option causes all SKILL callbacks defined on CDF parameters to be evaluated by default when you run the *Generate* and *Update* commands in Virtuoso Layout Suite XL.

Use significant digits when calculating parameter values option controls the number of significant digits used by Layout XL when calculating the value of layout CDF parameters.

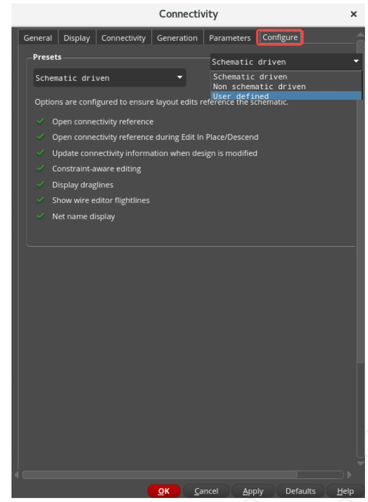
Layout XL Options Form: Configure Section

General › Display › Connectivity › Generation › Parameters › **Configure**

Layout XL Options Form

Use the **Configure** options to know the values that environment variables that are associated with the Preset options are set to when the Layout XL Options form is launched.

Presets	**Schematic driven** reflects that all the form fields associated with the Preset options have their value set to ON, implying the option is selected. **Non schematic driven** reflects that all the form fields associated with the Preset options have their value set to OFF, implying the option is deselected. **User defined** reflects that the form fields associated with the Preset options are neither Schematic driven (all fields set to ON) nor Non schematic driven (all fields set to OFF).



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Let us see the *Configure* section in the *Layout XL Options* form.

You can use the *Configure* options to know the values that environment variables that are associated with the Preset options are set to when the Layout XL Options form is launched.

In Presets field, *Schematic driven* option reflects that all the form fields associated with the Preset options have their value set to ON, implying the option is selected.

Non schematic driven option reflects that all the form fields associated with the Preset options have their value set to OFF, implying the option is deselected.

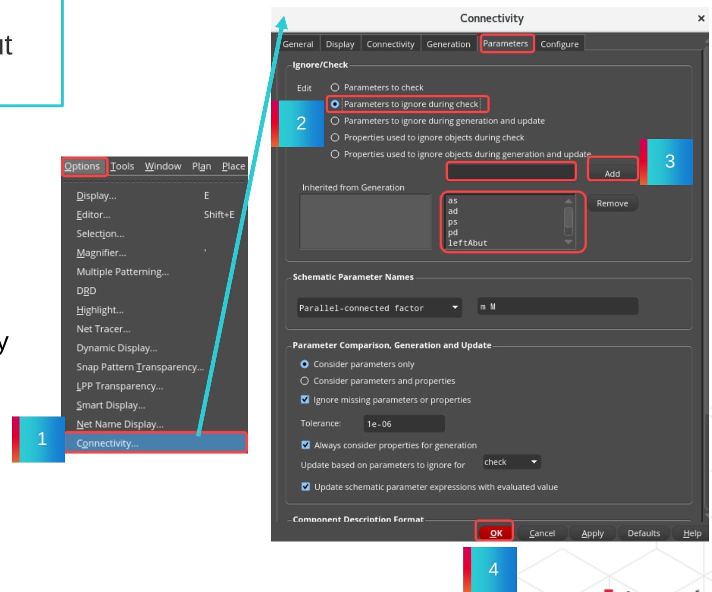
User defined option reflects that the form fields associated with the Preset options are neither Schematic driven, that is, all fields set to ON, nor Non schematic driven, that is, all fields set to OFF.

How to Use the Layout Connectivity Options Form

You can use the **Connectivity Options** form to set Layout options either for the current cellview or for the current Layout EXL session.

1. In the layout view, choose the **Options – Connectivity** menu command to open the connectivity form.
2. In the **Parameters** tab, select **Parameters to ignore** *during check* option.
3. **Add** the following seven parameters separated by a space into the *Ignore* field:
   - *as ad ps pd leftAbut rightAbut simW*
4. Click **OK** in the connectivity.

Result: These seven parameters are ignored during the check between schematic and layout.



Let us see how to use the Connectivity *Options* form.

You can use the Connectivity *Options* form to set the Layout options either for the current cellview or for the current Layout EXL session.

Firstly, in the design canvas, choose the *Options – Connectivity* menu command to open the Layout EXL Options form.

Next, in the *Parameters* tab, select the option: *Parameters to ignore during check*.

Then add these seven parameters separated by a space into the Ignore field.

And click *OK* in the form.

You can observe that these seven parameters are ignored during the check between schematic and layout.

Verifying the Design

You can use the **Check Against Source** command to show any differences between the schematic and the layout.

You can check the layout cellview against the schematic source cellview.

1. In the layout canvas, choose the **Connectivity – Check – Against Source** menu command or click the **Check Against Source (CAS)** icon in the Layout XL toolbar to open the Check Against Source form.
2. Keep the required options in the CAS form.
3. Click **OK** in the CAS form.

Result: The *Info* window opens with information about the differences between the layout and the schematic. You can check the *casLog* file as well from the current working directory.

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Let us see how to verify the design.

You can use the *Check Against Source* command to show any differences between the schematic and the layout.

You can check the layout cellview against the schematic source cellview.

To check against the source, firstly, in the layout canvas, choose the *Connectivity – Check – Against Source* menu command or click the *Check Against Source* icon in the Layout XL toolbar to open the CAS form.

Keep the required options.

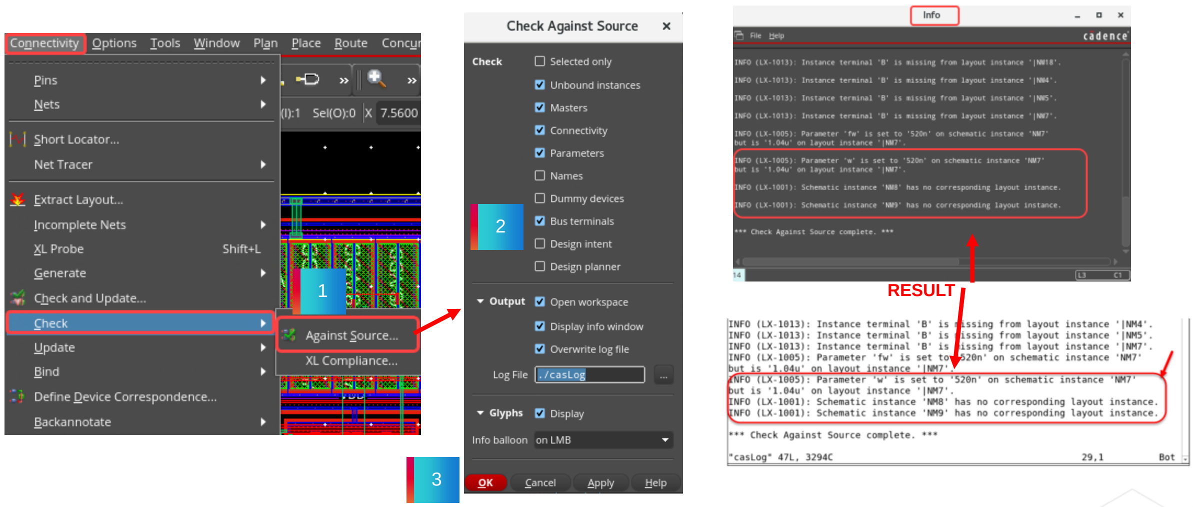
And click *OK* in the CAS form.

You can observe that the *Info* window opens with information about the differences between the layout and the schematic.

You can check the *casLog* file as well from the current working directory.

Verifying the Design (continued)

Scroll to the top and bottom of the report and notice that two instances (**NM8** and **NM9**) are missing in the layout and that the width parameter on the **NM7** device has changed.



After invoking the CAS form, set the required options and click *OK* in the form.

You can observe that the *Info* window opens with information about the differences between the layout and the schematic.

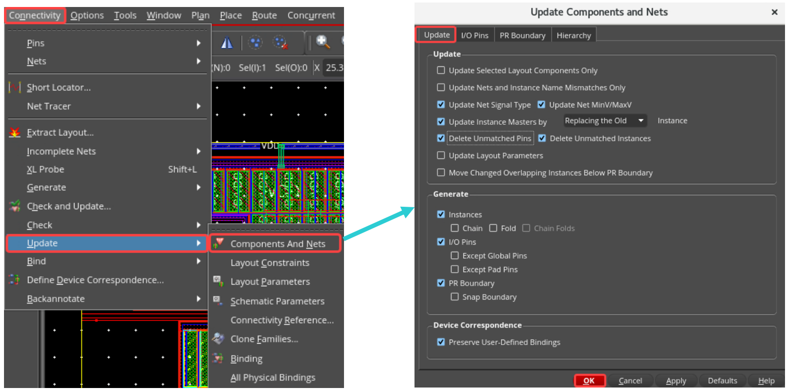
This information is available in the *casLog* file as well from the current working directory.

Scroll to the top and bottom of the report and notice that two instances (*NM8* and *NM9*) are missing in the layout and that the width parameter on the *NM7* device has changed.

Invoking the Update Components and Nets Form

To automatically update your layout in the design canvas, choose the **Connectivity – Update – Components And Nets** menu command or click the **Update Components and Nets** (**UCN**) icon in the Layout EXL toolbar to display the Update Components and Nets form.

Use the *Update Components and Nets* form to automatically update your layout to take account of the instances, pins, and connectivity that you have changed in the schematic.



Let us see how to invoke the *Update Components and Nets* form.

To automatically update your layout in the design canvas, choose the *Connectivity – Update – Components And Nets* menu command or click the *Update Components and Nets* icon in the Layout EXL Toolbar to display the Update Components and Nets form.

You can use this form to automatically update your layout to take account of the instances, pins, and connectivity that you have changed in the schematic window.

Using the Update Components and Nets Form

The **Update Components and Nets** form is split into the following tabs:

Update tab, *I/O Pins tab*, *PR Boundary tab*, and *Hierarchy tab*.

You can use these tabs to specify the required inputs to update your components and nets.

While the *Generate All From Source* command deletes any existing components in the layout view and regenerates everything from scratch, the *Update Components And Nets* command updates only the components that have changed.

The *Update Components and Nets* form is split into the following tabs: namely, *Update* Tab, *I/O Pins* Tab, *PR Boundary* Tab, and *Hierarchy* Tab.

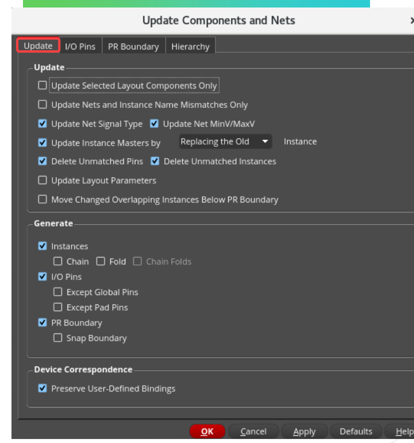
You can use these tabs to specify the required inputs to update your components and nets.

While the *Generate All From Source* command deletes any existing components in the layout view and regenerates everything from scratch, the *Update Components And Nets* command updates only the components that have changed.

Update Components and Nets Form: Update Tab

Update → I/O Pins → PR Boundary → Hierarchy

Update Components and Nets Form



Update	The **Update** group box lets you specify how much of the design is updated and what happens to existing layout components that need to be changed or removed from the design.
Generate	The **Generate** group box lets you specify which missing design objects are generated in the layout view.
Device Correspondence	**Preserve User-Defined Bindings** preserves user-defined bindings of devices between the schematic and the layout.

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Let us see the *Update* tab in the *Update Components and Nets* form.

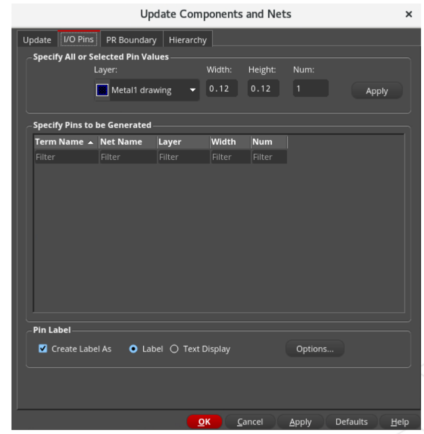
The *Update* group box lets you specify how much of the design is updated and what happens to the existing layout components that need to be changed or removed from the design.

The *Generate* group box lets you specify which missing design objects are generated in the layout view.

In *Device Correspondence* group box, *Preserve User-Defined Bindings* option preserves the user-defined bindings of the devices between the schematic and the layout.

Update Components and Nets Form: I/O Pins Tab

Update Components and Nets Form



Default Values for Electrical Pins	Lets you specify attribute values and *Apply* them to all the pins shown in the list box. Click *Apply* to apply the *Layer*, *Width*, *Height*, *Num*, and *Create* settings for all the listed pins.
Specify Pins to be Generated	Lets you select pins from the list box and update the attribute values used when those pins are generated in the layout. Click *Update* to update the *Layer*, *Width*, *Height*, *Num*, and *Create* settings for the currently selected pins.
Pin Label	Specifies the type of label generated when you create a pin. This setting is honored by the *Generate All From Source* and *Generate Selected From Source* commands.

Let us see the *I/O Pins* tab in the *Update Components and Nets* form.

Default Values for Electrical Pins option lets you specify the attribute values and *Apply* them to all the pins shown in the list box.

Click the *Apply* button to apply for the *Layer*, *Width*, *Height*, *Number*, and *Create* settings for all the listed pins.

Specify Pins to be Generated option lets you select the pins from the list box and update the attribute values used when those pins are generated in the layout.

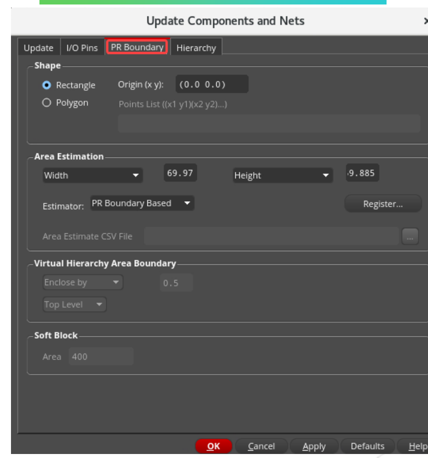
Click the *Update* button to update the *Layer*, *Width*, *Height*, *Number*, and *Create* settings for the currently selected pins.

Pin Label option specifies the type of label generated when you create a pin.

This setting is honored by the *Generate All From Source* and *Generate Selected From Source* commands.

Update Components and Nets Form: PR Boundary Tab

Update Components and Nets Form



Shape	The **Shape** group box specifies whether the place and route boundary is a rectangle or a polygon.
Area Estimation	The **Area Estimation** group box specifies how the system calculates the shape and size of a rectangular boundary.

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Let us see the *PR Boundary* tab in the *Update Components and Nets* form.

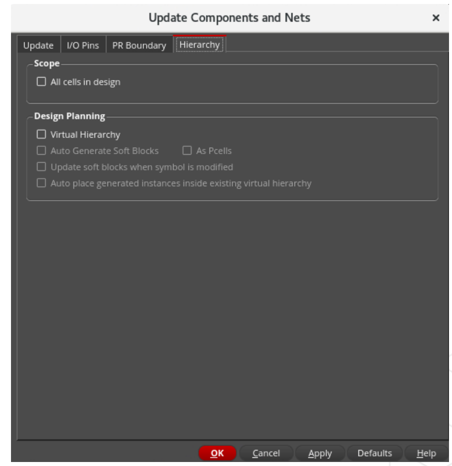
The *Shape* group box specifies whether the place and route boundary is a rectangle or a polygon.

The *Area Estimation* group box specifies how the system calculates the shape and size of a rectangular boundary.

Update Components and Nets Form: Hierarchy Tab

Update → I/O Pins → PR Boundary → Hierarchy

Update Components and Nets Form



Scope	The **Scope** group box specifies the extent in the hierarchy to which the design is updated. **All cells in design** checkbox specifies that the selected updates are carried out to all the non-transparent cellviews in the design hierarchy.

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Let us see the *Hierarchy* tab in the *Update Components and Nets* form.

The *Scope* group box specifies the extent in the hierarchy to which the design is updated.

All cells in design option specify that the selected updates are carried out to all the nontransparent cellviews in the design hierarchy.

Updating the Components and Nets

You can use the **Update Components and Nets** form to automatically update your layout to take account of the instances, pins, and connectivity that you have changed in the schematic.

1. In the design canvas, choose the **Connectivity – Update – Components And Nets** menu command or click the **Update Components and Nets** (**UCN**) icon in the Layout XL toolbar to open the Update Components and Nets form.
2. Keep the required options in the UCN form.
3. Click **OK** in the UCN form.

Let us see how to update the components and nets in your design.

You can use the *Update Components and Nets* form to automatically update your layout to take account of the instances, pins, and connectivity that you have changed in the schematic window.

To update the components and nets in your design, firstly, in the design canvas, choose the *Connectivity – Update – Components And Nets* menu command or click the *Update Components and Nets* icon in the Layout XL toolbar to open the Update Components and Nets form.

Keep the required options.

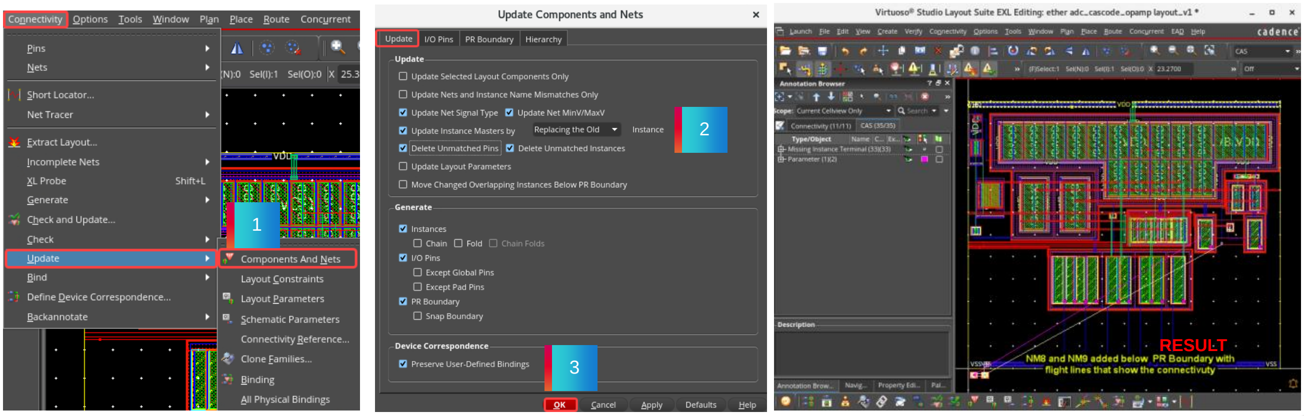
And click *OK* in the UCN form.

You can observe that there are two devices added below the *PR Boundary*: namely, *NM8* and *NM9*.

Also, there are flight lines now that show the connectivity.

Updating the Components and Nets (continued)

Result: Two devices **NM8** and **NM9** are added below the **PR Boundary** with flight lines that show the connectivity.

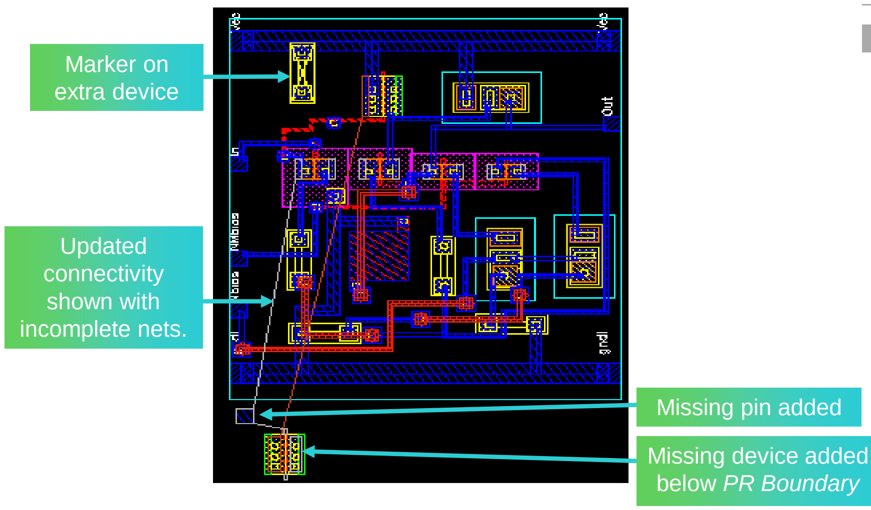


After invoking the *Update Components and Nets* form, set the required options and click *OK* in the form.

You can notice that two devices, *NM8* and *NM9*, are added below the *PR Boundary*, with flight lines that show the connectivity.

Updating the Components and Nets (continued)

Update Components and Nets command updates the layout with missing devices, I/O pins, markers on extra devices, and updated connectivity.



Update Components and Nets command updates the layout with missing devices, I/O pins, markers on extra devices, and updated connectivity.

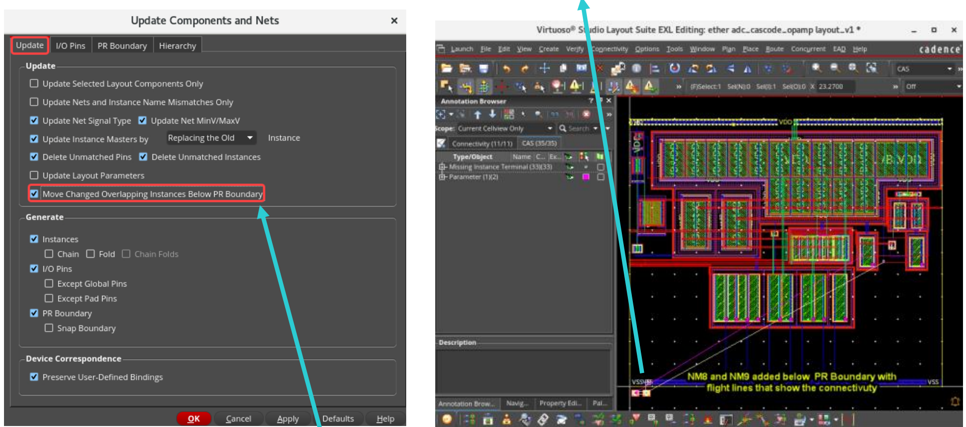
When the command is run, missing devices and pins are added below the *PR Boundary*.

Markers on extra devices are shown.

Updated connectivity are shown with incomplete nets.

Updating the Components and Nets: Advantage

Advantage: You can see clearly where the new or updated instances are located.



Choose this option to place new instances below the *PR Boundary* during an update.

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Choose *Move Changed Overlapping Instances Below PR Boundary* option to place the new instances below the *PR Boundary* during an update.

The advantage is that you can see clearly where the new or updated instances are located.

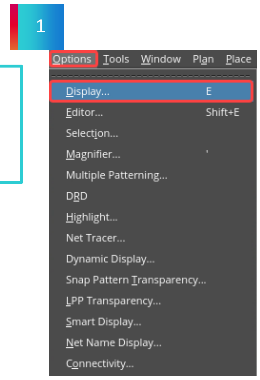
Setting the Snap Modes

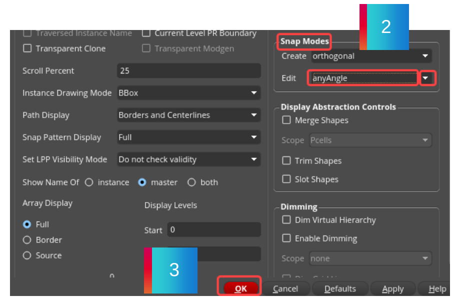
You can use the **Display Options** form to control the appearance of objects and the behavior of commands in the cellview.

1. To set the Snap Modes, choose the **Options – Display** menu command or press the bindkey **E** from the layout window to open the Display Options form.
2. In the **Snap Modes** section, change the *Edit* field from *orthogonal* to **anyAngle**.
3. Click **OK** in the form.

Result: Changing the snap mode to **anyAngle** makes it easier to move devices.

They will move at any angle rather than being locked into an orthogonal direction.





We will see how to set the *Snap Modes*, which makes it easier to move the devices.

You can use the *Display Options* form to control the appearance of objects and the behavior of commands in the cellview.

To set the *Snap Modes*, firstly, choose the *Options – Display* menu command or press the bindkey E from the layout window to open the Display Options form.

Then in the *Snap Modes* section, change the *Edit* field from *orthogonal* to *anyAngle*.

And click *OK* in the form.

You can observe that changing the snap mode to *anyAngle* makes it easier to move the devices.

They will move at any angle rather than being locked into an orthogonal direction.

Updating the Layout Parameters

You can use the **Update Layout Parameters** command to update the parameter values of devices in the layout to match the values in the schematic.

To update the layout parameters in the design canvas:

1. Choose the **Connectivity – Update – Layout Parameters** menu command or click the **Update Layout Parameters** icon in the Layout EXL toolbar.
2. An Info window appears with information about which devices have been updated. The parameter values of devices in the layout are updated to match the values in the schematic. Here fw and w parameter values on layout instance NM7 are set to 520n.

We will see how to update the layout parameters.

If you have changed any of the parameters in your schematic, you can use the *Update Layout Parameters* command to update the parameter values of devices in the layout to match the values in the schematic.

To update the layout parameters, firstly, in the design canvas, choose the *Connectivity – Update – Layout Parameters* menu command or click the *Update Layout Parameters* icon in the Layout EXL toolbar to open the *Update Layout Device List* form.

Then select all the devices in the Update Layout Device List form.

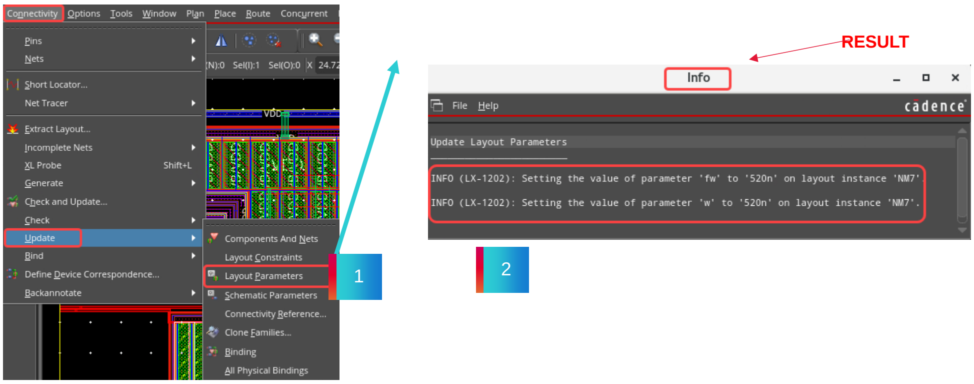
And click *Apply* in the form.

You can observe that an *Info* window appears with information about which devices have been updated.

The parameter values of devices in the layout are updated to match the values in the schematic.

Updating the Layout Parameters (continued)

The software checks the parameter values on the specified devices in the schematic against the values in the layout and updates the layout parameters when it finds differences (unless you have set the **lvsIgnore** or **ignore** property on a device).



Result: An Info window appears with information about which devices have been updated. The parameter values of devices in the layout are updated to match the values in the schematic. Here **fw** and **w** parameter values on layout instance **NM7** are set to **520n**.

The software checks the parameter values on the specified devices in the schematic against the values in the layout and updates the layout parameters, when it finds differences (unless you have set the *lvsIgnore* or *ignore* property on a device).

After invoking the *Update Layout Parameters* command in the displayed *Update Layout Device List* form, select all the devices and click *Apply* in the form.

Selecting all the devices in the list and applying the update to the layout parameters checks all devices in the design.

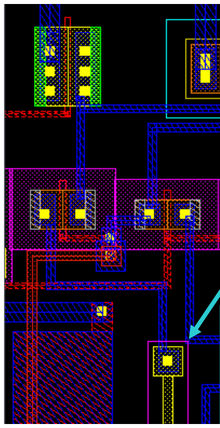
Since you might not be aware of which devices the schematic designer changed, this ensures all the devices are checked and updated.

You can notice that an *Info* window appears with information about which devices have been updated.

Here, *fw* and *w* parameter values on layout instance *NM7* are updated.

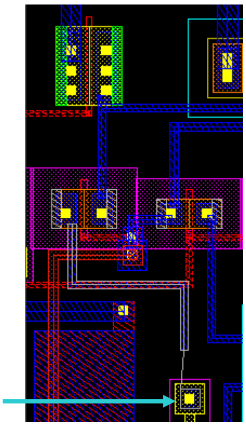
Updating the Layout Parameters (continued)

Result: The parameter values of devices in the layout are updated.



Resistor size reflects schematic parameter value.

Incomplete net flight lines update to show new open.



Before

After

The parameter values of the devices in the layout are updated.

The resistor size reflects the schematic parameter value.

Incomplete net flight lines update, which shows the new open.

The layout before and after updating the parameters are shown.

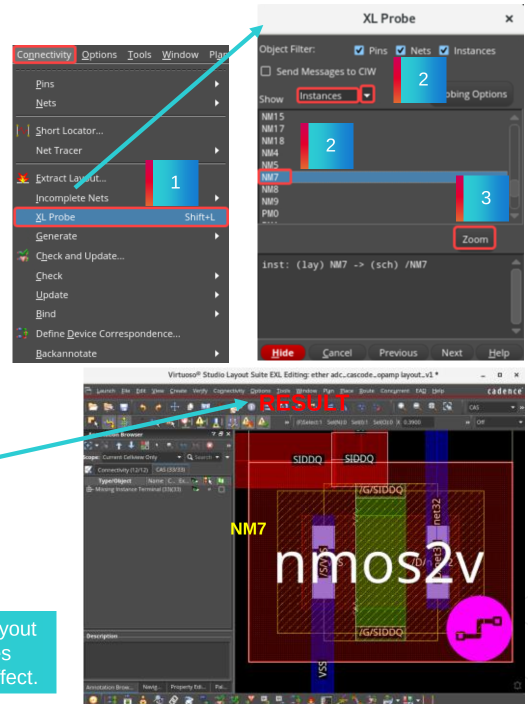
Probing the Devices

You can use the **XL Probe** command to probe the objects when you click in the layout or in the schematic window.

To probe the devices in the layout,

1. Choose the **Connectivity – XL Probe** menu command or press the bindkey **Shift+L** to invoke the XL Probe form.
2. Set *Show* criteria to, say for example, **Instances**, and click the device **NM7** in the XL Probe form.
3. Click the **Zoom** button in the XL Probe form.

Result: This action brings you right to the target device in this example **NM7**.



Object Filter specifies the types of objects that can be probed when you click in the layout or in the schematic window. For example, if you check only Pins, you can create probes only for pin objects by clicking in either window. Clicking on a net or instance has no effect.

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We will see how to probe the devices.

You can use the *XL Probe* command to probe the objects when you click in the layout or in the schematic window.

Object Filter option specifies the types of objects that can be probed when you click in the layout or the schematic window.

For example, if you check only *Pins*, you can create probes only for pin objects by clicking in either window.

Clicking on a net or instance has no effect.

To probe the devices in the layout, firstly, in the design canvas, choose the *Connectivity – XL Probe* menu command or press the bindkey *Shift+L* to invoke the XL Probe form.

Then set the *Show* criteria to, say, for example, *Instances* and click the device *NM7*.

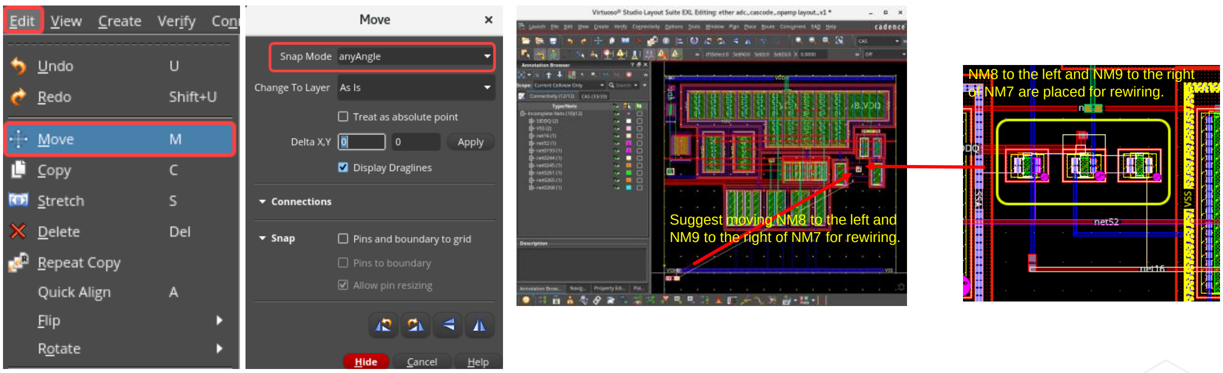
And click the *Zoom* button in the XL Probe form.

You can notice that this action brings you right to the target device.

Placing the Devices

To move and place the devices in the design canvas, choose the **Edit – Move** menu command or click the **Move** icon in the Edit toolbar or press the bindkey **M** to invoke the Move form.

Use the **Move** command to move and place the instances **NM8** and **NM9** from below the *prBoundary* into the lower right design area.



We will see how to move and place the devices.

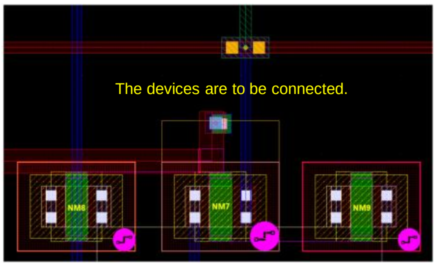
To move and place the devices, in the design canvas, choose the *Edit – Move* menu command or click the *Move* icon in the Edit toolbar or press the *M* bindkey to invoke the Move form.

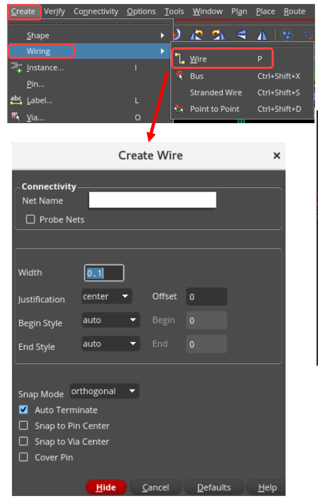
Here you have used the *Move* command to move and place the instances *NM8* and *NM9* from below the *prBoundary* into the lower right design area.

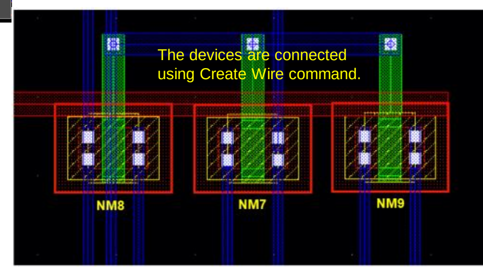
Wiring the Devices

To create wiring for the devices in the layout canvas choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.

Using the *Create Wire* form, you can create pathSegs and vias in routes.







To create wiring for the devices in the design canvas choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire form.

Using the *Create Wire* form, you can create pathSegs and vias in the routes.

In this example, the devices are connected using the *Create Wire* command.

Setting your design

You might want to complete a design that you started with the Layout Editor using Layout EXL. Use this procedure to set up your design.

- Use the **Connectivity Reference option** to update your layout using a schematic different from the default schematic.
- Use the **Connectivity – Check – Against Source** to analyze the differences between schematic and layout.
- Use the **Connectivity – Update – Components And Nets** to update the connectivity and identify any missing devices or pins as well as extra devices.
- Use the **Connectivity – Define Device Correspondence** to identify which layout device is associated with which schematic symbol.
- Use the **Connectivity – Update – Layout Parameters** to update the values of the parameters on the existing layout devices.

If Layout EXL did not create a device instance in your layout, the connectivity and parameters on that device will not be maintained. Choose **Connectivity – Define Device Correspondence** to specify the correspondence between a device instance in your layout and a symbol in the schematic.

First, you click the symbol in the schematic and then click the instance in the layout. If you have set the *ignore* or *lvsIgnore* property on a device, and then you try to set corresponding devices, a dialog box appears asking whether you want the *ignore* or *lvsIgnore* property to be removed. Click **YES**. Otherwise, the command is cancelled and the devices will not be corresponding.

Engineering Change Order (ECO)

Follow these steps to make changes to the schematic and then have them updated in the layout view.

1. Make the required changes to the schematic.
2. Run schematic extraction by running **File – Check and Save**.
3. Choose **Connectivity – Update – Components And Nets** to update the layout with the new schematic connectivity.
4. Choose **Connectivity – Update – Layout Parameters** to update the layout with the new schematic parameters.
5. Delete extra devices which are denoted with markers.
6. Move new devices and pins into place from below the prBoundary.
7. Delete wiring shapes associated with shorts.
8. Route the design with the Virtuoso® Space-based Router tool (optional).
9. Verify the design.

Editing the Schematic

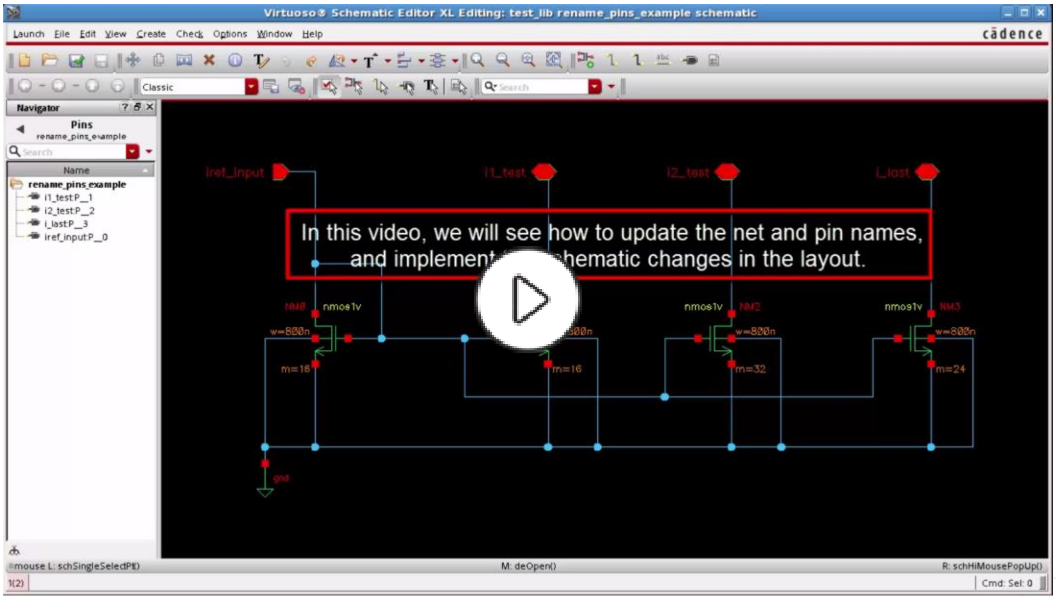
After you have completed your layout using Layout EXL, you might need to make changes to the schematic. These changes can include removing and adding devices, removing and adding wires, or rewiring nets. After you have edited the schematic, you must run extraction to verify that you have a clean schematic.

Instead of editing the schematic, you might receive changes to the schematic or a completely new schematic from another member of your design group. If you receive a completely new schematic, choose **Connectivity – Update – Connectivity Reference** to point your layout to the new schematic. Then follow the procedure above, starting from step three.

After step three, you can use the **Annotation Browser** assistant to see any resulting changes in connectivity.

You can use the **Edit – Undo** command to undo any changes that you made with the **Connectivity – Update – Components And Nets** command.

Demo: How to Update the Net and the Pin Names and Implement the Schematic Changes in the Layout



Video Play Time: 2.22 minutes

Click the Play button to start the video.

Click the link https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1O3w000009bgvoEAA&pageName=ArticleContent to play the video on support.cadence.com.

Module Summary

In this module, you learned how to

- Update the Connectivity Reference of the layout using a schematic different from the default schematic
- Examine the differences between the original and the new schematic views
- Verify data between schematic and layout views
- Update components and nets to implement an Engineering Change Order (ECO)
- Update the layout parameters with the new schematic parameters
- Place the new devices within the prBoundary using the Annotation Browser assistant
- Create wiring for the components
- Update the layout to reflect the pin names changes between the schematic and layout using the Update Components and Nets (UCN) command

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Labs

Lab 3-1 Implementing an Engineering Change Order (ECO)

Lab 3-2 Updating the Net and the Pin Names

This brings you to lab exercise on Module 3.

In Lab 1, you will explore and implement an Engineering Change Order.

In Lab 2, you will update the Net and the Pin Names.

FAQ: Schematic/Layout Views

When you make changes to the schematic and then have them updated in your layout view or associate a new schematic with your layout view, what is it called?

When you make changes to the schematic and then have them updated in your layout view or associate a new schematic with your layout view, it is called an Engineering Change Order (ECO).

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FAQ: Check Against Source Form

How do you invoke the Check Against Source form?

In the design canvas, choose the **Connectivity – Check – Against Source** menu command or click the **Check Against Source** icon in the Layout EXL toolbar to invoke the Check Against Source form.

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FAQ: Updating Layout View Using Different Schematic View

Which command can you use to update your layout using a schematic different from the default schematic?

You can use the **Connectivity – Update – Connectivity Reference** menu command to update your layout using a schematic different from the default schematic.

This page does not contain notes.

FAQ: Probing the Devices

Which command can you use to probe the devices in your layout view?

You can use the XL Probe command to probe the devices in your layout view.

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FAQ: Updating Components and Nets Form

What is the advantage of choosing Move Changed Overlapping Instances Below PR Boundary option in the Update Components and Nets form?

When you choose Move Changed Overlapping Instances Below PR Boundary option in the UCN form, new/updated instances are placed below the PR Boundary during an update. The advantage is that you can see clearly where the new/updated instances are located.

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Quiz: Updating the Schematic View

Which command can you use to update the source schematic for your layout view and to change the physical configuration which specifies how the layout is generated from the source schematic?

[Select the single correct choice]

A. Update Layout Parameters

B. Update Connectivity Reference

C. Update Components And Nets

D. Check Against Source

E. None of the above

Answer: B

You can use the **Update Connectivity Reference** command.

This page does not contain notes.

Quiz: Viewing the Violations

Which assistant can be used to view the violations in the design?

[Select the single correct choice]

A. Search
B. Wire Assistant
C. Navigator
D. Annotation Browser
E. None of the above

Answer: D

Violations in the design are reported in the **Annotation Browser** assistant.

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Quiz: Setting the Snap Modes

Which form can you use to set the **Snap Modes** for **Create** / **Edit** commands?

[Select the single correct choice]

A. Display Options

B. Layout XL Options

C. Define Device Correspondence

D. Check Against Source

E. None of the above

Answer: A

You can use the **Display Options** form.

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Quiz: ECO

What is **ECO** stands for? ------ ------ ------

[Enter the answer in the blank field]

Answer: **Engineering Change Order**

ECO stands for **Engineering Change Order**.

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Quiz: Updating the Layout

Which command can you use to update the connectivity of all existing components in the layout to match the schematic connectivity?

[Select the single correct choice]

A. Connectivity – Check – Against Source
B. Connectivity – Update – Connectivity Reference
C. Connectivity – Update – Components And Nets
D. Connectivity – Define Device Correspondence
E. None of the above

Answer: C

You can use the **UCN** command.

This page does not contain notes.

Quiz: Updating the Parameters Between Schematic/Layout

Which commands can you use to update the parameters between the schematic and layout views?

[Select all the correct choices]

A. Connectivity – Define Device Correspondence

B. Connectivity – Update – Components And Nets

C. Connectivity – Update – Layout Parameters

D. Connectivity – Update – Schematic Parameters

E. All of the above

Answer: C and D

Update Layout Parameters and **Update Schematic Parameters** commands can be used.

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Quiz: VLS EXL

If your schematic is not opened for editing, then **Connectivity – Update – Schematic Parameters** menu command is grayed out in the Virtuoso Layout Suite EXL pull-down menu.

[Select the single correct choice from the two choices]

A. True

B. False

Answer: A

You cannot update schematic parameters when schematic cellview is in read-only mode.

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Quiz: GFS and UCN Commands

Generate All From Source and **Update Components and Nets** commands delete any existing components in the layout view and regenerates everything from scratch.

[Select the single correct choice from the two choices]

A. True

B. False

Answer: B

While **GFS** deletes any existing components in the layout view and regenerates everything from scratch, **UCN** updates only the components that have changed.

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Quiz: UCN Form

Which are the tabs available in the **Update Components and Nets** form?

[Select all the correct choices]

A. Update
B. I/O Pins
C. PR Boundary
D. Hierarchy
E. General
F. All of the above

Answer: A, B, C, and D

UCN form is split into **Update** tab, **I/O Pins** tab, **PR Boundary** tab, and **Hierarchy** tab.

This page does not contain notes.

Quiz: Layout EXL Options Form

You can use the **Layout EXL Options** form to set Layout EXL options either for the current cellview or for the current Layout EXL session.

[Select the single correct choice from the two choices]

A. True

B. False

Answer: A

You can use the **Layout EXL Options** form to set Layout EXL options either for the current cellview or for the current Layout EXL session.

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Module 4

Course Conclusions

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Summary

In this course, you learned to

- Generate clones as Free Objects, Grouped Objects, and Synchronized family
- Generate synchronous copy clones of wires
- Examine synchronous behavior in the synchronous copy clones of wires created by the copy command
- Generate synchronous copy clones of modgen
- Examine synchronous behavior in the synchronous copy clones of modgen created by the copy command
- Generate mutant clones with exact connectivity turned ON/OFF with relax match options
- Create a repetitive layout cell using the clone and the Copy commands
- Implement an Engineering Change Order (ECO)
- Update the net and the pin names

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References

Refer to the link below for further details on the topics in this course

Virtuoso Layout Suite XL: Basic Editing User Guide IC25.1

Virtuoso Layout Suite EXL Reference IC25.1

Virtuoso Layout Suite XL: Connectivity Driven Editing User Guide IC25.1

Virtuoso Layout Suite XL User Guide Product Version IC6.1.8

Virtuoso Layout Suite XL: Basic Editing User Guide IC25.1

Virtuoso Layout Suite EXL Reference IC25.1

Virtuoso Layout Suite XL: Connectivity Driven Editing User Guide IC25.1

Virtuoso Layout Suite XL User Guide Product Version IC6.1.8

Module 5

Next Steps

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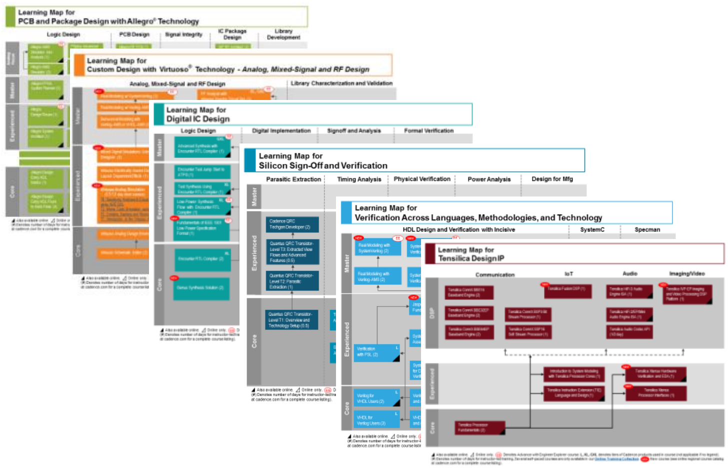
Cadence® Training Services learning maps provide a comprehensive visual overview of the learning opportunities for Cadence customers.

Click here to see all our courses in each technology area and the recommended order in which to take them.



Go here to view the learning maps:

http://www.cadence.com/Training/Pages/learning_maps.aspx

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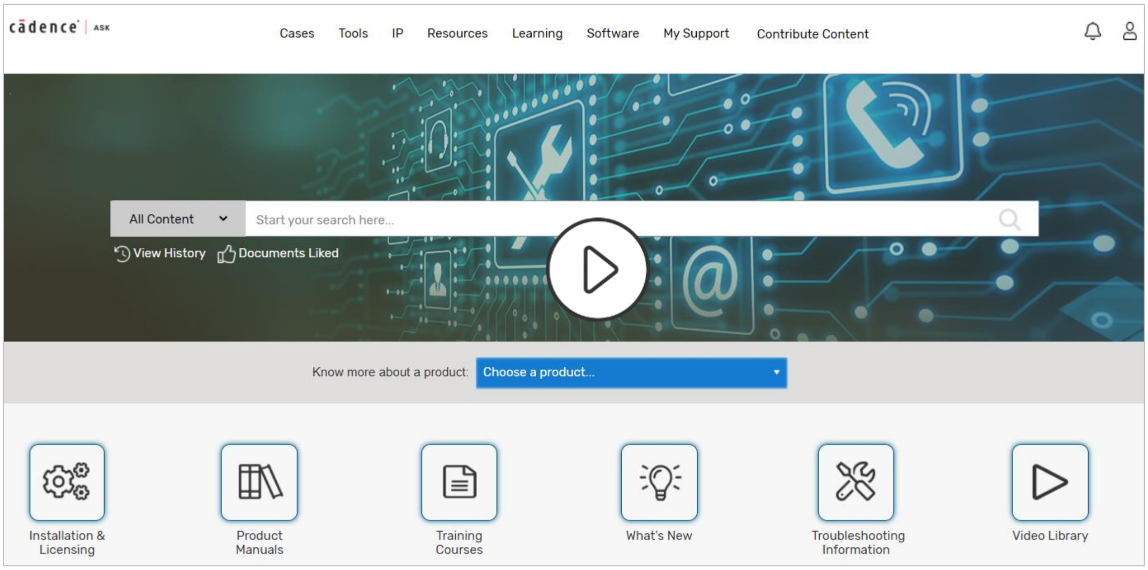

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Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion
- Complete the Course's Digital Badge exam to earn your Badge

Thank you!

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